Bengaluru's suburban woes: Studying the impacts of an expanding industry on green spaces of the I. T. belt, an ecosystem services approach

Dissertation submitted in partial fulfilment of

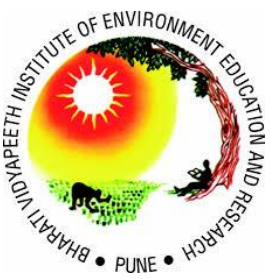
Master's Degree in Wildlife Conservation Action

Rajat Rao

Under the guidance of

Prof. Dr. Kranti Yardi





Institute of Environment Education and Research

Bharati Vidyapeeth Deemed University

Pune, Maharashtra, India

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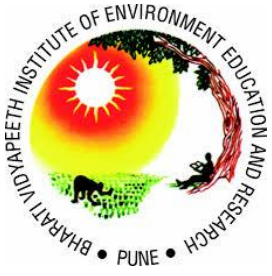
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CERTIFICATE

This is to certify that **Mr. Rajat Rao**, final year student at the Bharati Vidyapeeth (Deemed to be University) Institute of Environment Education and Research, Pune has successfully completed the dissertation titled, **'Bengaluru's suburban woes: Studying the impacts of an expanding industry on green spaces of the I. T. belt, an ecosystem services approach'** under the guidance of **Prof. Dr. Kranti Yardi**, in partial fulfilment of the Master's Degree in Wildlife Conservation Action during the academic year 2022-23.

The work has been completed to our satisfaction and standards. The said dissertation has not been submitted for any other degree or diploma and any such material which has been obtained from other sources has been duly acknowledged in the dissertation.

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ABSTRACT

The city of Bengaluru wears many hats, what started off as the “Garden City” went on to quickly become the “Silicon Valley of India.” However, this transition has not been smooth and organic. The unabated growth of high-rise buildings and urban sprawls in what was once forested and a wetland-rich surrounding of the city, is a cause for major concern. The city’s expansion patterns accompanied with the loss of green spaces seems hurried. This study explores the rate of expansion in the taluks that contains the famed IT belt of the city over three time periods (2002, 2012 and 2022). It also uses InVEST Habitat Quality Model to find habitat quality and rarity for the species present in the region. The findings are as follows: The rate of urban expansion has increased rapidly by taking up agricultural spaces and vegetation cover over time. The vegetation cover used to be continuous and now is highly fragmented. The habitat quality and rarity of the study area are deteriorating and regions that have good ecological assets are very few, and present in patches. The purpose of this type of a study is to use the processed-interpreted data to galvanize policy change. It paints a complete picture of the ecological profile of the region so that policymakers can understand the amount of stress it experiences and take necessary steps to remediate it.

Keywords: Ecosystem services, Land use land cover, InVEST, MOLUSCE, Artificial Neural Networks, Urban expansion, Trade-offs.

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I am grateful for the brainstorming sessions with my course mate – Tanaya Rele, who was an excellent sounding board, and helped with the refinement of my thesis.

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Last but not the least, I would like to thank Gullu (my late pet cat), the long nights filled with monologues with him, made me switch lanes from being an IT professional to a conservationist. All that I ever conserve, will be for him.

Rajat Rao

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1. INTRODUCTION

The present study is conducted to understand the services provided by the ecological assets in the South and South-Eastern part of Bengaluru city, assessing the rate of urban expansion and the ecological damages that it causes. At a time when biodiversity is becoming a necessity for urban centres, because without it most cities are experiencing heat islands and other climate related issues (1). It is important to make a justified and a regulated scale of development in cities especially in a rapidly growing one like Bengaluru. This study looks at the expansion of the city over a span of 20 years to identify patterns about land use change and looks to provide recommendations to improve management.

1.1. Bengaluru – India's Silicon Valley – A biodiversity conundrum

Bengaluru used to be a small village around the 12th century and became the capital only after post-independence era (2). It is now one of the largest megacities in the world, going through unprecedented growth and transformation (3). The name of the city is said to be derived from *benga*, which is the local name for *Pterocarpus marsupium*, an indigenous, deciduous tree species, and *ooru*, meaning village or town (4).



Figure 1.1. Pterocarpus marsupium tree. Locally known as benga which gives the city its name. The older parts of the city still have many avenues with this indigenous species growing around temples and parks (5).

However, Kempe Gowda is attributed to the founding of modern Bengaluru in the year 1537. He is also known for building the four towers in different directions from centre of the city known as *Petta*, which defined the boundaries of the city (4).

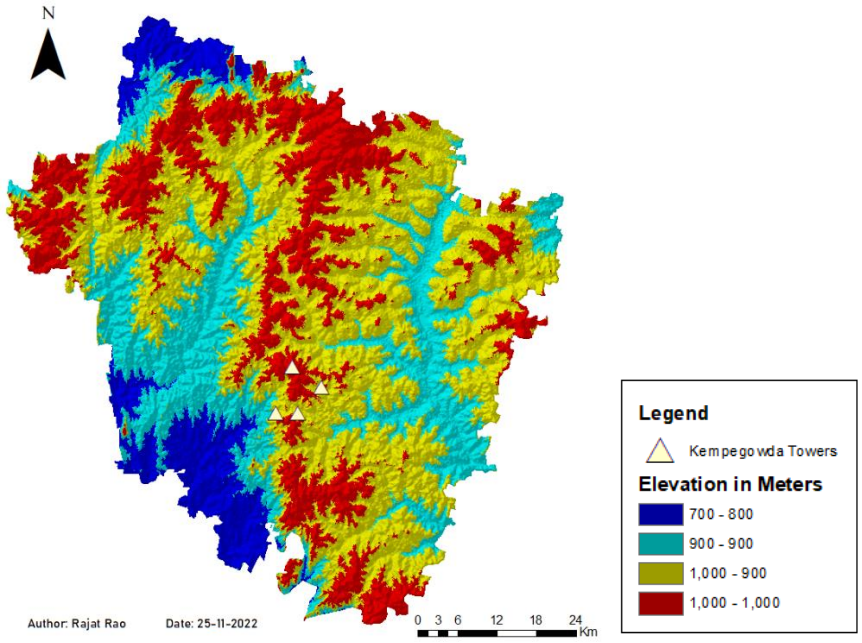


Figure 1.2. Map depicting the planned site selection by Kempe Gowda. He selected the high lying areas for the city. The four towers which are present even today marks the four corners of the erstwhile Bengaluru city. The city was planned with a vision of civilization being on the higher regions and lower elevations in the surroundings to be used for agriculture due to accumulation of rainwater.

Situated at almost an equal distance from the East and West coast of the Indian peninsula, Bengaluru, located at 12°59’ North, 77°57’ East, with an altitude of 920 m, and mean annual rainfall of 880 mm, enjoys a pleasant weather throughout the year except for a couple of summer months (6). The city has several hundreds of lakes, the Bannerghatta National Park at the southern end, and has rocky and boulder regions all around it, which forms a prime Leopard habitat, with its neighbouring district of Tumkur housing one of the highest leopard densities outside a protected area in India (7). The onset of industrial estates began in the periphery of this, almost circular city and became a part of the urban sprawl as the city expanded (3).

At present, with a population of 13 million, Bengaluru is the Information Technology (IT) capital of India and has become the fastest-growing technology

based work ecosystem in the world since 2016, and it is followed by cities like London, Munich, Berlin and Paris in terms of growth (8). In the last few decades, the city has transformed in an observably unprecedented manner into a global centre for information technology companies. Since Bengaluru hosts an excellent start-up atmosphere and an extremely high number of Special Economic Zones (SEZs), it houses more than 67000 registered IT companies within the city (9). Such ambitious developments will only be met with an equal ecological trade-off and can spell a threat to this beautiful garden city.

1.2. Urbanization trends – Trade-offs and limits

Urbanization is a process in which a city grows, and it involves a perpetual competition between demand and supply of resources. It usually occurs when people start moving to a city for better opportunities (10). In India, the process gained momentum from the 1980s when towns and cities gradually transitioned into megacities, this also led to a subsequent opening of the Indian markets to the world (11). Some cities like Mumbai began expanding vertically with construction of high-rise buildings due to lack of space, while cities like Bengaluru began its expansion horizontally as well as vertically, decimating the surrounding green habitats along with depletion of the water tables.

A United Nations (UN) study has concluded that the urbanization rate of the global cities will only increase from 56.2% today to 60.4% by the year 2030, and by 2050, 2.5 billion more people will be living in cities (12). Urbanization rates in developing countries will see only an increase while it decreases in most developed countries. It is predicted that, 96% of growth will occur in urban regions of East Asia, South Asia and Africa, with three countries: China, India and Nigeria, causing maximum growth of 35% between 2018-2050 (13).

A 2020 report by the World Bank emphasizes on the fact that the land use patterns are permanently concretized once a city gets built, therefore it is essential to plan the expansion of cities in a sustainable manner (14). With the established megacities of the world hailing from developed countries like US and UK, we

have developing countries wanting similar amenities and the governments end up selling extravagant visions of smart cities with excellent infrastructure without ideally assessing the ground reality. The major Indian cities of Mumbai, Delhi, Chennai, Pune and Bengaluru do not have the infrastructure to handle rapid urban growth, this leads to ecological habitat related compromises (15).

Bengaluru has grown in an almost amoebic manner, with the city being at its core and engulfing the surrounding satellite towns of Yelahanka, Kengeri, Jigani and Yeshwanthpur to become a part of the city. The same development is expanding along nodes connected to highways, for example: the urbanization of Anekal region on Hosur Road, Hoskote village on Old Madras Road, Devanhalli on New Airport Road, Nelmangala and Dobbaspete on Tumkur Road, Bidadi and Ramnagara on Mysuru Road and Kanakpura on Kanakpura Road (16). The city's population has transitioned to become greater in the periphery as compared to the central region, this means that the outskirts and the green spaces in the periphery are most at risk with increased amount of natural and economic stressors (16).

1.3. Historical, social, and cultural impact of the IT revolution in Bengaluru

Despite companies like Infosys and Wipro setting up shop in Bengaluru since the 1980s, the initial headway towards the expansion of the IT industry occurred in the year 1991, when the Ministry of Electronics and Information Technology set up the very first Software Technology Parks of India (STPI). This STPI provided the right infrastructure for companies to enter the global market and also helped India during the global 'Y2K' crisis (16). The 2000s saw the growth of Indian companies (TCS, Wipro, Infosys) at a rapid pace and the past decade has seen over 35% of India's global centres for international companies being set-up within the city (17). The choice of Bengaluru was not a co-incidence for these companies, but it comes due to the establishment of a unique industrial economy and a vast presence of public sector undertakings. All of this, traced back to the network of institutions established by the first ever engineer of India, Sir M. Visvesvaraya, who was the diwan of the princely state of Mysore since 1911 (18).

Bengaluru is a city which has accepted technology with open arms since the beginning. Even before the pandemic hit, contactless service at restaurants and UPI payments were happening in every part of the city. Most workspaces are now co-workspaces, i.e., spaces where people from different organization just go to for work without any connection to the building (19). Nowadays, most people order grocery and every other item though mobile applications rather than visit a supermarket (20). Most things in this city runs through apps, this shows how ingrained technology and the supporting industries are into the social and cultural aspects of the city (21). This unfortunately, is the type of centralized culture that is influencing the need for more and more resources to make better and better things for the comfort of urban citizens.

1.4. Urban expansion – Where are we headed?

In 2018, a story done by BBC put Bengaluru in a global list of cities that will soon run out of water in the near future; This threw an emphasis on the underlying and often ignored water crisis in the city (16). The IT sector has caused an unprecedented growth in the population leading to increase in the accompanying resources and structures within the city. Eventually this leads to urban planning challenges and improper land management (16). Bengaluru's development has become haphazard and inefficient because the system is not able to manage the large influx of population into the city, which only causes the city to expand more and more, and there comes the question of where is the limit? (16).

It is necessary to establish a balance between green spaces and grey spaces within the city and the response to every problem should not be cutting down another avenue of trees or encroaching another lake bed in the outskirts of the city (22). As the following literature review will highlight, there are very little studies done on the requirement of green spaces around a city like Bengaluru.

Ecosystem Services (ES) is an essential tool in quantifying the necessity of biodiversity and green spaces (23). Ecosystem Services (ES) introduces the concept of natural capital and it includes: land, forests, minerals, fossil fuels,

fisheries, etc., and it is one of the most important sources of human well-being (24). The IPCC's working group 2 report stresses on the importance of studying the ecosystem costs of urbanization, and there is a great level of scientific intrigue worldwide in understanding how urban settlements affect ecosystem services so that sustainable development can be attained (25). Therefore, it becomes important to have an ecological study in the surroundings of an expanding urban centre like Bengaluru.

2. LITERATURE REVIEW

The following review glances at the different types of research from all over the world, specific to the present study. It also looks to critique the concepts and methodologies followed in these literatures. Larger themes are identified, made into sub-headings and the flow of information will be narrative-based under them; Therefore, this review attempts to follow a topical organizing principle.

Around 85 papers were reviewed for knowledge gains about concepts, techniques and methodologies which lead to identification of a research gap. The literature search tool used is Google Scholar and the database library referred to is Scopus. The following keywords were used in search, for identifying broad themes: "Ecosystem Services," Ecosystem Services Bangalore, Ecosystem Services India, "inVEST model," Ecosystem Services Bangalore "inVEST model," Intitle: "inVEST model." *"India," Intitle: "inVEST model" AND "Bangalore," Intitle: "Peri-Urban" AND "Bangalore."

2.1. Concept of Ecosystem Services (ES) and its significance in an urban setting

Even though the Ecosystem Services (ES) concept has been popularized because of the Millennium Ecosystem Assessment (MEA) in 2005 (23), the concept was first introduced in the early 1970s as 'Environmental Services' (26). It was later re-named to 'ecosystem services' during the mid-1980s (27). The MEA's definition of ecosystem services is the most popular one and in use at present. It defines ecosystem services as "The benefits people obtain from ecosystems and it covers provisioning services such as food and water; regulating services such as regulation of floods, drought, land degradation, and disease; supporting services such as soil formation and nutrient cycling; and cultural services such as recreational, spiritual, religious and other non-material benefits" (23). The concept has grown beyond the original intent it was created for, and now forms a basis for immediate expansion of literature in ways and means of assessment and valuation of these resources (28).

Ecosystem services became a binding concept that brought economists and ecologists together to quantify resources provided by nature (28). The theory of ecosystem services led to major policy-related studies and introduced newer concepts. In terms of policy studies: It inspired 'The Economics of Ecosystems and Biodiversity' (TEEB) study commissioned by the European Union, and an 'Inter-governmental Platform on Biodiversity and Ecosystem Services' (IPBES) was established between 90 governments. In terms of newer concepts, it made way for the concept of 'Payment for Ecosystem Services' (PES), which looks into payment for services such as watershed services, carbon sequestration, biodiversity conservation and many more (29). Ecosystem services benefits started to be viewed as unique, and at the same time, essential in terms of value addition to biodiversity conservation (30).

2.2. Critical evaluation of Ecosystem Services (ES)

Even though the concept of ecosystem services is novel, and the most championed one when it comes to quantification of usefulness of ecosystems, it becomes essential to investigate the drawbacks that it may have as well. There has been a persistent trend in literature to treat 'internal processes' of ecosystems, synonymously with 'ecosystem functions' and 'ecosystem services.' A double counting can be caused in such a scenario, for example: Nutrient cycling is a process that is essential for the ecosystem service known as timber production, but nutrient cycling by itself is also considered as an ecosystem service (31). It is noticed that there is a more complex and deeper relationship between biodiversity and provisioning and regulating services (32). In certain cases, there is an attempt to oversimplify the concept of ecosystem services and their literature has minimum ways of documenting them.

Along with gaps in implementation and definitions, certain aspects essential to ecosystems have also been ignored. Ecosystem disservices needs to be included in a broader framework where ecosystems cause more harm than good, for

example: human wildlife conflicts and flood damages (33–35). The MEA has been vague in highlighting the importance of trade-off measurements, for example: An increase in timber ecosystem service would lead to a decrease in biodiversity cover and hence causing a decline in recreational services (36–38). Keeping this in mind, the present study attempts to address the trade-off gap of MEA, using the, 'Integrated Valuation of Ecosystem Services and Trade-offs' (InVEST) model, which is a suite of software models used to map and evaluate goods and services obtained from ecosystems along with their trade-offs (39).

2.3. Global research efforts

The global research on InVEST is looked at through a spatial scope, which makes it accessible to a compare-contrast analysis with the geography and demography of countries that this review covers. Out of the 45 global papers reviewed on InVEST: 30 of them are from China, 2 from South Korea, 2 from Portugal, 2 from Sri Lanka, 2 from Iran, 2 from Italy, 1 from Pakistan, 1 from Morocco, 1 from Belgium, 1 from Chile and 1 from Ethiopia.

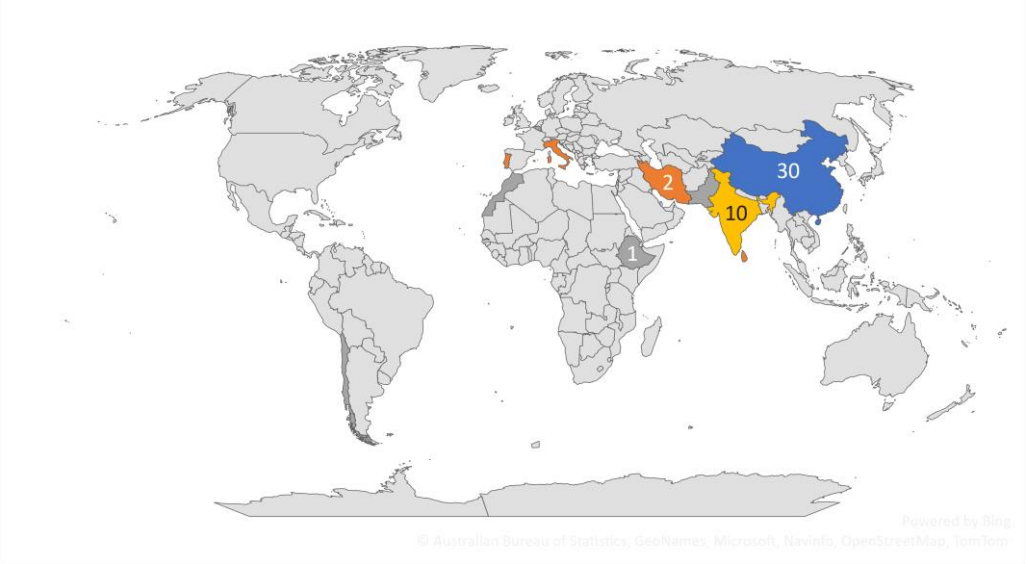


Figure 2.1. Spatial distribution of Ecosystem Services studies conducted using inVEST

Out of the 30 studies conducted in China, 10 were from an urban setting and 20 were from a non-urban setting. The studies in urban China have been conducted between 2017-2022 and all of them use the Habitat Quality model (40) of InVEST

and address the following aspects of urban issues: Green policy initiation, urban versus greens – trade-off mitigation, future urban planning measures and green space connectivity (41–50). In non-urban China between 2013-2022, inVEST has been explored through Carbon Storage and Sequestration model, Habitat Quality model, Coastal Vulnerability Model and Seasonal Water Yield model (40). This research led to the following observations specific to individual studies: sustainable water management plans proposed, simulated water yield in a data scarce region, gave evidence for coordinated development as the way forward, to help policymakers overcome knowledge gap, and risk assessment for coastal disasters (51–70).

Countries other than China that have researched on InVEST modelling for Ecosystem Services studies, have limited work done. These investigated the Habitat Quality Model, Habitat Risk Assessment model, Seasonal Water Yield model, Carbon Sequestration model and Sediment delivery model. The studies were conducted mostly in an urban setting and they aimed to influence policy change through their work (71–85). The global trend shows that China, due to their ever-expanding development has looked at InVEST as a tool to assess a large amount of data about their ecosystem changes in a short duration of time. India shows trends of following suite and can only increase the number of studies done using InVEST.

2.4. Local research efforts

2.4.1. National

The national level review of literature is specified by design to studies done using InVEST modelling, so that: similarities, differences, and the range of applicability to the proposed study, can be explored. Six different studies were conducted between 2016-2021 on the Seasonal water yield (40) model of InVEST (86–91). The focus of these studies was ranging from influencing policies for better planning and predicting flood vulnerabilities to economic valuation of floods. One

study from the city of Nagpur in the year 2021, looked at the Urban Cooling Model (40) of InVEST (92) in heatwave affected zones. Over here, a heat mitigation model was created using this research and ecological structures were proposed in urban areas. Six more studies were done using InVEST modelling between 2017-2022, in the field of carbon sequestration using the Carbon storage and Sequestration model (40) . The studies addressed the following: Carbon emission estimation, demonstrating deforestation potential, helping out forest dependent communities by influencing Payment for Ecosystem Service (PES) (23) scheme and influence policies for conservation and management of natural resources (93–98).

Literature shows national trends of most of the studies on InVEST have happened only in the past few years. This gives further impetus for studies to be conducted in more regions so that it can eventually be incorporated in the national government policy for ecosystem services assessments.

2.4.2. Regional

The scope of this review at a regional level is for identifying a fundamental gap in research towards InVEST modelling. The studies about ecosystem services by researchers in Bengaluru are limited to recent years, almost as if in response to the changes in their surrounding environments. A study that used the Biofin approach came up with an evidence-based finance plan. It advocated a circular urban governance system with a focus on biodiversity (99). A study conducted on peri-urban agriculture in Bengaluru looked at using participatory mapping exercises to evaluate ecosystem services, it proposed an integrated landscape approach for agricultural livelihood to exist sustainably (100). A study conducted in Lalbagh Botanical Garden looked to estimate the cultural Ecosystem Services, some green space management and ecotourism policies were suggested (101). Tourist sites such as Nandi Hills, near Bengaluru were assessed for recreational Ecosystem Services using Travel Cost Method (102). The trees of riparian zones of Cauvery

river was assessed using Ecosystem Services approach, dominant trees were determined and how they form the major part of the ecosystem (103).

A study conducted on the Recreational Open Spaces (ROS) in Mumbai, Bengaluru and Chennai were looked at as a part of a comparative study that considered the ecosystem services of these spaces, a framework was proposed to the government in order to improve policies (104). Recently in the year 2020, two studies in Bengaluru looked at a purely GIS based approach to: (1) Assess environmentally damaging urbanization occurring in different parts of the city. (2) Spatial and temporal analysis of agricultural land in the peri-urban regions of Bengaluru over a span of 15 years (105,106). There were other studies in Bengaluru which focused purely on lake ecosystem revival programs, but that is outside the purview of this review. Hence, the gap observed here is the lack of usage of inVEST modelling in the city of Bengaluru and very little ecosystem services studies conducted.

2.5. Role of geospatial technologies in understanding ecological landscapes

With the advent of geospatial technologies, studies that would have required a large amount of man power and lots of years to do physically, have now become easier as the remote sensing data takes less time, is more reliable and effective (107). It is good for studying urban and peri-urban regions because vulnerability patterns can be analysed spatially, and results can be compared or displayed (108). The application of geospatial technology, apart from being used in diverse human endeavours, has made immense contribution towards monitoring and evaluating the changing pattern of vegetation ecosystems, making it one of the most essential tools for habitat studies over a period of time (109). The geospatial assessment of land use over a span of years is an important way of identifying the root cause of ecological problems in a region. Therefore, given the time constraint and the need to attain impactful outputs, geospatial technology is the tool selected for the study.

2.6. Potential research options. Why LULC analysis and InVEST?

This review found multiple ways in which ecosystem services were assessed. Most of them were interview based, finding out what value a particular service has to humans, and economic aspects of the study such as Payment for Ecosystem Services (PES). InVEST is a suite of models used for ecosystem services and trade-off mapping and valuation. They are spatially-explicit as they take maps as inputs and produce maps as outputs (39). The land use land cover change tells a story of the ecological landscape transformation over time (Objective 1). The biodiversity profile along with InVEST Habitat Quality Model (HQM) (Objective 2) gives an ecological description of the species with respect to the surroundings, in turn highlighting the importance of the region, which is the end goal of this thesis.

3. AIM AND OBJECTIVES

3.1. Aim

To assess the impact of rapid urbanization on ecological assets along the IT belt of Bengaluru.

3.2. Objectives

1. Spatial-temporal change analysis of land-use, land-cover along the IT belt of Bengaluru between the years 2002 to 2022.

2. A species and habitat quality overview of the study region.

4. METHODOLOGY

The approach of methodology used in this study is inductive reasoning. In an inductive reasoning approach, the hypothesis of the study is never established at the beginning of the study (110). It is a more dynamic study when compared to the studies involving a null and an alternative hypothesis. When inductive reasoning is used in a quantitative study like the present one, it is called exploratory data analysis (111). The main reason for selection of inductive reasoning is that the outcome of the study is familiar.

For example: we already know that urbanization is causing habitat loss, degradation of the surrounding nature, etc. But - “To what scale?” “How intensely is it happening in this specific region?” “What policymakers need to do to course-correct?” Are all the important questions this study looks to address. It involves two steps: First, recognizing ideas and concepts coming out of the study. Second, linking these ideas to established theories with due explanation (112). The attempt here is to do just the same so that a cognizable change can be made using the data presented.

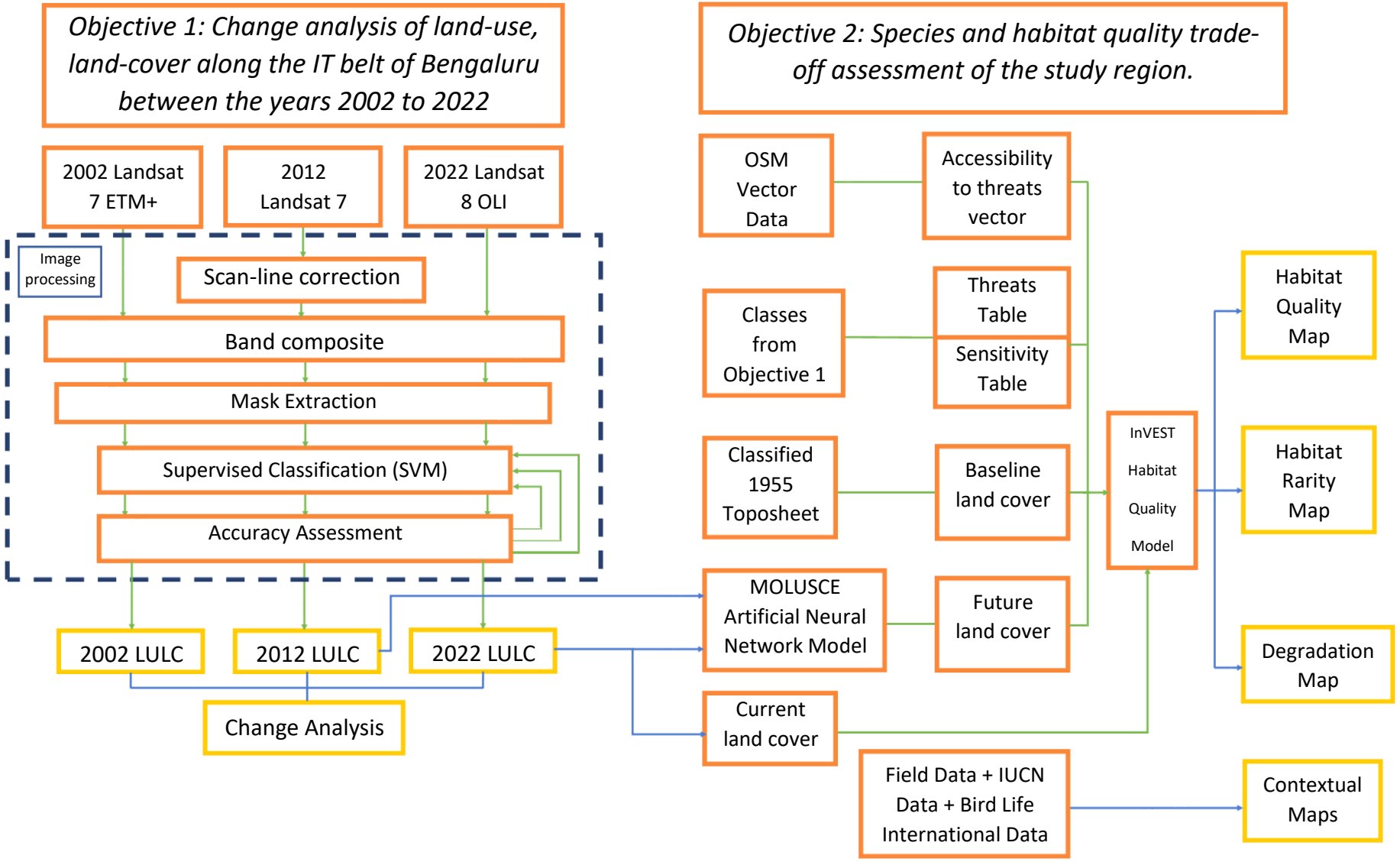


Figure 4.1. Methodology block diagram

4.1. Study area

The study area selected is as shown in the map below, it is the region containing most of prominent technological parks and SEZ zones of Bengaluru popularly known as the “IT belt.” Having an extent of 13.3 °N at the top to 12.6°N at the bottom, and 77.25°E on the left to 78.04°E to the right. It occupies parts of Bengaluru within the city as well as the areas in the outskirts so that the industry has a scope for expansion. This includes – Bangalore East, Bangalore South and Anekal tehsils of Bangalore Urban district and Hoskote tehsil of Bangalore Rural district (113). Total study area is 1949.5 sq. km. at an altitude varying between 697 - 993 m.

The population of Bengaluru has been growing at an alarming rate of 3.35-3.87% per year, in the past few years (114), one of the major reasons for this is the IT industry. Despite of this, the study area has vastly varying ecosystems ranging from – the forests of Bannerghatta National Park, the agricultural spaces of rural regions, to the lakes and park system of the city.

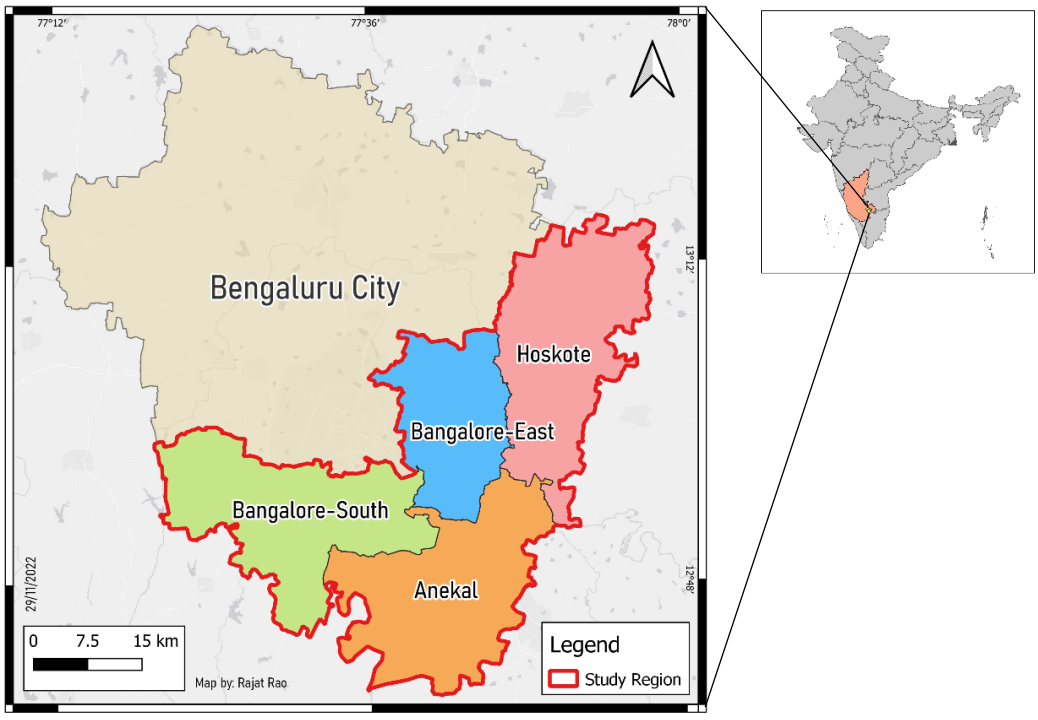


Figure 4.2. The study area map shown in the red border, is a group of four tehsils which contain most of the IT companies of Bengaluru, locally referred to as the IT belt of the city.

4.2. Site selection

After multiple recces on field, a thorough Google Earth survey of the whole city, and going through several government documents, the following conclusions were drawn about selection of this specific site for study:

- The selected taluks have most of the IT companies of the city with all the major technological parks present in them.
- There are more than 126 lakes and green spaces within this region of the city.
- Bannerghatta National Park starts from the South side which includes parts of Anekal and Bangalore South.
- The outskirts are majorly an agricultural and a rocky outcrop belt. The study of expansion of urban spaces into these areas is interesting to understand trends of ecosystem eradication around urban centres all over the world.
- The four tehsils were selected due to the unique nature of each one that would be beneficial to understand the rate of expansion. Bangalore-East has a proximity towards the main city and is an example of a region which will not show too much variation. Bangalore-South is laden with the forests bordering Bannerghatta National Park. Anekal is a region which houses a part of the Electronic City and still does not come under the BBMP (municipality) limits and is governed by panchayats due to lack of expansion of the wards with the expansion of the city. Hoskote is a rural part of the city also seeing rapid urbanization due to proximity of Whitefield, which is one of the major IT hubs.

The complexities mentioned above makes the region a good candidate for an ecological degradation study.

4.3. Data

The design of the following section is such that the data sources for specific objectives are listed in the form of individual tables containing a list having the data itself, their corresponding sources and the software used.

Objective 1: Spatial-temporal change analysis of land-use, land-cover along the IT belt of Bengaluru between the years 2002 to 2022.		
Data	**Source**	**Software**
February-March Satellite Image from the year 2002	Landsat 7 ETM+	USGS Earth Explorer for GeoTIFF download, ERDAS Imagine for atmospheric correction and ArcGIS and QGIS for further image processing.
February-March Satellite Image from the year 2012	Landsat 7 ETM+	
February-March Satellite Image from the year 2022	Landsat 8 OLI	
Karnataka Thesil, District, and village boundaries shapefile	Karnataka State Remote Sensing Application Centre (KSRSAC) website (115).	ArcGIS for viewing and vector processing.

Table 4.1. Data for Objective 1.

Objective 2: A species and habitat quality trade-off assessment of the study region.		
Data	**Source**	**Software**
Current land cover	2022 LULC map of Objective 1	ArcGIS, ERDAS Imagine and QGIS used to prepare map.

Future land cover	2032 map predicted from 2012 and 2022 LULC map	Artificial neural network of MOLUSCE plugin in QGIS 2.18.15.
Baseline land cover	Toposheets from 1955 (116), mosaicked and classified into LULC categories of the study.	ArcGIS and QGIS.
Threats table	LULC categories	MS Excel
Sensitivity table	LULC categories	MS Excel
Accessibility to threats vector	Open Street Maps data	ArcGIS
Lakes and greens data	Field surveys, Google maps, Google Earth Pro	Chrome, ArcGIS, MS Excel
IT Companies data	Field surveys, Google maps, Google Earth Pro	Chrome, ArcGIS, MS Excel
High-rise Buildings and Gated Communities data	Field surveys, Google maps, Google Earth Pro, Real-estate websites	Chrome, ArcGIS, MS Excel
Species Data	IUCN, Birdlife International	MS Excel

Table 4.2. Data for Objective 2

4.4. Software used

As the study is a purely GIS based, data processing and analysis is done by software like ArcGIS, QGIS, ERDAS Imagine, Google Earth Pro and Google Maps. As a rule, geodatabases were not used during the entire study as shapefiles ensures cross-platform compatibility. Each platform was used for its unique functionality as listed below:

ArcGIS: Used for image pre-processing, classification, and accuracy assessment. ArcMap has quite a large geoprocessing repository in ArcToolbox. Therefore, it is the software used for most of data the processing done in the study.

QGIS: This software is used mainly for publishing maps because of its easy-to-use layout view feature and the variety of functions made available during publishing. Another important task QGIS helps perform is the creation of future maps using the MOLUSCE plugin present in QGIS 2.18.15 which is one of the inputs of the second objective.

ERDAS Imagine: Despite its vast functionality, the use of ERDAS Imagine is limited in the current study. It has only been used for performing scanline correction of Landsat 7 2012 image, as this is one of the software that has a built-in feature to perform the correction with minimum noise.

Google Earth Pro: Used for performing surveys of the study area to identify patterns in land cover.

Google Maps: Used for extracting location information of lakes, green spaces, High-rise apartments, and IT companies for the creation of contextual maps. The Google Takeout feature is used from maps to export data as a JSON.

4.5. Methodology for Objective 1

The 1st Objective of the study focussed on the spatial-temporal change analysis of land-use, land-cover along the IT belt of Bengaluru between the years 2002 to 2022. It is an essential part of the study which gives a land cover overview and the way it has changed over time. The output land use maps of this section serve as the input for the next section. The following were the steps taken to achieve the objective:

4.5.1. Satellite image procurement and correction

The satellite images acquired for the study are high resolution Landsat images from three different time spans – 2002, 2012 and 2022 which explores the different timelines of IT industry growth in the city. 2002 was a time when the growth was still in its nascent stages, 2012 was after the first global recession and

right before the start-up boom in the city and 2022 shows the present situation. The following table gives information about the various satellites and sensors used; These form the base data for the whole study.

Satellite	Sensor	Path/Row	Date	Cloud Cover	Bands
LANDSAT-7	ETM+	144/051	18-02-2002	0%	1-8 (Used 7)
LANDSAT-7	ETM+	144/051	01-03-2012	0%	1-8 (Used 7)
LANDSAT-8	OLI	144/051	05-03-2022	0%	1-11 (Used 7)

Table 4.3. Satellite source data. A table showing information about the satellite image obtained from USGS Earth Explorer.

The satellite images selected were of the months of February and March since these are post monsoon months with little to no cloud cover in the region.

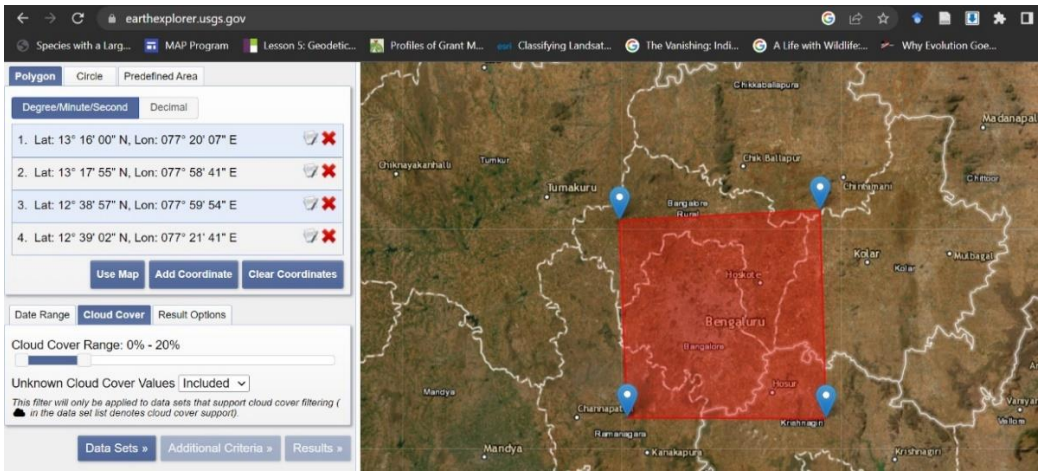


Figure 4.3. USGS study site selection. The range of the study site is selected as shown above and the cloud cover range is restricted to below 20% to obtain clear images.

Once the image range and the satellite collection are specified, the image is downloaded. For ease of usage and to acquire error-free images, *Landsat Collection 2 Level-2 Science Products* were used for both Landsat 7 and Landsat 8 images (117). To understand what it is, a brief theory about satellite errors needs to be understood along with the working principle of satellites which will be covered in the following section.

Satellite sensors are optical instruments which use reflected light to capture an image and works on the same principle as a camera. There are two ways in which it gets reflected light: Active (source of radiation is the satellite itself) and Passive (source of radiation is the sun).

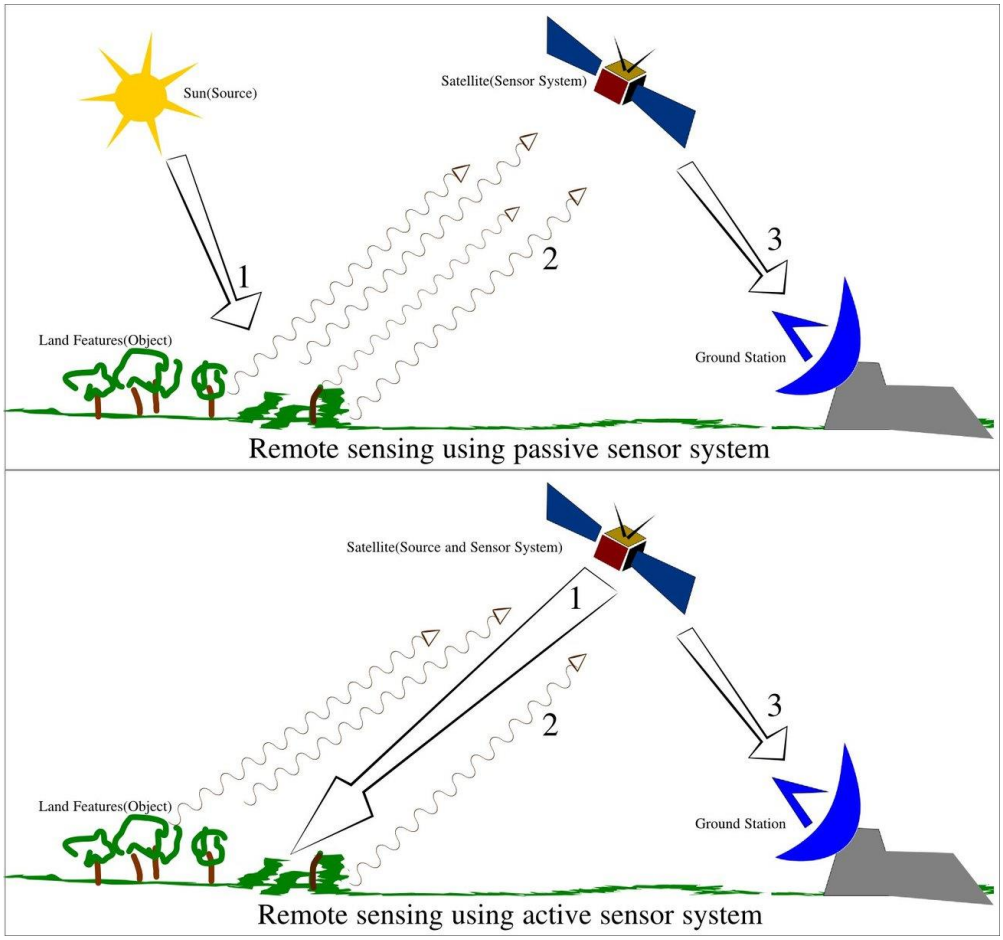


Figure 4.4. Satellite image formation. This figure shows the different ways in which satellite images are generated (118).

During the process of recording of the image at the satellite sensor, there are several ways in which the image accuracy is compromised, and this induces errors into the images. When these images are used as the base factor for a study, it can induce a bias due to the erroneous data.

The main types of errors that can be seen during remote sensing are as follows:

- Radiometric errors: Errors caused due to satellite sensors. They are the errors caused at the time of recording the image. Sometimes, as we will see ahead, some sensors of the satellite go bad and stops working, this

leads to that part of the image not being recorded, this appears as blank pixels in the form of lines or dots. It is the most common radiometric error and is called a scan line error. It is rectified with the help of image processing.

- Geometric errors: These errors can be caused due to any of the following reasons: Change in perspective of the sensors, motion of the scanning system, instability of the satellite platform, curvature, and rotation of the Earth, etc.

- Atmospheric errors: The errors caused by the variation in the atmospheric conditions are atmospheric errors. These are usually caused due to the absorption of the reflected light coming from Earth's surface, by suspended particles in the atmosphere (mostly clouds).

Landsat Collection 2 Level-2 Science Products:

This is a collection of GEOTIFF images which are geometrically and atmospherically corrected and readily available for download. It has a CEOS CARD4L stamp (119) which is given by the Committee on Earth Observation Satellites (CEOS) and is an internationally recognized certificate. The stamp implies that the data is analysis ready with minimum user data processing requirement. It essentially means that the scope for data integrity is high.

Then where is correction required?

On 31st May 2003, the Scan Line Corrector on board Landsat 7 satellite stopped working due to a malfunction. This was an important component which compensated for the forward motion of the satellite. There were efforts made to fix it remotely, but the failure remains permanent till date (120).

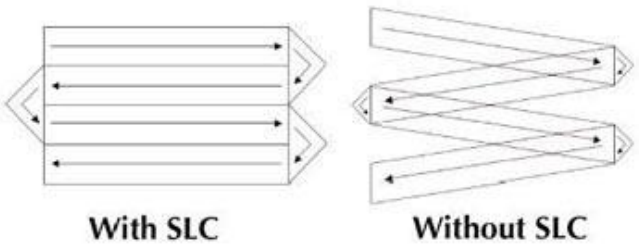


Figure 4.5. SLC correction. The figure shows the motion of image formation with SLC and without SLC (120).

This created problems in the formation of image and all images are still formed with the following gaps as shown in the figure below. 78% of the original image remains and 22% is lost to the error. Despite this, 78% is a good data and hence image processing modifications were made to compensate for this loss of data.

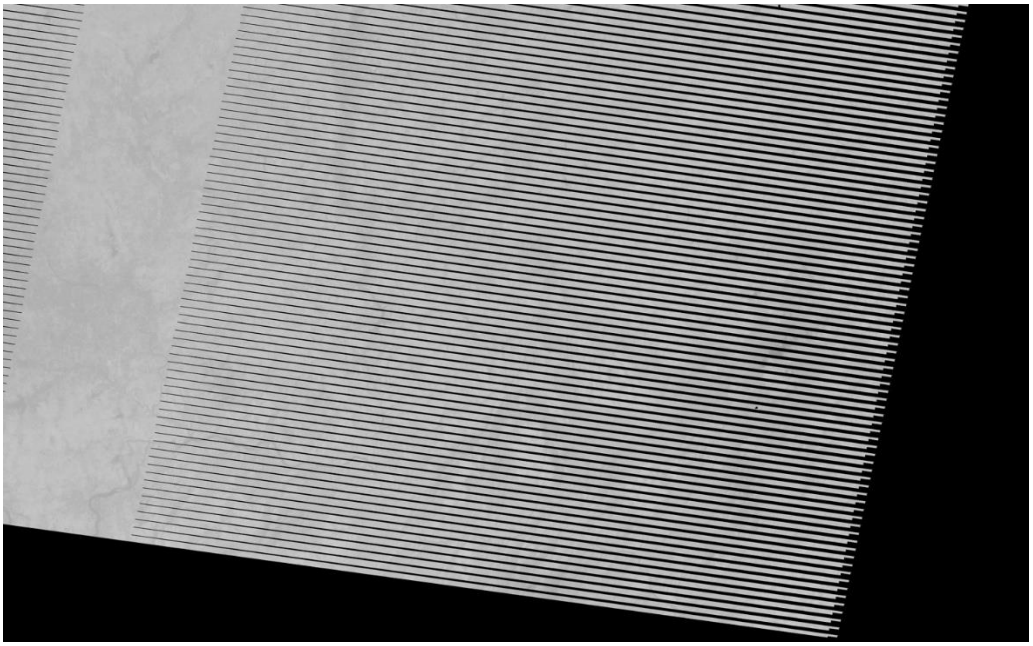


Figure 4.6. Scanline error correction. The figure showing scanline error in Band 6 of the 2012 image obtained from USGS.

Elimination of scanline error:

The focal analysis tool under ERDAS Imagine was used to perform scanline correction. The correction does not happen immediately but is an iterative process with subsequent result correcting the band a little better than the previous iteration.

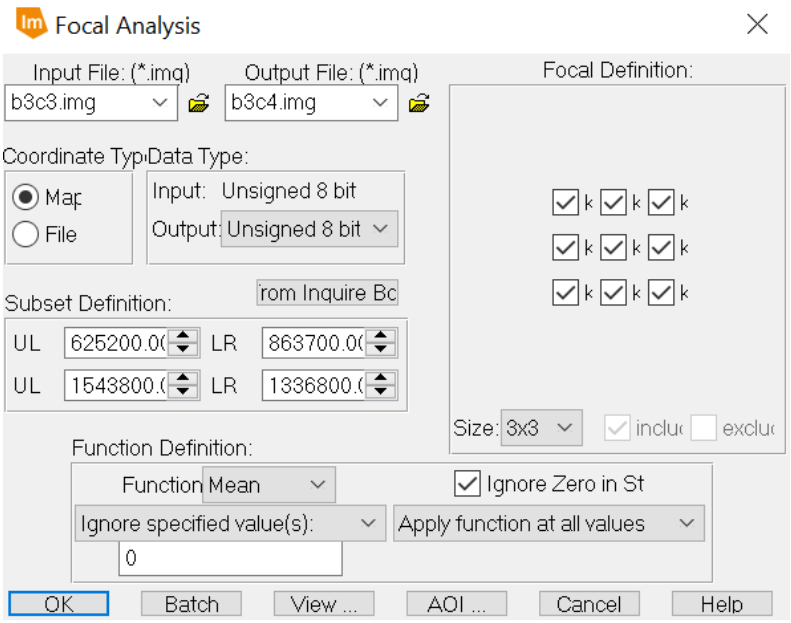


Figure 4.7. Focal analysis. The image shows the Focal Analysis tool of ERDAS Imagine used for scanline correction. Each band is sent through at least 7 iterations to obtain the final image.

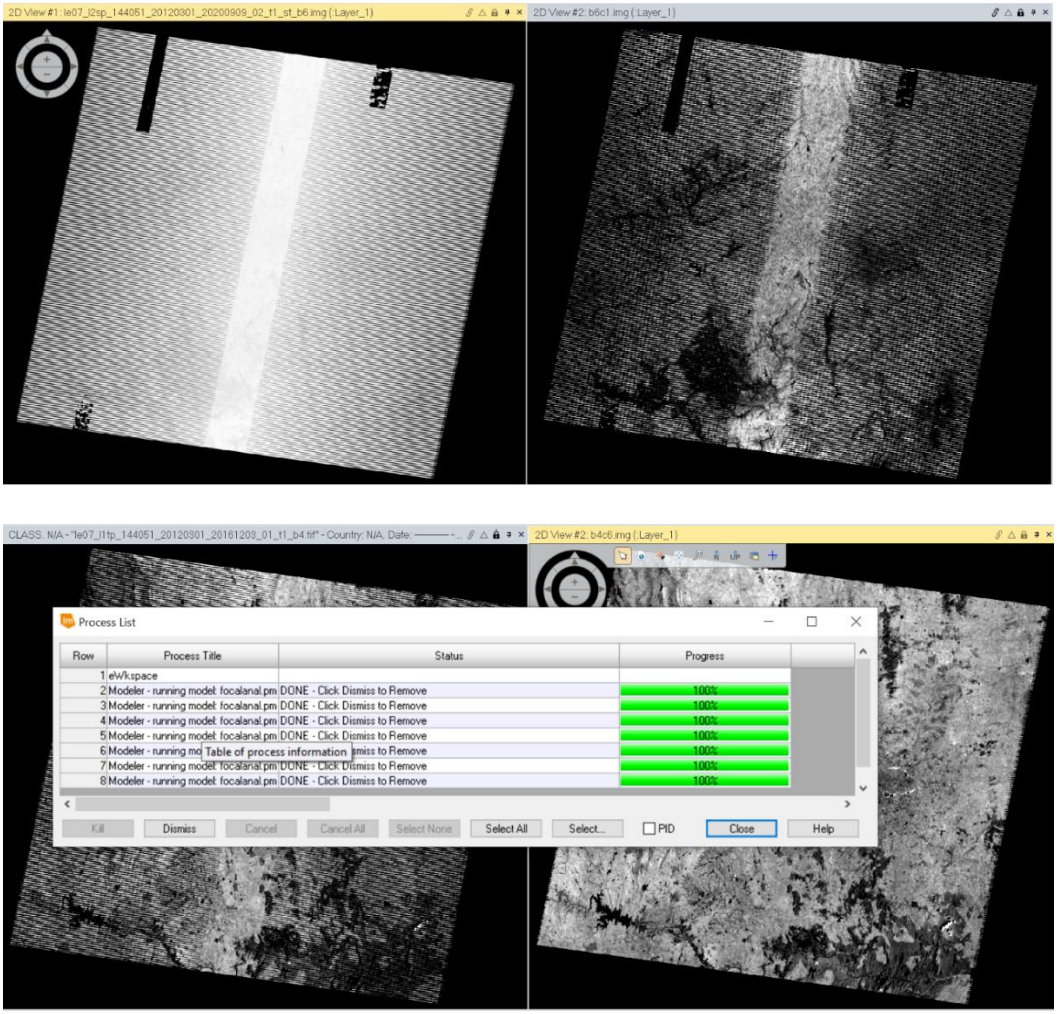


Figure 4.8. Error improvement. Showing improvement in error with each iteration.

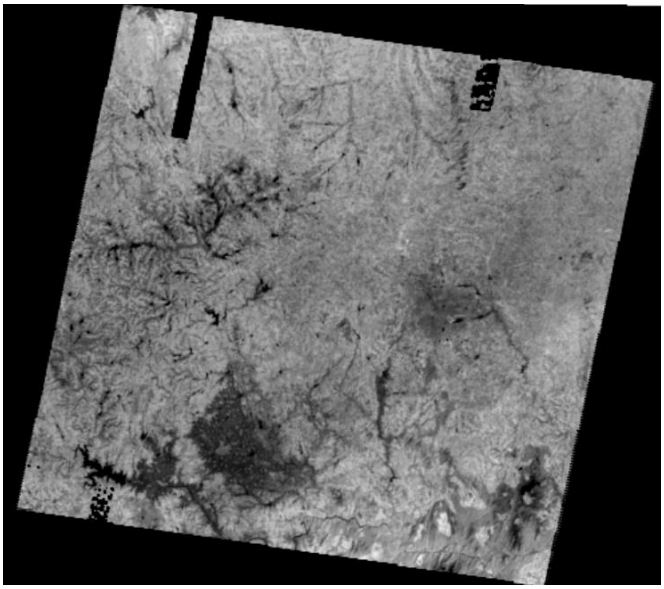


Figure 4.9. Error corrected image. Showing final error corrected image of Band 6 for the study area.

This method is repeated for all the seven bands of the 2012 image individually so that on performing Band composite will have all the features in it. Once these errors are rectified, the image is ready for pre-processing.

4.5.2. Image pre-processing

The corrected satellite images need to be pre-processed and brought down to a uniformity before performing classification operation. The following operations were performed to make the satellite images ready for classification.

Band Composition:

The raw satellite images of Landsat are always made available in separated bands because that is how the satellite records data for better accuracy. When we use the data of the satellite, we use it by combining the necessary bands into one band for interpretation and further processing. Landsat 7 and Landsat 8 have different band compositions because they both use different sensors.

Landsat 7 uses an Enhanced Thematic Mapper Plus (ETM+) which is a fixed, "whisk-broom," 8 band sensor apparatus onboard the satellite. It has the following specifications (121):

Sensor Type	Opto-mechanical
Spatial Resolution	30 meters
Spectral Range	0.45 – 12.5 micrometres
Number of Bands	8
Temporal Resolution	16 days
Image Size	183 km x 170 km
Swath	183 km
Programmable	Yes

Table 4.4. Specifications of ETM+ sensor of Landsat.

Band No.	*Wavelength (um)*	*Bands*	*Resolution*
1	0.45 – 0.5	Blue	30m
2	0.525 – 0.6	Green	30m
3	0.63 – 0.69	Red	30m
4	0.77 – 0.9	Near-Infrared	30m
5	1.55 – 1.75	Short wave Infrared (SWIR)	30m
6	10.4 – 12.5	Thermal	60m
7	2.08 – 2.35	Mid-Infrared	30m
8	0.52 – 0.9	Panchromatic	15m

Table 4.5. Landsat 7 ETM+ Band description. *The above table shows the different bands and what information each band carries for Landsat 7 ETM+ sensor (120).*

Landsat 8 uses the Operational Land Imager (OLI) sensor which instead of using the scan mirrors like older Landsat satellites it uses long detector arrays which has around 7,000 detectors assigned per spectral band (122). It also incorporates a "push-broom" design for data reading. It has similar specifications as the ETM+

but it is more efficient in data accumulation. The following are the band specifications of Landsat 8 OLI sensors.

Band No.	*Wavelength (um)*	*Bands*	*Resolution*
1	0.43 – 0.45	Coastal Aerosol	30
2	0.45 – 0.51	Blue	30
3	0.53 – 0.59	Green	30
4	0.64 – 0.67	Red	30
5	0.85 – 0.88	Near Infrared (NIR)	30
6	1.57 – 1.65	SWIR 1	30
7	2.11 – 2.29	SWIR 2	30
8	0.50 – 0.68	Panchromatic	15
9	1.36 – 1.38	Cirrus	30
10	10.60 – 11.19	Thermal Infrared 1	100
11	11.50 – 12.51	Thermal Infrared 2	100

Table 4.6. Landsat 8 OLI Band description. Table shows the various band specifications of Landsat 8 OLI (123).

For both, the Landsat 7 and 8 images, bands 1-7 are combined using the Band Composite tool of ArcGIS.

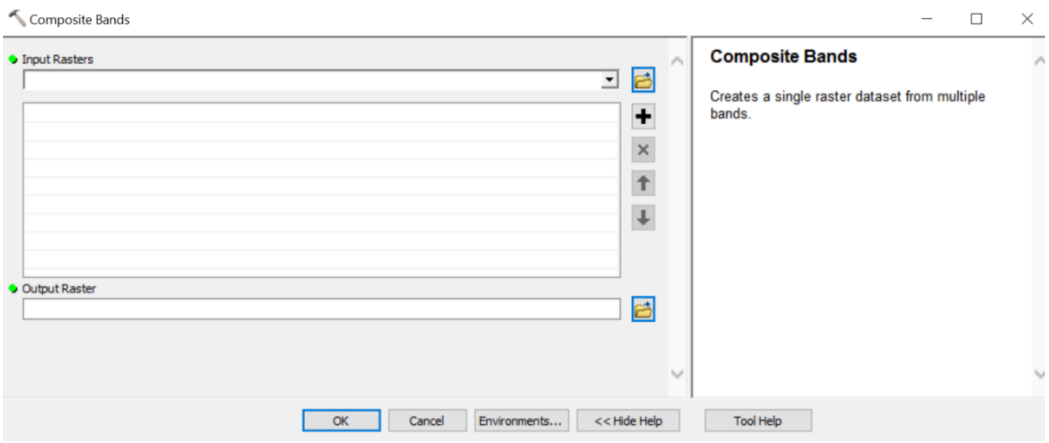


Figure 4.10. Band composite tool. The figure shows the Band Composite tool of ArcGIS that combines several different band images into a single image.

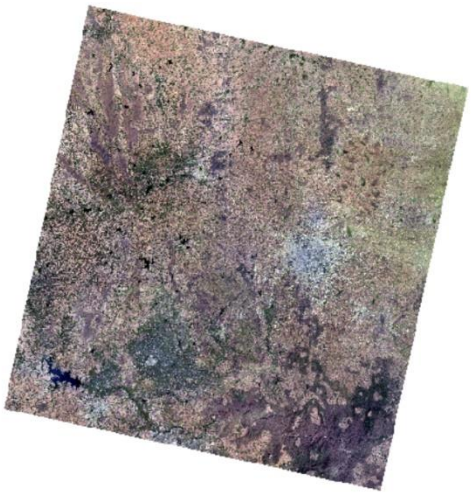


Figure 4.11. Band composite image. Band composite image containing data of all the 7 bands that were combined.

The final image is as shown above. It shows the bands in the combinations of red, blue, and green colours. This is the band composite image that is then narrowed down to our study site by a process of extract by mask in ArcGIS.

Mask Extraction:

In this stage, a shape file of the study area is created by combining the tehsils maps of the study region using dissolve function. This becomes the primary study area shapefile also known as Area of Interest (AOI).

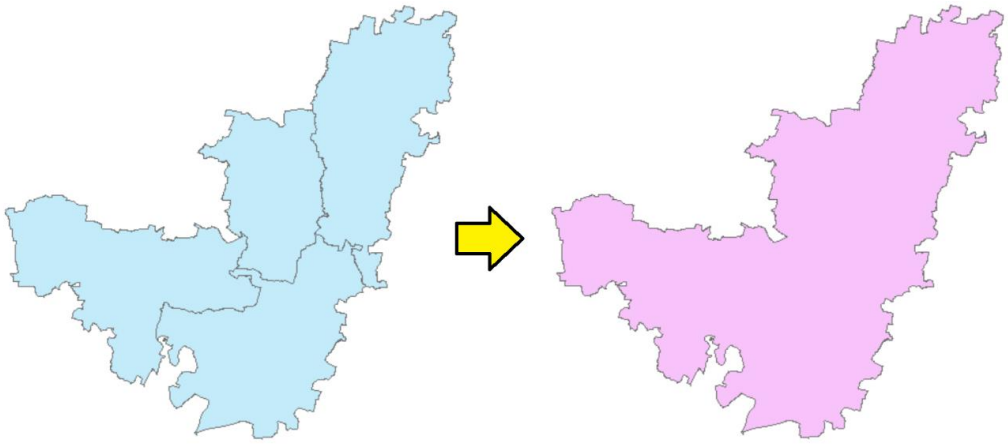


Figure 4.12. Dissolving layers. Using dissolve geoprocessing tool in ArcGIS to create the Area of Interest.

The AOI is then overlayed with the composite image and the extract by mask geoprocessing tool of ArcGIS is used to extract the part of the image which is a part of the study region.



Figure 4.13. Extract by mask. Image extracted by mask forming the study region map that is used for classification and analysis.

This procedure is followed for obtaining the masked images of all three time periods of data – 2002, 2012 and 2022.

4.5.3. Image classification

The different classification techniques provided by ArcGIS are as follows:

- Interactive Supervised Classification
- Maximum Likelihood Classification
- Iso Cluster Unsupervised Classification
- Class Probability
- Principal Components
- Support Vector Machine (SVM)

Out of these, Support Vector Machine Classification is used in this study. The main reason for selection of Support Vector Machine is that accuracy of

classification is much better than the commonly used Maximum Likelihood Classification (124).

Support Vector Machine is an efficient supervised machine learning algorithm which is mostly used for classification. SVM does classification by creation of a hyperplane (125). Hyperplane is a line in 2D or a plane in 3D which divides the data sets into two separate components, hence classifying them.

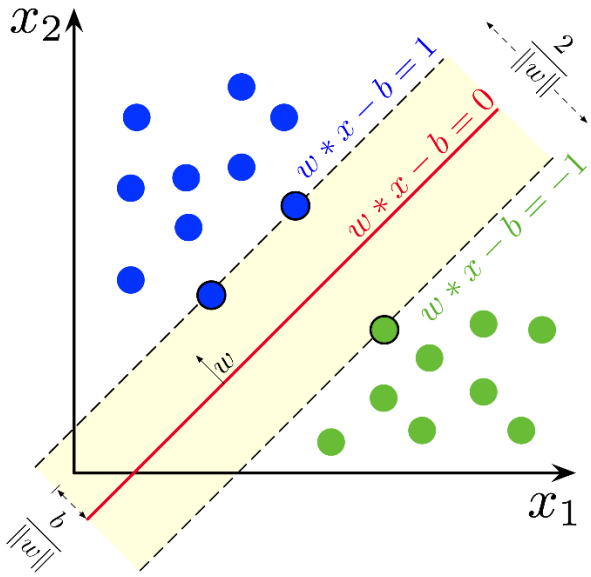


Figure 4.14.: SVM Hyperplane depiction. (126)

For example, if there is a data and one needs to classify them into a blue class or a green class then when we run a SVM algorithm it segregates the objects into different parts of the plane by drawing a hyperplane (line shown in red in the graph above). It is important however to note that the algorithm performs the classification based on a set of training data, therefore it is important to familiarize the classes to the algorithm by training it with multiple samples. This is done in ArcGIS through the Training Sample Manager of the Classification Tool.

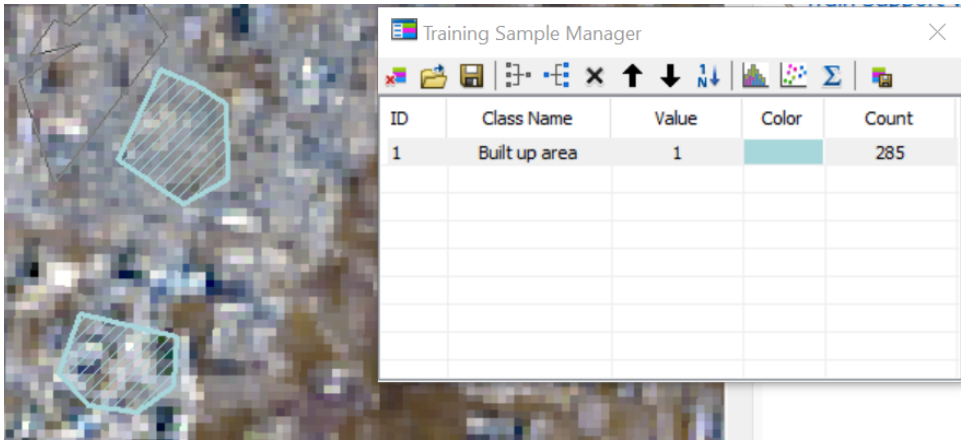


Figure 4.15. Sampling for SVM classification. The figure shows the samples taken on the left and the training sample manager on the right which has the division of the samples into classes that will be used for training the SVM algorithm.

Due to diverse land use data in different parts of the study area and certain classes looking very similar to other classes, many samples were taken in classification and reclassification of data. Samples were taken from 4048 pixels for classes that got classified with accuracy to 49403 pixels for classes that showed a lower accuracy for classification.

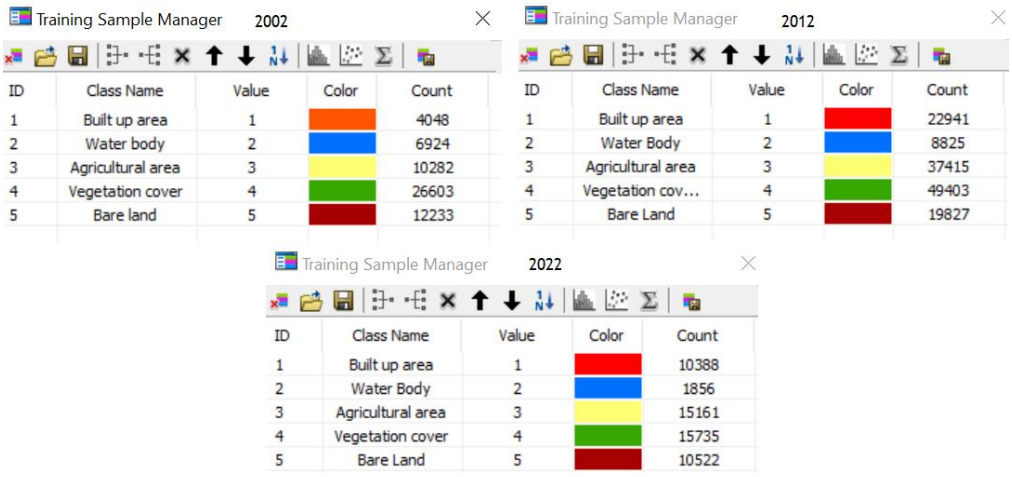


Figure 4.16. Training sample manager values. The figure shows the different samples that were taken for each year.

Once the samples are taken, they are saved as a shape file and the same sample set is used for training the SVM classifier. After the classifier is trained, a signature file is generated which contains the details and instructions for classification. The signature file is then used by the ArcGIS classify raster tool to classify the raster into different classes.

4.5.4. Accuracy assessment

The output of supervised classification needs to be tested for accuracy always before proceeding on to the next stage of analysis, if accuracy is low then it is reclassified with more samples. This is done through the process of accuracy assessment which are components of the spatial analyst tools of the ArcToolbox.

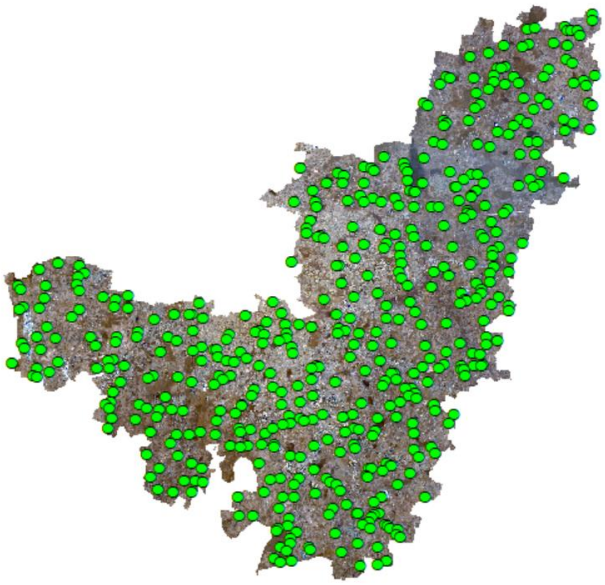


Figure 4.17. Ground truth points. A set of randomly generated ground truth points used for accuracy assessment

Every map (2002, 2012 and 2022) is verified through the process of accuracy assessment with a set of 500 points created for each. These points are further assigned numbers from 1 – 5 as follows:

1	Built up area
2	Water body
3	Agricultural area
4	Vegetation cover
5	Bare land

Table 4.7. The list of land use classes. It has the unique ID assigned to each.

Table

AA2022_New

FID	Shape *	Classified	GrndTruth
13	Point	1	1
27	Point	1	1
56	Point	1	3
57	Point	1	1
65	Point	1	1
66	Point	1	1
70	Point	1	3
73	Point	1	1
75	Point	1	1
78	Point	1	1
80	Point	1	1
82	Point	1	1
84	Point	1	1
96	Point	1	1
98	Point	1	1
99	Point	1	1
100	Point	1	1
104	Point	1	1
106	Point	1	1

Figure 4.18. Manual validation process.

The classified numbers are manually validated for all 500 points for each year (2002, 2012 and 2022). Google Earth Pro's historical imagery was used as a base map for validation of 2002 and 2012 features. The present ESRI 2022 imagery base map was used for validation of 2022 points. For each classified point, the ground truth was changed manually if it appeared to be different at the pixel point as shown in the table above. This table is then used for calculation of the *confusion matrix* which provides the user and producer accuracy of the classification along with the Kappa Co-efficient.

		Reference Data							
		Built up area	Water body	Agricultural area	Vegetation cover	Bare land	Total	User Accuracy	
Classified Data	Built up area	101	0	4	0	2	107	94%	0%
	Water body	1	9	0	0	0	10	90%	0%
	Agricultural area	2	0	169	15	9	195	87%	0%
	Vegetation cover	1	0	8	117	0	126	92%	0%
	Bare land	1	0	4	8	51	64	79%	0%
	Total	106	9	185	140	62	502		
	Producer Accuracy	95%	100%	91%	84%	82%	0	89%	0%
Kappa		0%	0%	0%	0%	0%	0	0%	85%

Table 4.8. Classification of data within a Confusion Matrix.

Errors during map making (127)**:**

Error of Omission and Producer's Accuracy: This is a type of error that occurs when a particular reference class was left out or omitted from the class in the map. This leads to the concept of Producer's Accuracy which is the complement of Omission error (100% - Omission Error).

Error of Commission and User's Accuracy: This error is generated when the classified data gets incorrectly classified. User's Accuracy concept is derived from this which is the complement of the Commission Error (100% - Commission Error).

Kappa Coefficient: It is the result of a statistical test that is used to check how accurate the classification has occurred. It compares the accuracy of classification in comparison to randomly assigning values (ground truth points). The range of Kappa values and its meaning are as follows (128):

<0	No Agreement
0 – 0.20	Slight Agreement
0.21 – 0.40	Fair Agreement
0.41 – 0.60	Moderate Agreement
0.61 – 0.80	Substantial Agreement
0.81 – 1.0	Perfect Agreement

Table 4.9. Kappa coefficients.

Overall Accuracy: It is the ratio of the number of correctly classified sites to the total number of sites. In the example above of 2022's confusion matrix, the overall accuracy is the total of the diagonal elements in grey divided by 502, which is 89% accuracy.

4.6. Methodology for Objective 2

Objective 2 of the study looks at species and habitat quality trade-off assessment of the study region. This is done in two ways:

- InVEST Habitat Quality Model (HQM) which looks to assess the trade-off in habitat with respect to the growing urban population.
- Contextual maps – Apartment complexes, IT companies, Lakes, Green Spaces species presence data from IUCN and Birdlife International.

The following sections will explain in depth the methodology involved in data accumulation and analysis of the sub-topics mentioned above.

4.6.1. Habitat Quality Model

InVEST stands for – Integrated Valuation of Ecosystem Services and Trade-offs. It is a collection of models used to evaluate and map the services that are provided by nature for betterment of human life. It tries to provide a variation in ecosystem perspective to trade-off with human needs and requirement so that a balance can be maintained between requirement and destruction (39). The Habitat Quality Model of InVEST looks to use habitat quality and rarity as proxies for understanding the biodiversity of a region. The following factors influence the development of the model (129):

- *Relative sensitivity of habitat:*
  Habitat sensitivity is the inverse of habitat resistance (130). The sensitivity of habitat type *j* to threat r, is defined by $S_{jr} \in [0,1]$ where values closer to 1 indicate greater sensitivity. The model assumes, more the sensitivity to a threat, more will be the degradation from the threat.

$$D_{xj} = \sum_{r=1}^{R} \sum_{y=1}^{Y_r} \left(\frac{w_r}{\sum_{r=1}^{R} w_r} \right) r_y i_{rxy} \beta_x S_{jr}$$

The above equation gives the total threat level in a cell x with habitat type j. Where Y_r indicates the set of cells on the raster map of threats and y indexes all the cells on the same map. Q_{xj} gives the quality of habitat in a parcel of image x which is in the habitat j.

$$Q_{xj} = H_j \left(1 - \left(\frac{D_{xj}^z}{D_{xj}^z + k^z} \right) \right)$$

Where, z = 2.5 and k are the constants.

Habitat rarity is also a feature that Habitat Quality Model explores in the given data set. This can be calculated if the baseline and future maps of the region are developed. The following two equations are used for

calculating the rarity of habitat in the same way habitat quality is calculated.

$$R_j = 1 - \frac{N_j}{N_{jbaseline} + N_j}$$

$$R_x = \sum_{x=1}^{X} \sigma_{xj} R_j$$

Where, N_j is the area of the habitat on the current or future map and $N_{jbaseline}$ is the area of the same on the baseline map. R_j is the R score of each habitat. This R score is used in the next equation to calculate the habitat rarity for the region.

- *Relative impact of threat:*

  A score (degradation source weight w_r which indicates the degree of destruction of a threat to all habitats) is given to each threat because some threats are more intense than the others. Urban regions are more damaging than the agricultural regions where certain reptiles thrive.

- *Degree of legal protection of habitat:*

  There is a need to check the protection status of the habitat region, because if the space is protected then the chance of it getting overrun by human activities is going to be rare. This basically measures the difficulty to access a green space to public or private use, hence even areas that are having extreme slope are also considered. Hence, the level of protection is classified as follows:

  - Legal
  - Institutional
  - Social
  - Physical

The model assumes that if any of these protections are present then it will be less affected from the nearby threats without regard to the type of threat. An accessibility to threats vector is created along the same principle where the accessibility to threats, $\beta_x \in [0, 1]$, with 1 indicating complete accessibility and 0 indicating no accessibility.

- *Distance between threats and habitats:*
  The impact of threat on habitat is inversely proportional to the distance of the habitat from the threat. This is explained by the following formula which is used for assessment of impact of threat. For a threat r that is present in cell y and has an impact on the habitat present in cell x, the impact is given by:

$$i_{rxy} = 1 - \left(\frac{dxy}{d_{rmax}}\right) \text{ , when linear}$$

$$i_{rxy} = \exp\left(-\left(\frac{2{\cdot}99}{d_{r\max}}\right) d_{xy}\right) \text{ , when exponential}$$

where d_{xy} is the linear distance between grid-cells x and y and d_{rmax} is the maximum effective distance of the reach of threat r. The relationship between the rate decay of threat intensity with respect to distance is an exponential one.

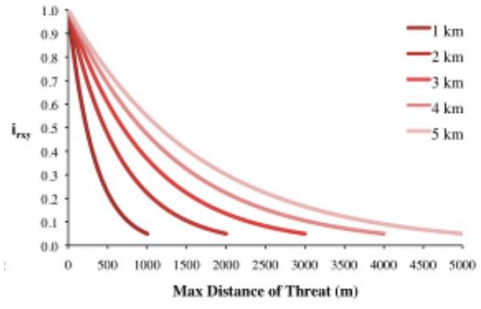


Figure 4.19. Habitat decay rate. The figure shows the relation between the rate of decay of threat impact with increase in distance from source (129).

Priorities to conservation areas can be applied through a fine filter or a coarse filter based on the outcome needed for the region (131). A coarse filter uses a habitat or

a vegetation approach where the justification for conservation is that if there is a green space, then the animals and birds will come. A fine filter is specific to species and its niche of a habitat which it needs. The Habitat Quality Model uses a coarse filter to identify the regions that are good habitats for species and in the study, the contextual maps show the context of degradation along with the distribution of the species sighted in the region. The need for mapping the habitat quality in the outskirts of a city is of great necessity, since it allows to identify the win-win areas where nature and urban centres can co-exist, as this seems to be the future of urban regions in the country. The following image shows InVEST Habitat Quality Model and the provided inputs:

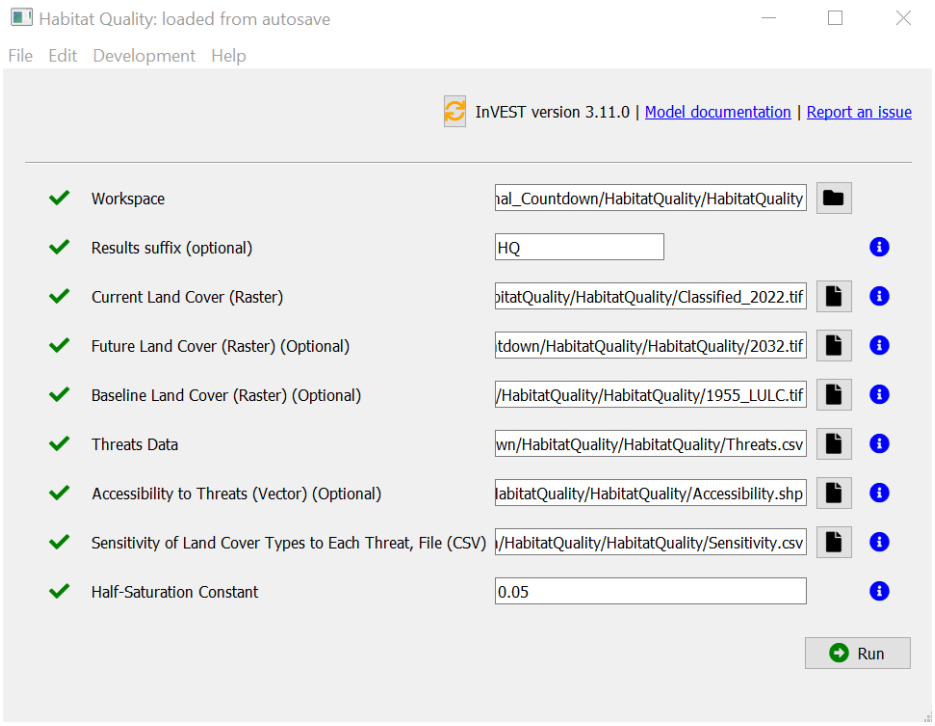


Figure 4.20. InVEST inputs. Inputs going into the InVEST Habitat Quality Model

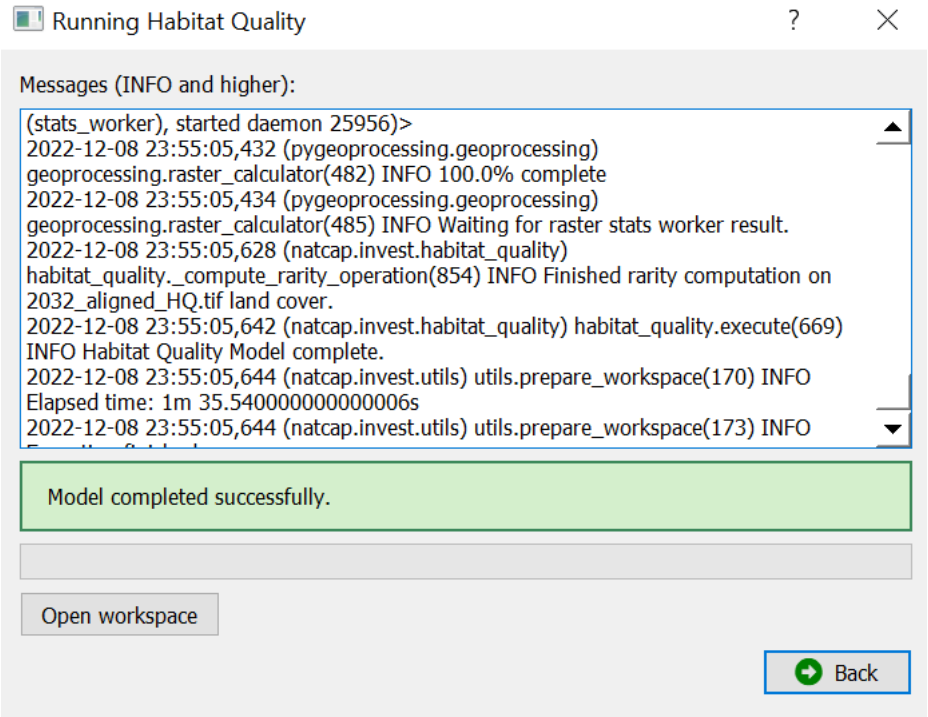


Figure 4.21. Model completion. The display shown on running the model.

4.6.2. Current land cover

The current land cover input to the Habitat Quality Model, is the 2022 land use land cover map that is classified, this is one of the results of the first objective. The map is one of the primary inputs of the model as it serves as the base for the entire model. The detailed methodology for obtaining the map is mentioned in the previous section titled – Methodology followed for Objective 1.

4.6.3. Future land cover

The future land cover map is one of the non-mandatory inputs needed for the InVEST Habitat Quality Model. The model does not specify how many years future it needs, but the map is used just to provide a perspective to the data that is being processed under the current scenario along with the parameters mentioned in the accompanying tables. Software like TerrSet and QGIS offer ways of

creation of future land use maps. For the present study, the MOLUSCE plugin of QGIS 2.18.15 is used for making the future land cover map.

MOLUSCE – Methods of Land Use Change Evaluation

The main operations performed by the plugin are simplistically summarized into the following points (132):

- Uses two land use rasters of different time periods as the main inputs. The auxiliary inputs consist of rasters such as Dynamic Elevation Model, slope, aspect, population data, distance from road, distance from water body, distance from urban region, etc.
- Trains one of the 4 different models:
    - Artificial Neural Network
    - Weights of Evidence
    - Multi-Criteria Evaluation
    - Logistic Regression
- The future map is predicted by one of the models mentioned above.

The following image shows the different inputs used in the MOLUSCE plugin of QGIS. The spatial variables mentioned below are the auxiliary inputs used for better training the model.

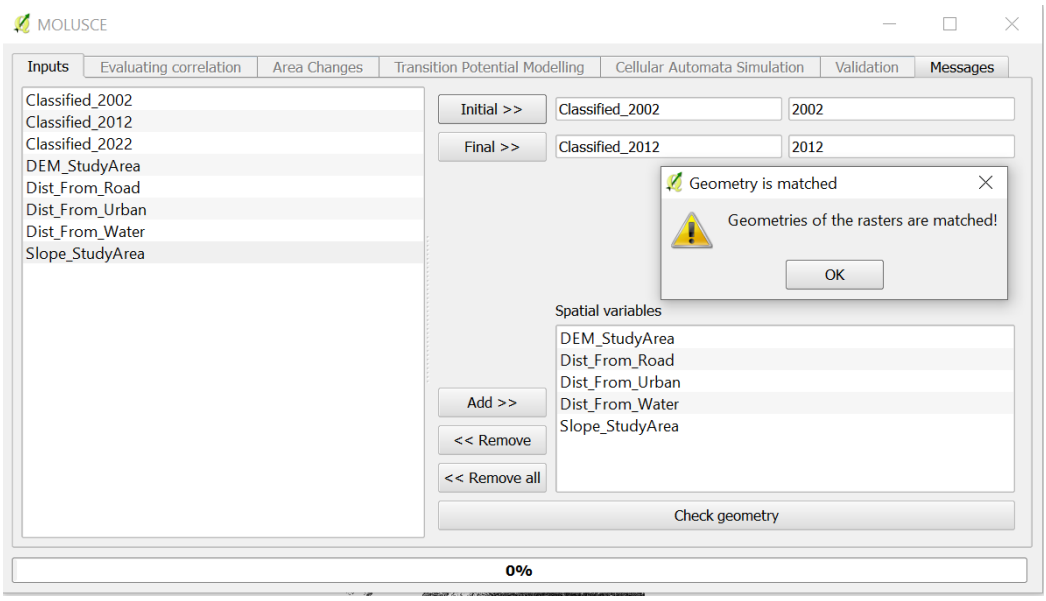


Figure 4.22. Input tab of MOLUSC plugin.

Artificial Neural Network (ANN) is the training model used for this study as it is one of the most efficient models for prediction of future land use. ANN is a computational model that is based on the working of neurons of the brain. It uses algorithms to learn and adjust based on the inputs received. Based on its dynamic learning process that works to predict the best output for the given inputs, It has at least three interconnected layers for processing data (133). The first layer is the input layer which provides data to the subsequent inner layers. The inner layers are hidden layers and are also known as neural layers because it serves as both an input and output layer to subsequent layers, so that there is better understanding at every step. Backpropagation is one way in which neural networks do error correction by assigning weights to individual errors and the same errors are used to recalibrate the data (134).

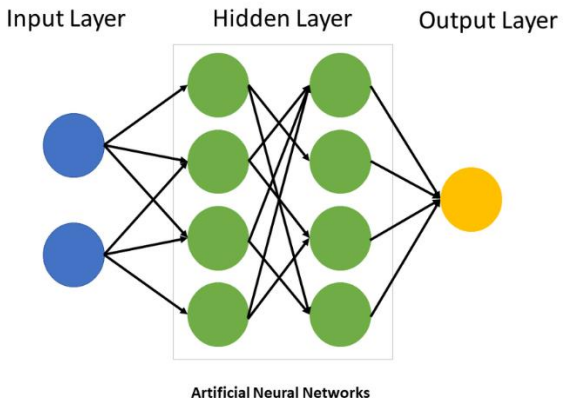


Figure 4.23. ANN working. Figure shows the structure of an artificial neural network highlighting the transfer of information between different layers within it (135).

The following image shows one of the iterations where the most efficient combination of sample size and iterations were finalized for the best output of the function.

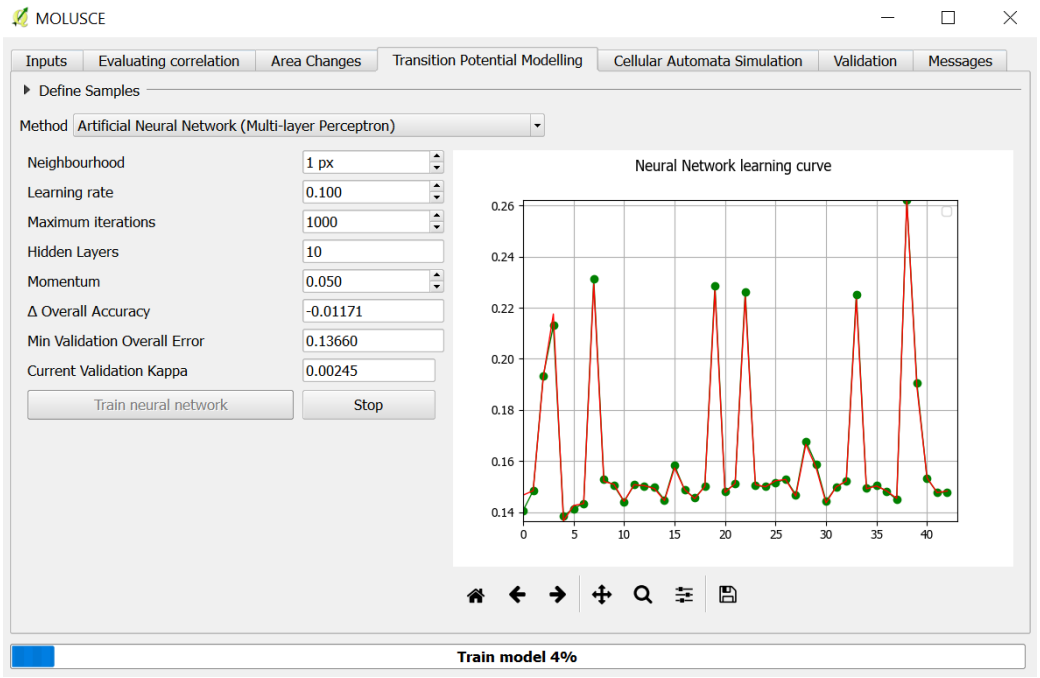


Figure 4.24. ANN learning curve. The figure shows the various parameters that need to be set for the machine learning of ANN to be efficient and the best predicted map is obtained for the inputs.

The validation of the predicted map is done against another land use map of the same time period with the help of cross tables (132). The validation process uses the Kappa coefficient (covered in the previous section) for assessing the accuracy of the predicted map. Once validation is complete, 2012 and 2022 maps are used as inputs and a 2032 prediction map is made for the study.

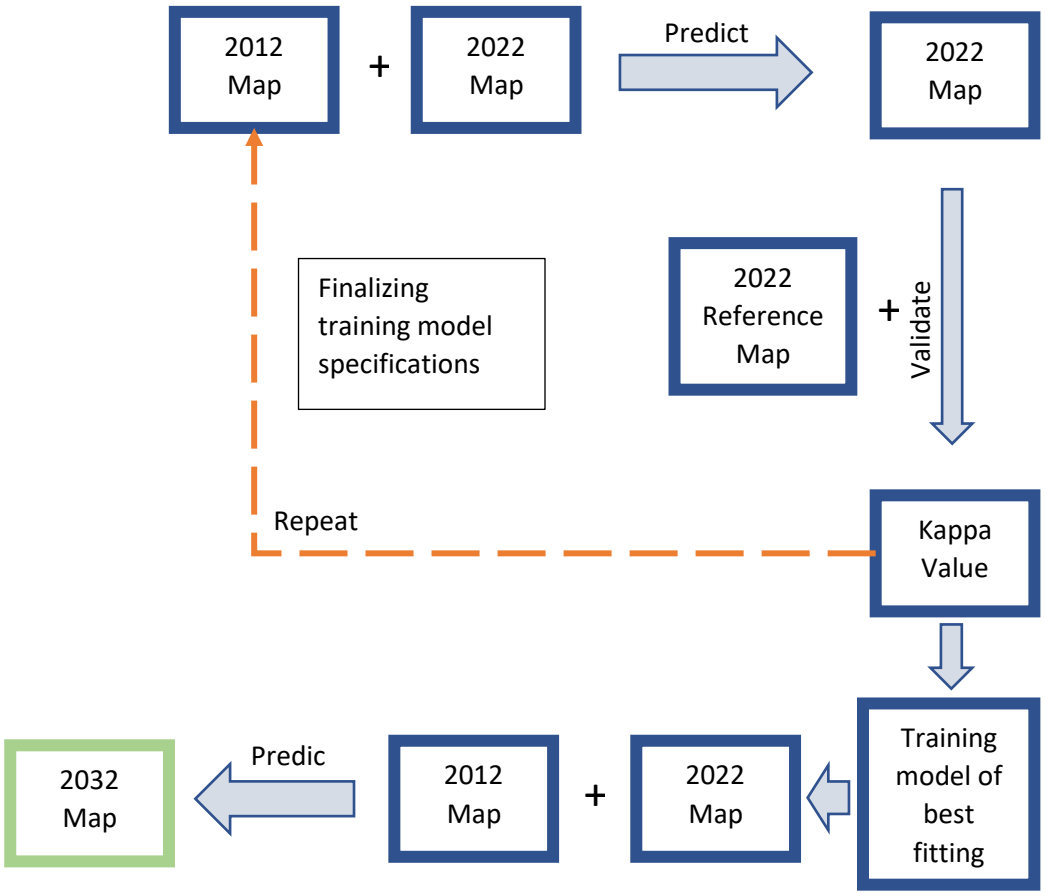


Figure 4.25. Future map block diagram. The block diagram shows the workflow for prediction of the future land cover map.

4.6.4. Baseline land cover

As per the InVEST Habitat Quality Model documentation (129), the baseline land cover map is one of when the intensive land management of the region was low compared to present day scenario. It is mentioned to use maps as old as 1851 for a baseline. Therefore, the assumption of a baseline map is that everything was at one point a forest. The oldest map of topography of the region was of a 1955 U.S. army service map, with data from surveys conducted in 1920s – 30s. A classified form of this map is used as the baseline map for the study. The following toposheets were mosaicked, masked and classified into land use land cover classes as done in some earlier studies (136).

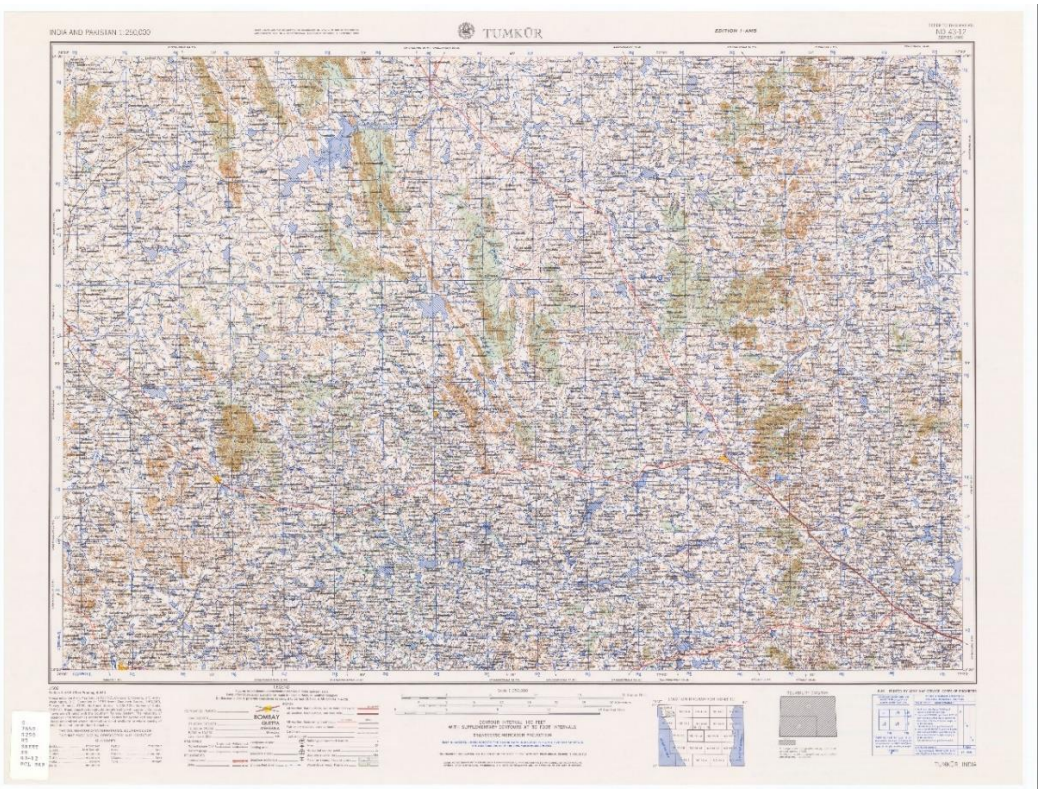


Figure 4.26. 1955 toposheet of Tumkur region.

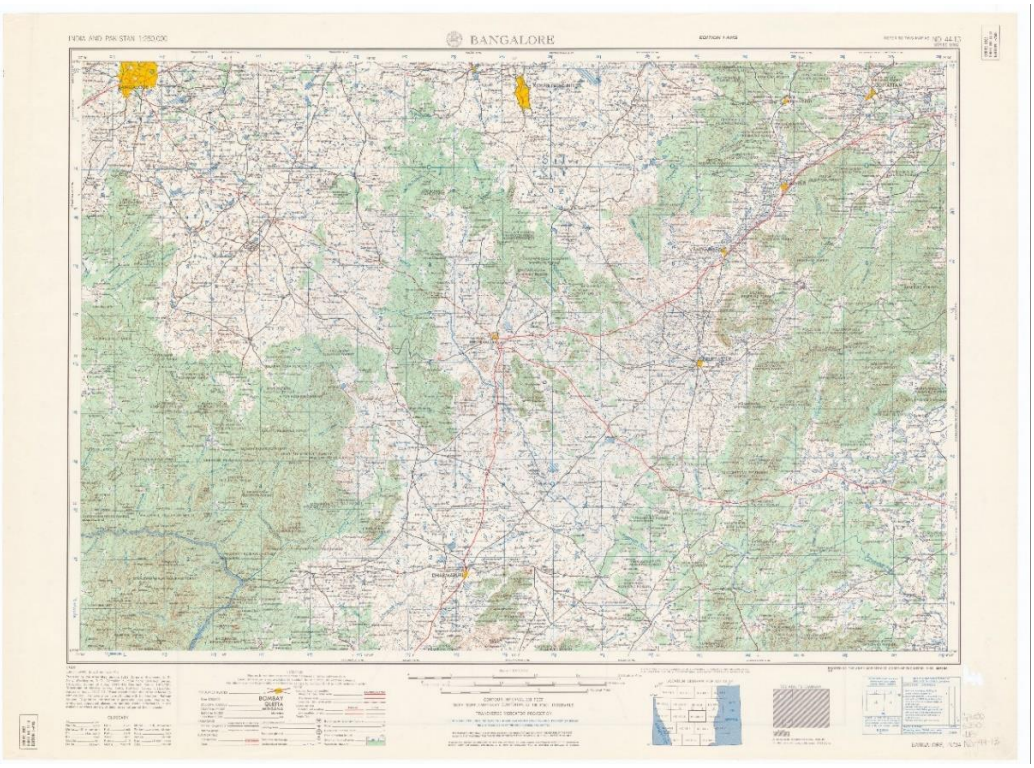


Figure 4.27. 1955 toposheet of Bangalore region

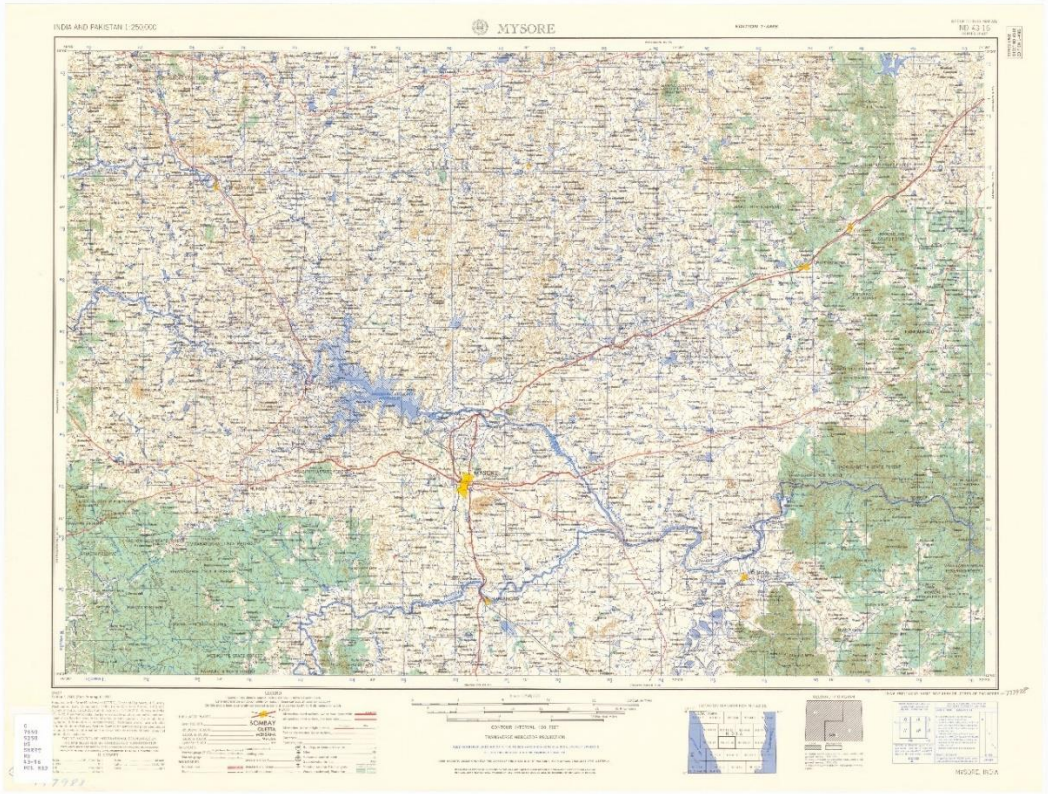


Figure 4.28. 1955 toposheet of Mysore region

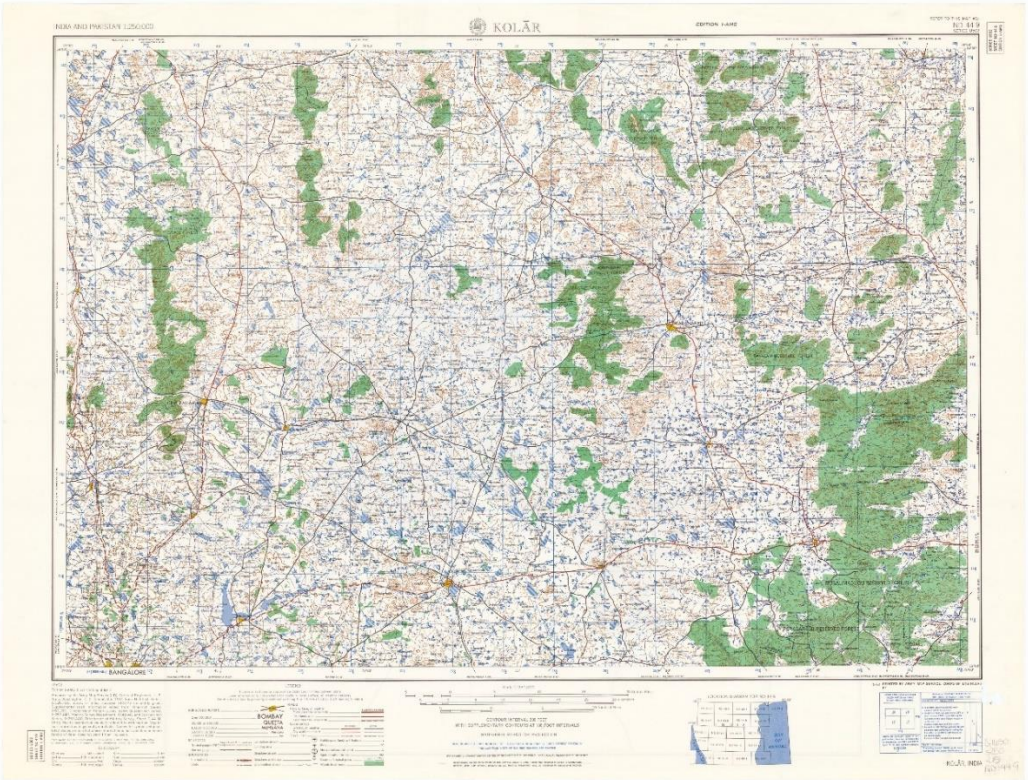


Figure 4.29. 1955 toposheet of Kolar region

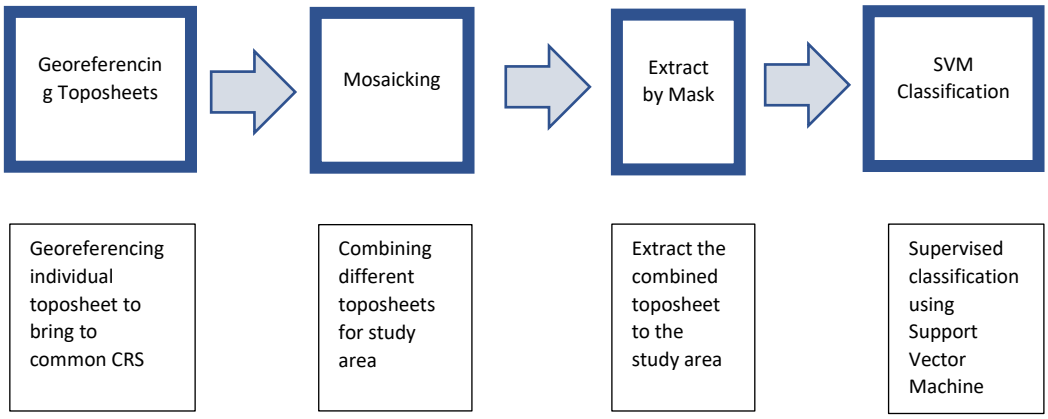


Figure 4.30. Baseline map block diagram. Block diagram showing the flow of events for creation of baseline map.

4.6.5. Supporting tables and vectors

Along with baseline, present and future land use maps, some additional data needs to be prepared for InVEST Habitat Quality Model to provide a desired output. The additional data is listed as follows (129):

- *Threats table*

  A table mapping each threat to its individual properties along with their distribution maps. It has the following components: **threat** – Name of the threat, **max_dist** – The maximum distance over which the threat has its influence, **weight** – How much impact does the threat have over habitat quality when compared to other threats? **Decay** - rate of decay of habitat, it can be linear or exponential. **Cur_path, base_path, fut_path** are the paths to the various rasters of present, baseline and future threat maps.

THREAT	MAX_DIST	WEIGHT	DECAY	BASE_PATH	CUR_PATH	FUT_PATH
URBAN	4	0.8	exponential		urban.tif	urban_f.tif
ROADS	2	0.4	linear		roads.tif	roads_f.tif

Table 4.10. Threats table.

- *Sensitivity table*

It is a table that maps habitat presence to threat. The mapping happens land use land cover class-wise. It has the following components: **LULC** – The land use codes corresponding to each class. **Habitat** – Habitat suitability, 0 means not suitable and 1 means suitable. **Threat –** The relative sensitivity of LULC class to a type of threat which are listed as separate columns.

LULC	NAME	HABITAT	URBAN	ROADS
1	Built up area	0	0	0
2	Water body	1	0.9	0.4
3	Agricultural area	1	0.4	0.4
4	Vegetation cover	1	0.9	0.7
5	Bare land	0	0	0
0	Null	0	0	0

Table 4.11. Sensitivity Table. The table depicting the sensitivity table used in the present study.

- *Accessibility to threats vector*

This vector is designed in a way that military areas, forest regions and lakes are given the status of protected ecological assets because they come under the protection of various institutions such as the Indian Army, Karnataka Forest Department, and the BBMP (Municipality of Bengaluru), respectively. The polygons with protection are given a status

less than 1 and the polygons that do not have protection are given status of 1.

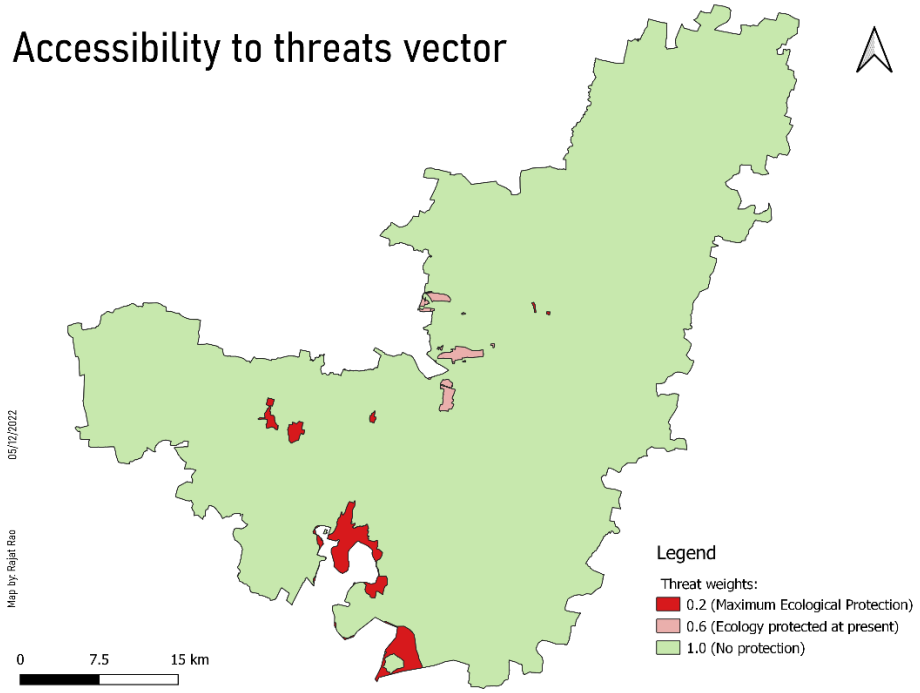


Figure 4.31. Accessibility map. The vector map that shows accessibility to threats for the study region.

These three data sources throw more light on the study region and help steer the model towards an accurate direction.

5. RESULTS

The following section is a detailed documentation of intermediate results, end results and all the findings of the study. The observances are presented in the form of maps and graphs for easy interpretation of data. The first section (till *Change at Tehsil level*) lays out the results pertaining to objective 1 and the section that follows has results pertaining to objective 2.

5.1. Confusion matrix and Kappa co-efficient

2002 data:

For the year 2002, the number of ground truth samples taken are 500 and the method of sampling is – *Stratified Sampling*, since the area of classes are not evenly distributed. Out of the 500 points, the following were the *Producer's Accuracy* and *User's Accuracy* derived from the confusion matrix.

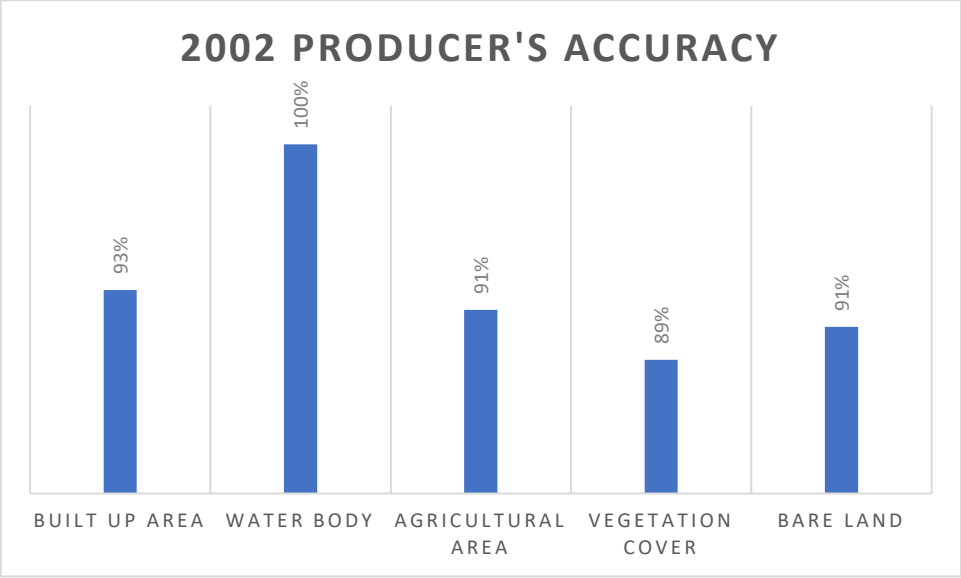


Figure 5.1. Graph of Producer's Accuracy of 2002 classified data

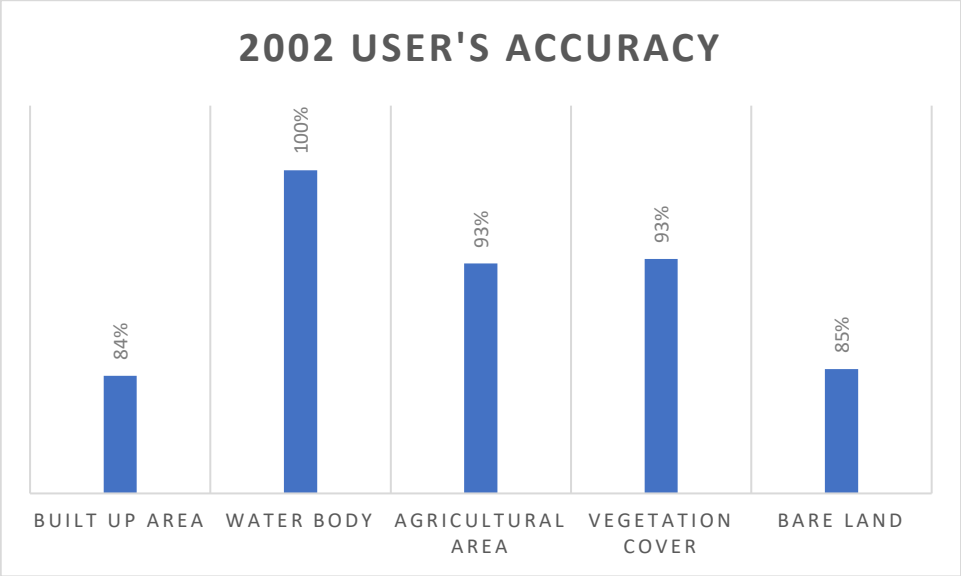


Figure 5.2. Graph of User's Accuracy of 2002 classified data

The final *Kappa Coefficient* obtained from this is – **86.79 %**, making the classified data highly accurate.

2012 data:

For the year 2012, the number of ground truth samples taken are 501 and the method of sampling is – *Stratified Sampling*, since the area of classes are not evenly distributed. Out of the 501 points, the following were the *Producer's Accuracy* and *User's Accuracy* derived from the confusion matrix.

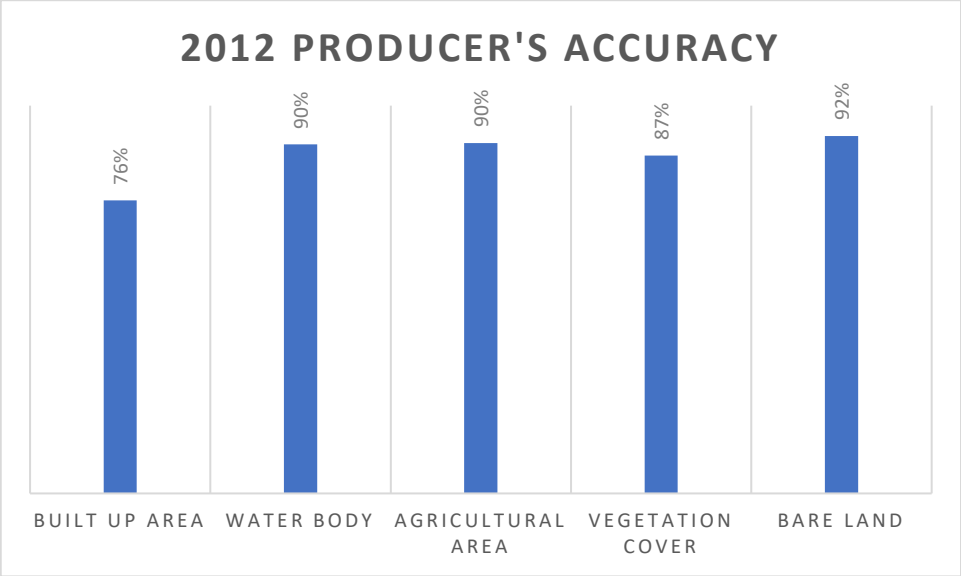


Figure 5.3. Graph of Producer's Accuracy of 2012 classified data

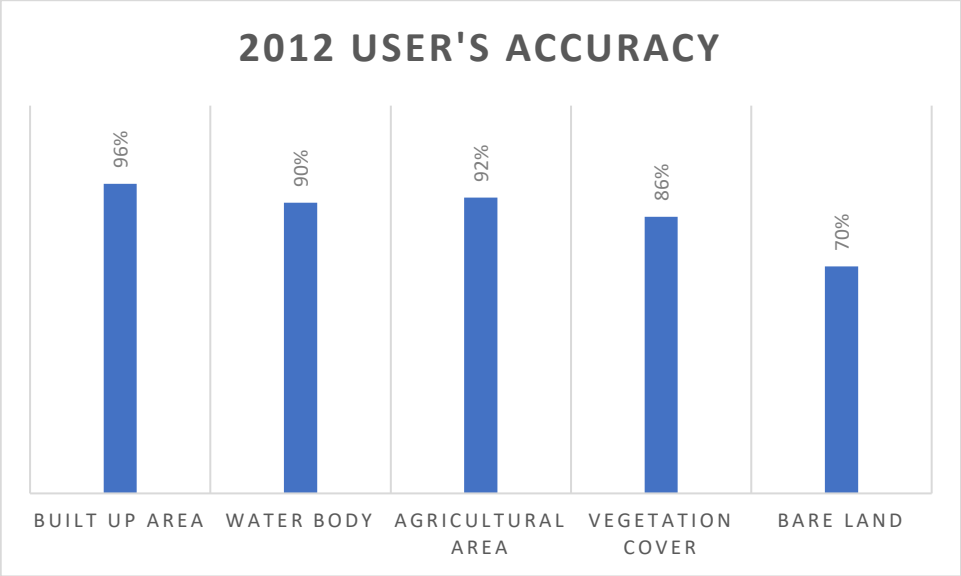


Figure 5.4. Graph of User's Accuracy of 2012 classified data

The final *Kappa Coefficient* obtained from this is – **81.75 %**, making the classified data highly accurate.

2022 data:

For the year 2022, the number of ground truth samples taken are 502 and the method of sampling is – *Stratified Sampling*, since the area of classes are not

evenly distributed. Out of the 502 points, the following were the *Producer's Accuracy* and *User's Accuracy* derived from the confusion matrix.

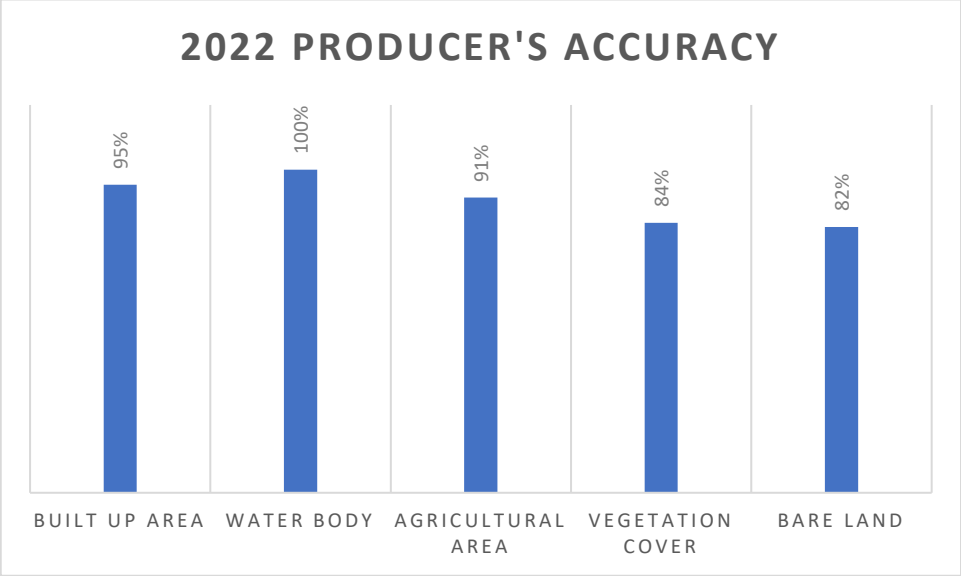


Figure 5.5. Graph of Producer's Accuracy of 2022 classified data

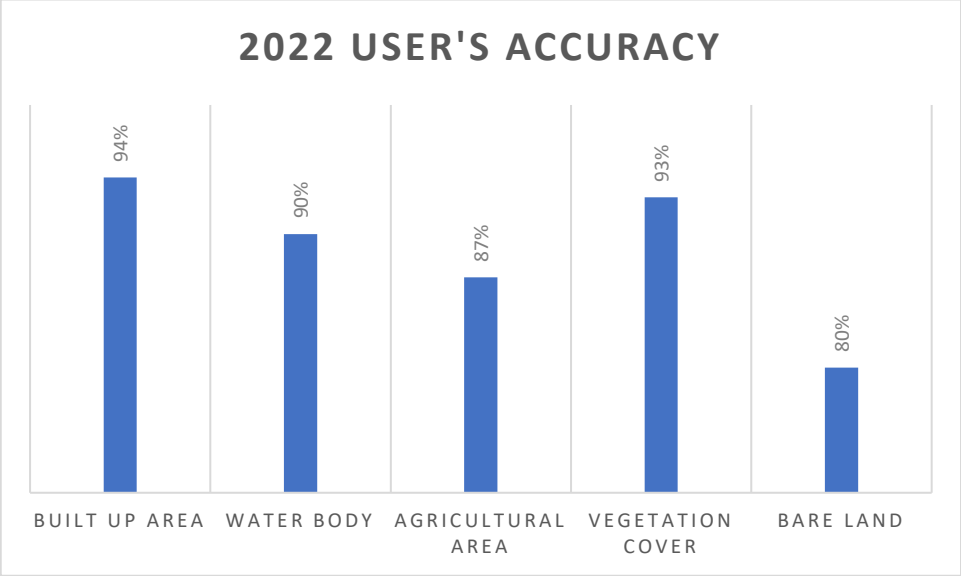


Figure 5.6. Graph of User's Accuracy of 2022 classified data

The final *Kappa Coefficient* obtained from this is – **84.90 %**, making the classified data highly accurate.

5.2. Area statistics

This section shows changes in different landform classes associated with an urban setting over three spans of time 2002, 2012 and 2022. The classes are categorized are as follows:

- Built up area – region showing urban areas such as buildings, roads, cemented surfaces, etc.
- Water body – areas showing different water bodies all over the study area. In Bengaluru region, the water bodies are usually lakes and lakes with hyacinth growth.
- Agricultural area – These are cultivated or cultivable lands.
- Vegetation cover – Shows the green growth in the study area, it includes – empty sites, unchecked undergrowth, forests, parks, and roadside avenues.
- Bare land – This includes exposed rock surfaces, barren land, and regions where the land is completely exposed.

5.2.1. Study region statistics

This region shows the variation in area statistics of the region over the span of two decades as shown below:

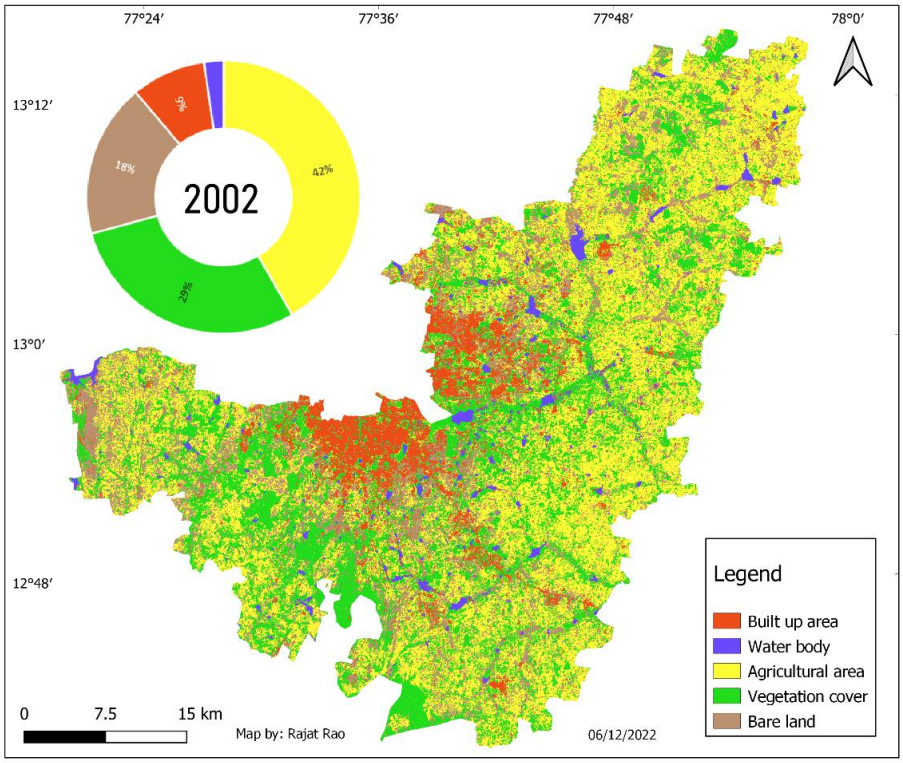


Figure 5.7. 2002 LULC classes showing areas occupied by different classes. Urban growth is localized to one region of the study area with good amount of green cover.

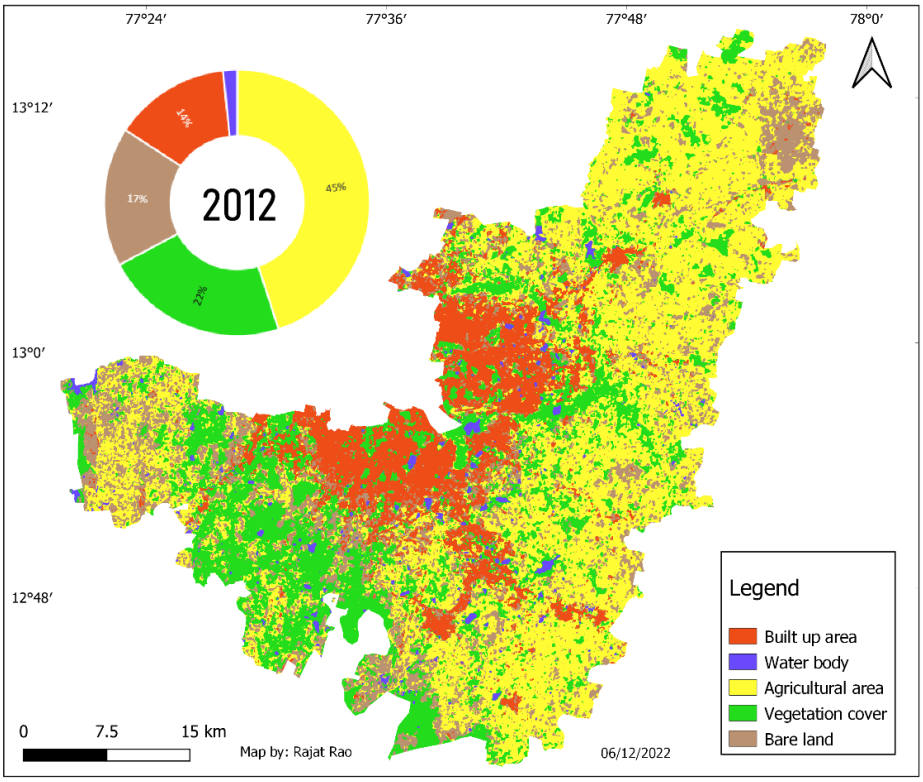


Figure 5.8. 2012 LULC classes showing areas occupied by different classes. Urban growth looks to be spreading. Regions of Electronic city, Sarjapur road and Whitefield are prominent.

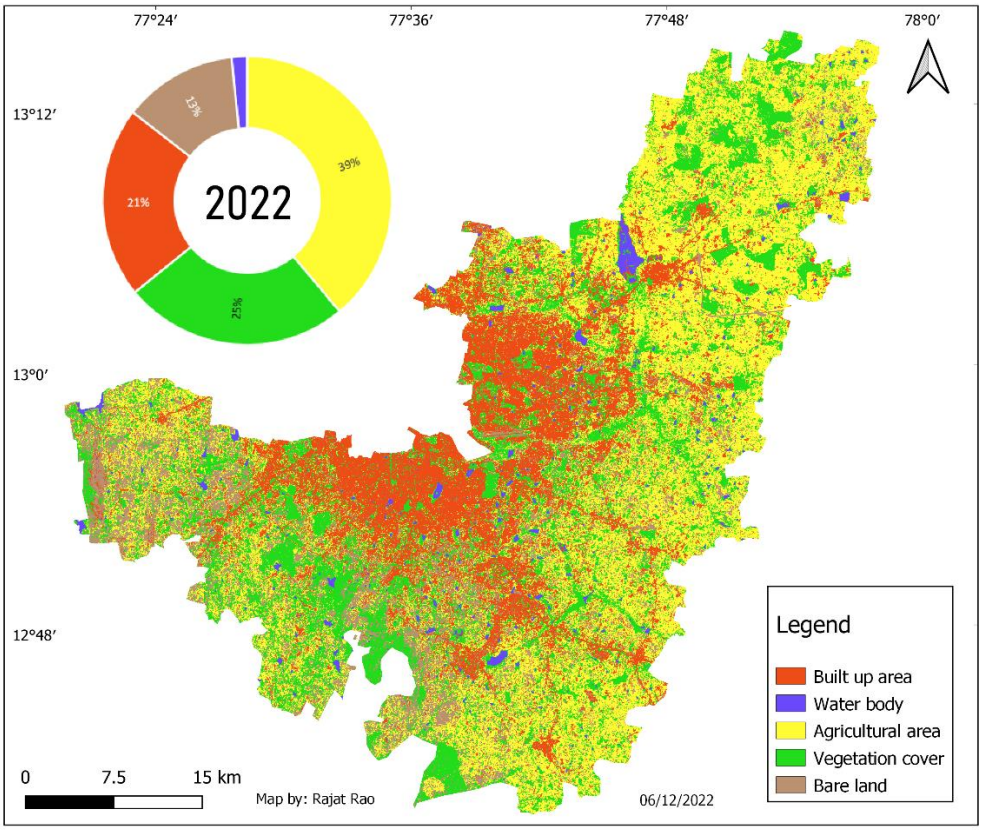


Figure 5.9. 2022 LULC classes showing areas occupied by *different classes. The urban regions are seen not to be localized to the three regions but growing in a fragmented manner from there.*

The following graph effectively summarizes the variation in land use classes of 2002, 2012 and 2022.

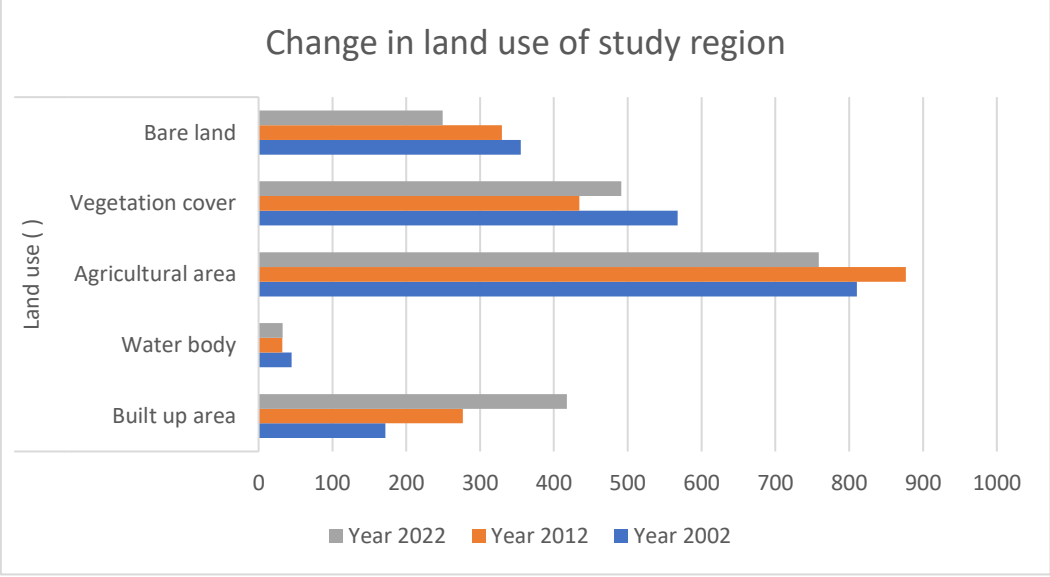


Figure 5.10. Overall summary of study region variation over time

5.2.2. Tehsil-wise statistics

This section documents the results associated with respect to tehsils. These are used in answering a few conservation questions during discussion.

Anekal:

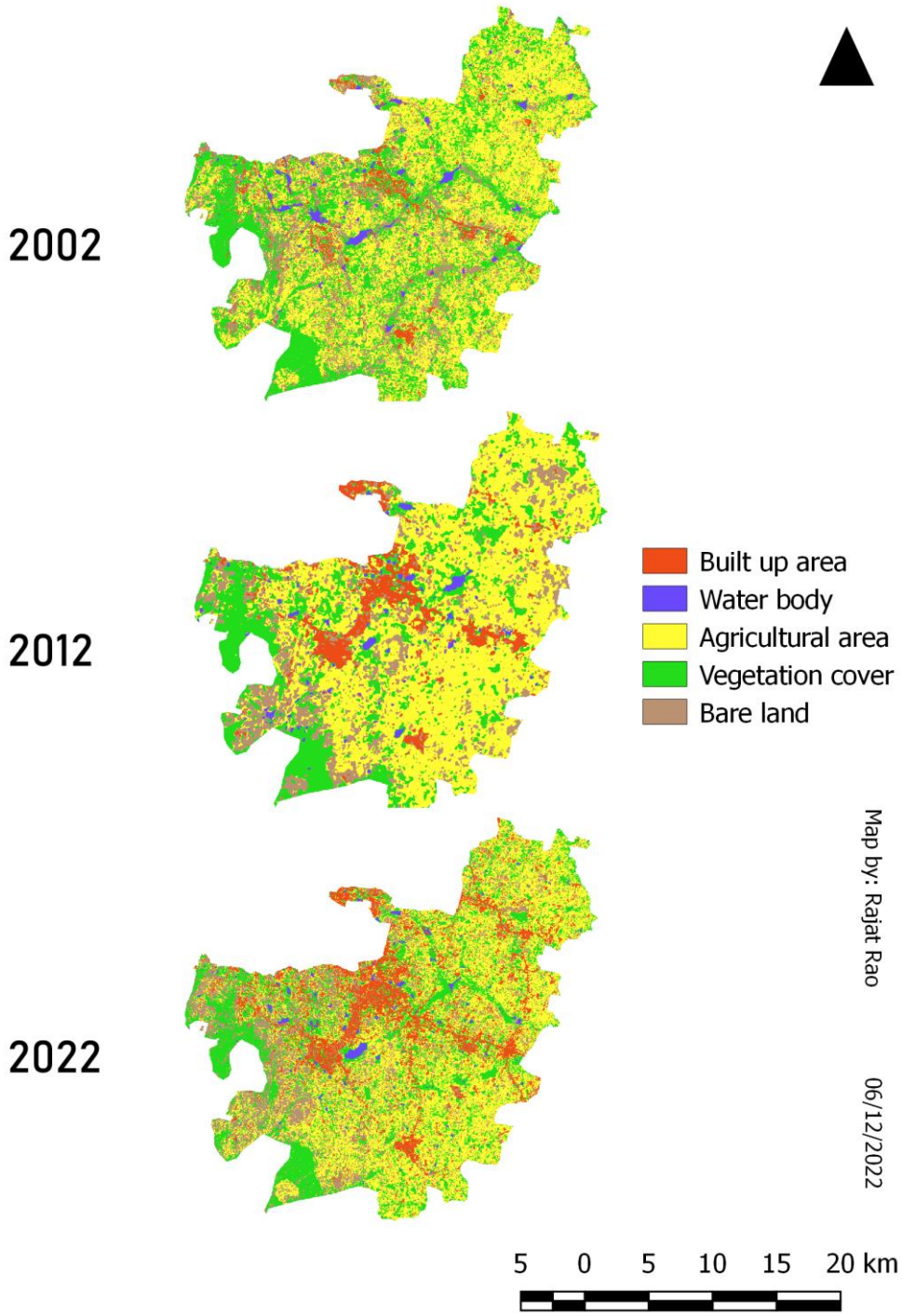


Figure 5.11. Spatial-temporal analysis of Anekal tehsil. Consists of Electronic City – Phase II patch shown as the prominent patch in the map.

Bangalore-East:

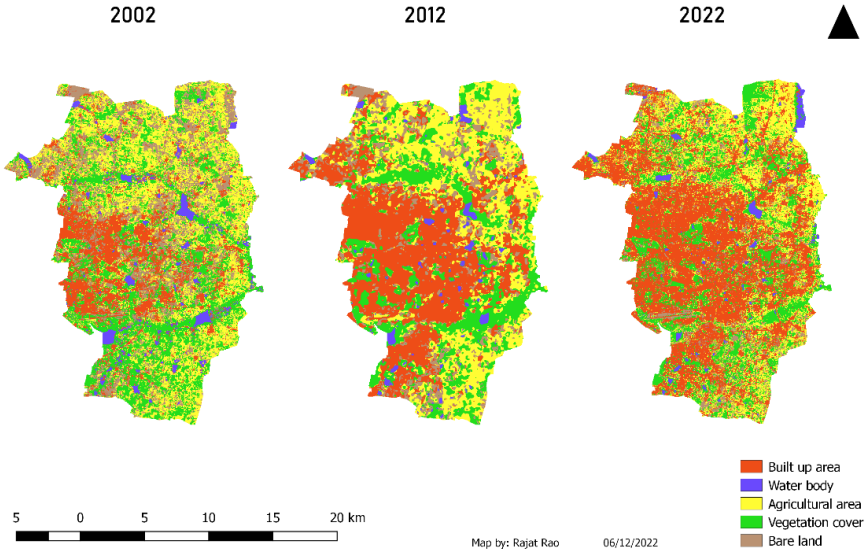


Figure 5.12. Spatial-temporal analysis of Bangalore-East tehsil. Houses the Sarjapur and Whitefield belt which has a lot of IT companies. The related development is shown on the map.

Bangalore-South:

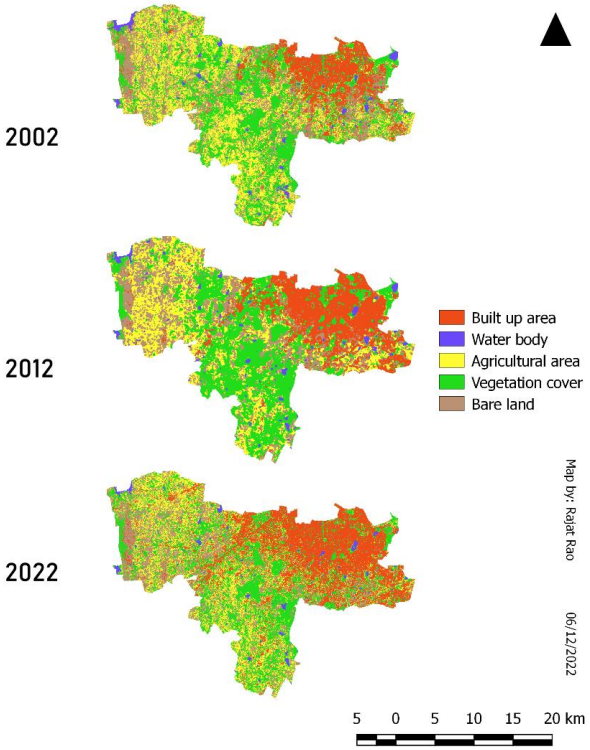


Figure 5.13. Spatial-temporal analysis of Bangalore-South tehsil. This has Electronic City Phase I and related development.

Hoskote:

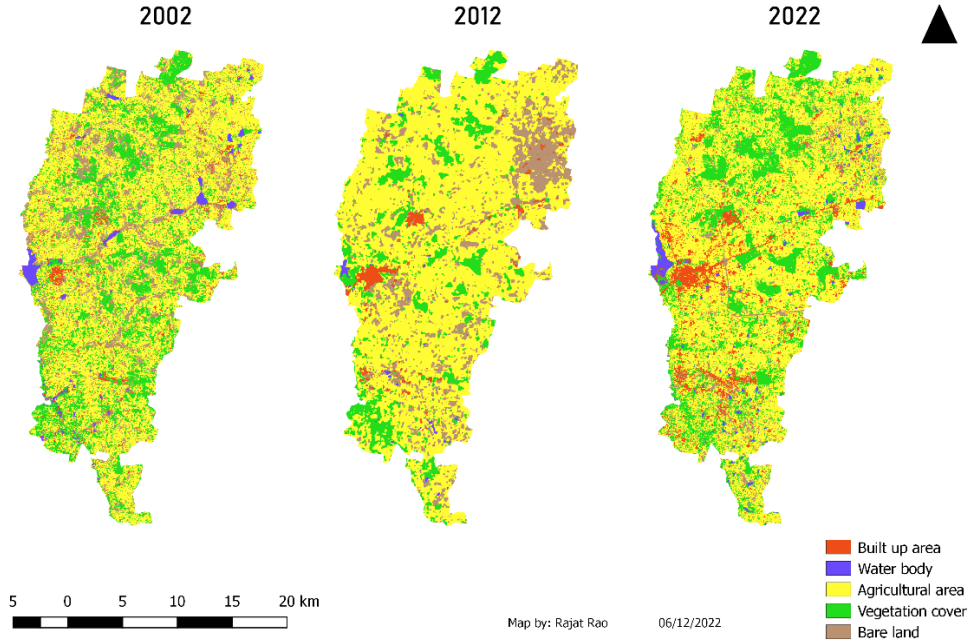


Figure 5.14. Spatial-temporal analysis of Hoskote tehsil. A part of the Bangalore rural tehsil which contains very little built up areas.

5.3. Change analysis

The results of change analysis are covered in two parts under this section. The first being in terms of the entire study region and the second explores results specific to each tehsil.

5.3.1. Change in study region

2002 – 2012 Changes

The change analysis results for the decade between 2002 to 2012 is as follows, the colour-coded map shows details of transformation from one land use class to another.

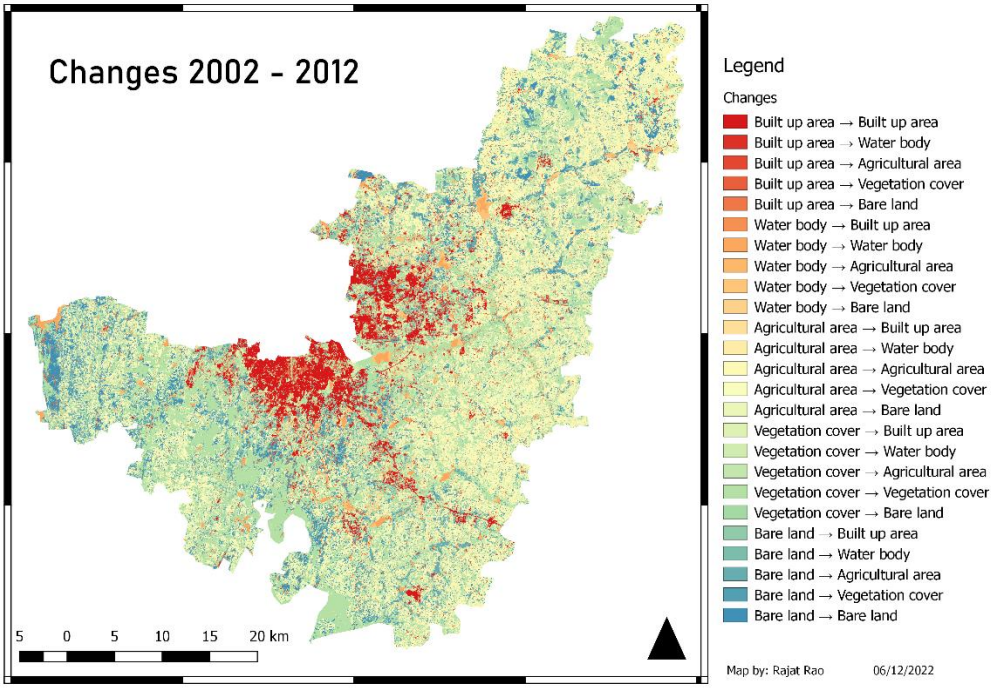
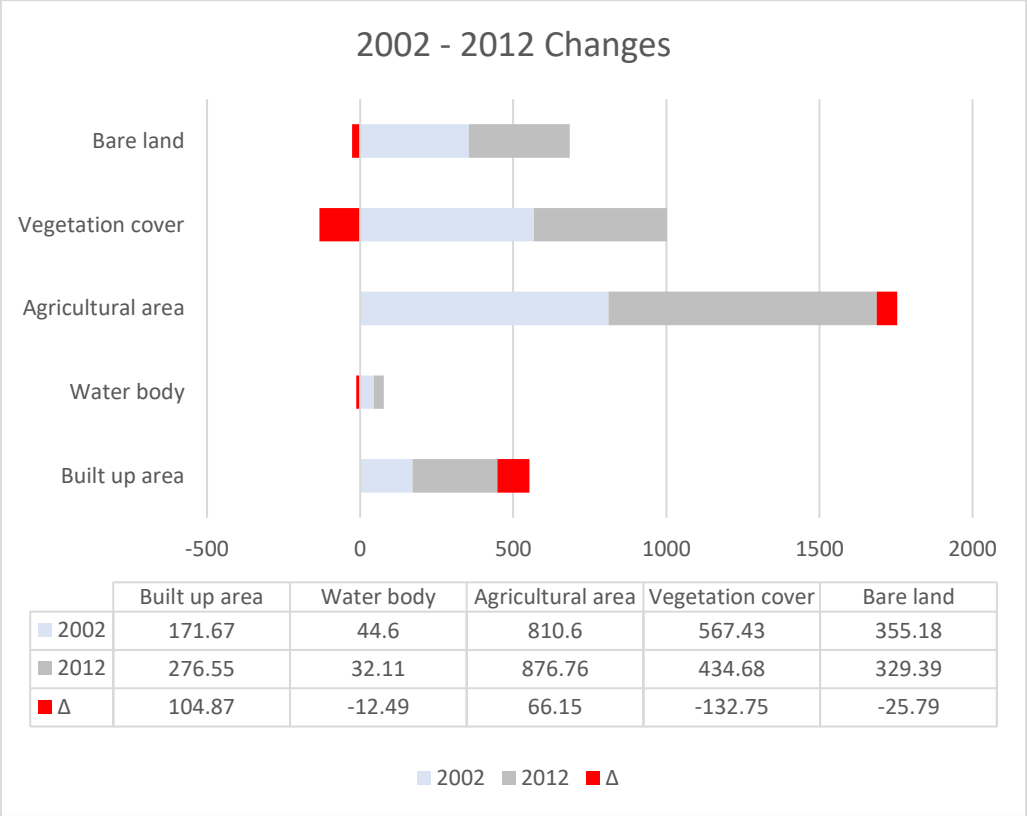


Figure 5.15. Transformation between various LULC classes 2002 – 2012

The following graph depicts the statistics of change that is shown in the above map. The variation in land use class is depicted by red and a reduction is shown with a negative sign.



	Built up area	Water body	Agricultural area	Vegetation cover	Bare land
2002	171.67	44.6	810.6	567.43	355.18
2012	276.55	32.11	876.76	434.68	329.39
Δ	104.87	-12.49	66.15	-132.75	-25.79

Figure 5.16. Graph of changes in LULC 2002 – 2012

2012 – 2022 Changes

The change analysis results for the decade between 2012 to 2022 is as follows, the colour-coded map shows details of transformation from one land use class to another.

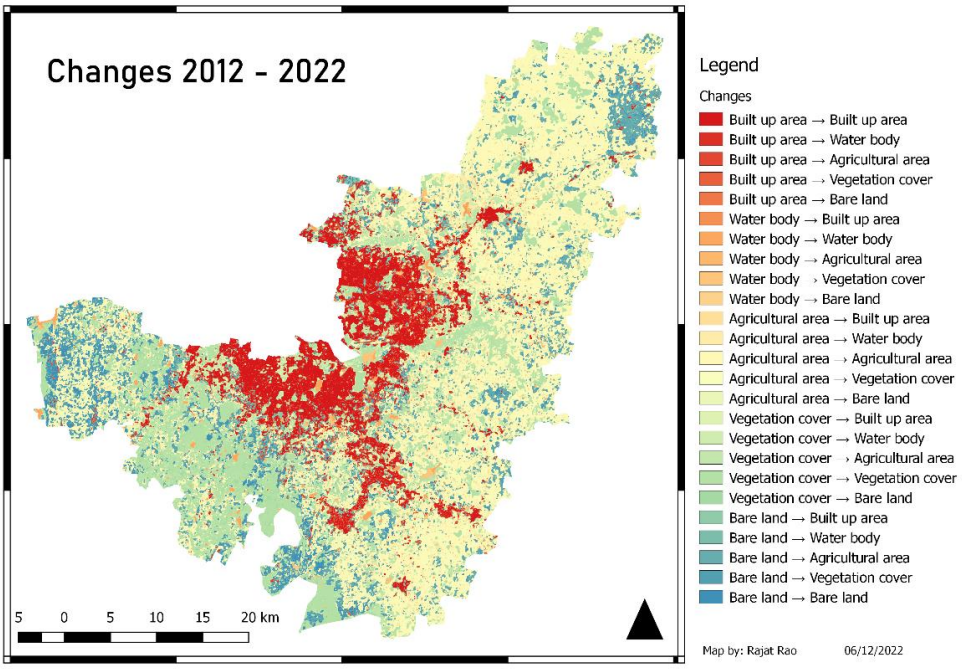
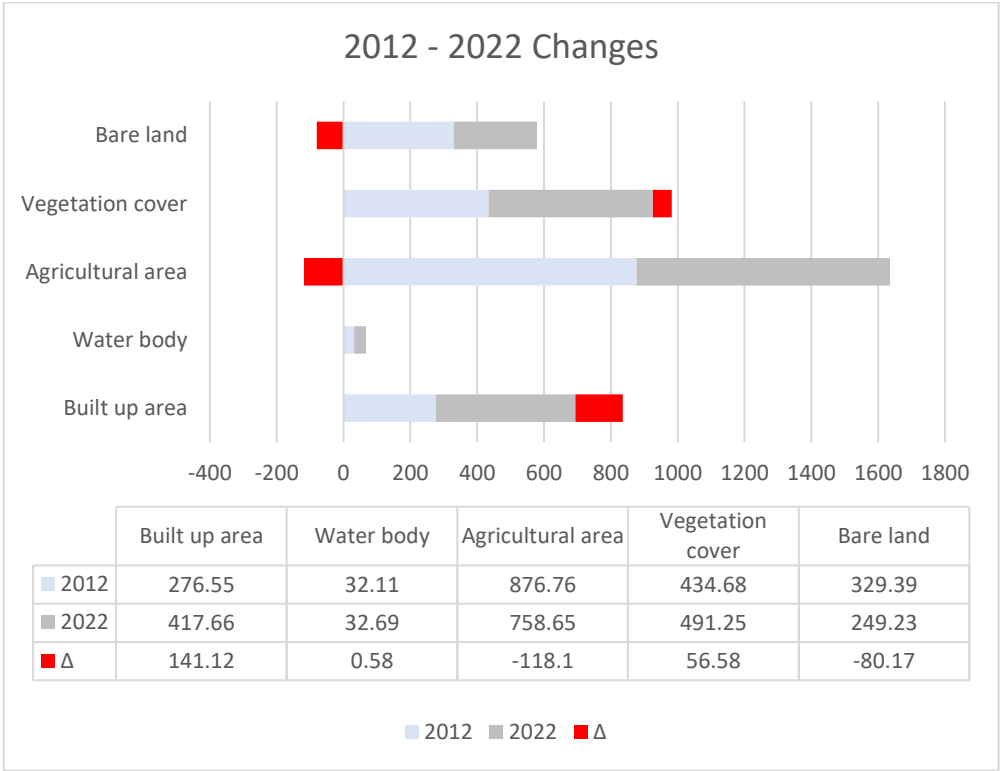


Figure 5.17. Transformation between various LULC classes 2012 – 2022

The following graph depicts the statistics of change that is shown in the above map. The variation in land use class is depicted by red and a reduction is shown with a negative sign.



	Built up area	Water body	Agricultural area	Vegetation cover	Bare land
2012	276.55	32.11	876.76	434.68	329.39
2022	417.66	32.69	758.65	491.25	249.23
Δ	141.12	0.58	-118.1	56.58	-80.17

Figure 5.18 Graph of changes in LULC 2012 – 2022

5.3.2. Change at tehsil level

When looked at a narrower scope of a tehsil, instead of the big picture results of the study region, there is an attempt to focus on more specific problems. In this case, we consider the problem of trade-offs between green spaces and urban region within each tehsil.

Bangalore East Tehsil

Having an area of 329.77 sq. km. it is the part of Bengaluru that used to be the erstwhile outskirts of the city. These days, they have become very much a part of the heart of the city having most commercial spots present here. The following graph shows the variation between urban spaces and green spaces over three time periods.

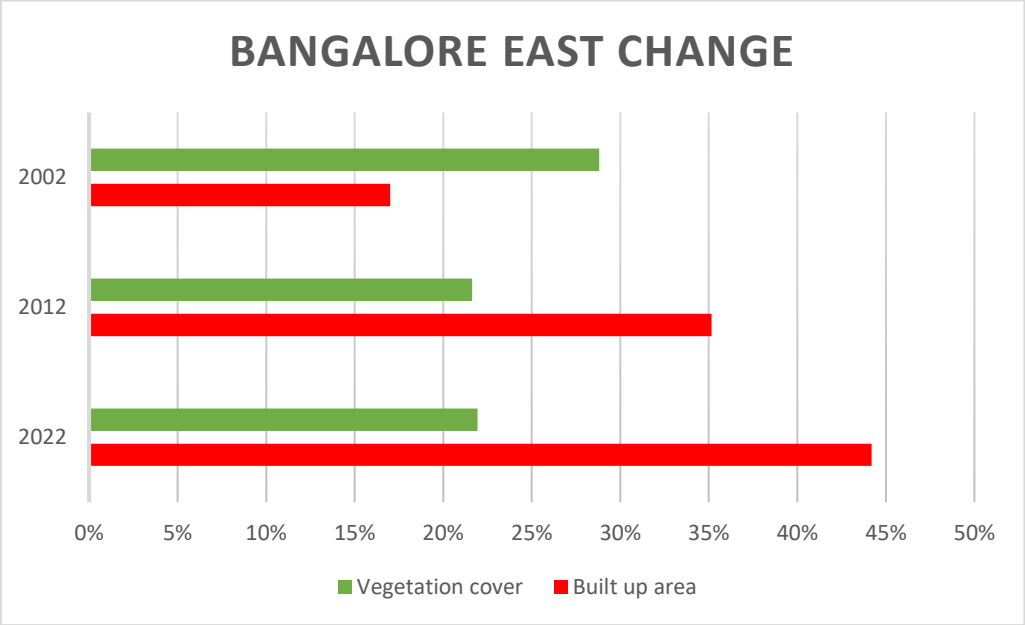


Figure 5.19. Graph showing tehsil level change for Bangalore East region

Bangalore South Tehsil

Having an area of 540.25 sq. km. it is the part of Bengaluru that is connected to Bannerghatta National Park. The lowest half of this region contains the start of the

national park. The following graph shows that there is not much decrease in vegetation cover from 2002 to 2022 but, the built up areas have increased.

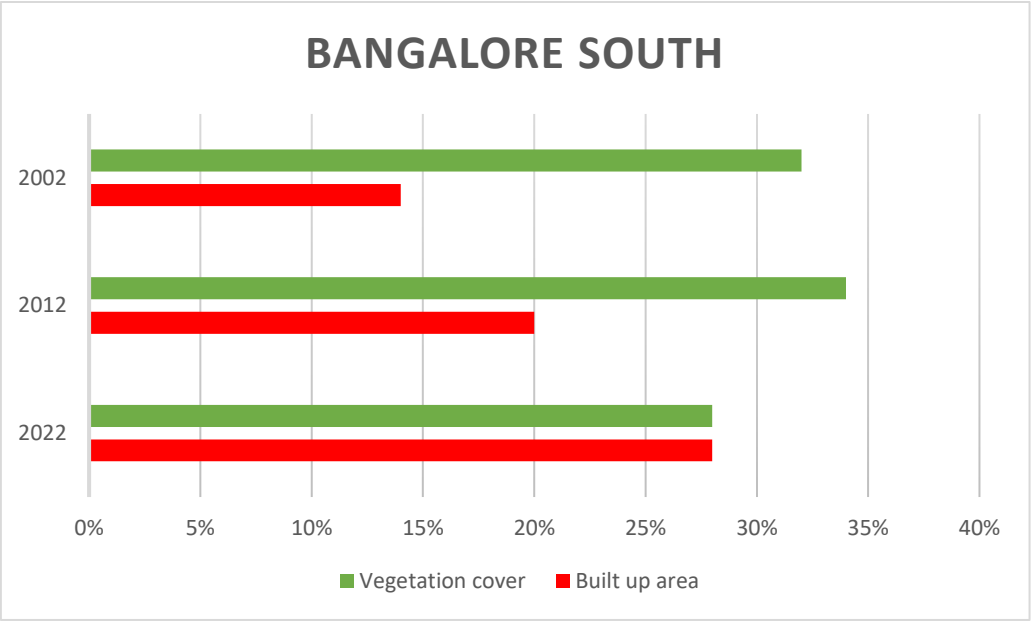


Figure 5.20. Graph showing tehsil level change for Bangalore South region

Anekal Tehsil

Having an area of 533.40 sq. km. it is the part of Bengaluru that houses the Electronic City (One of the earliest SEZs of the city). This region is also unique because, despite being a part of the Bangalore Urban district, some of the constituencies do not come under the BBMP (the city's municipality) but rather it is covered under the Gram Panchayat. The following graph shows the variation between urban spaces and green spaces over three time periods for the region.

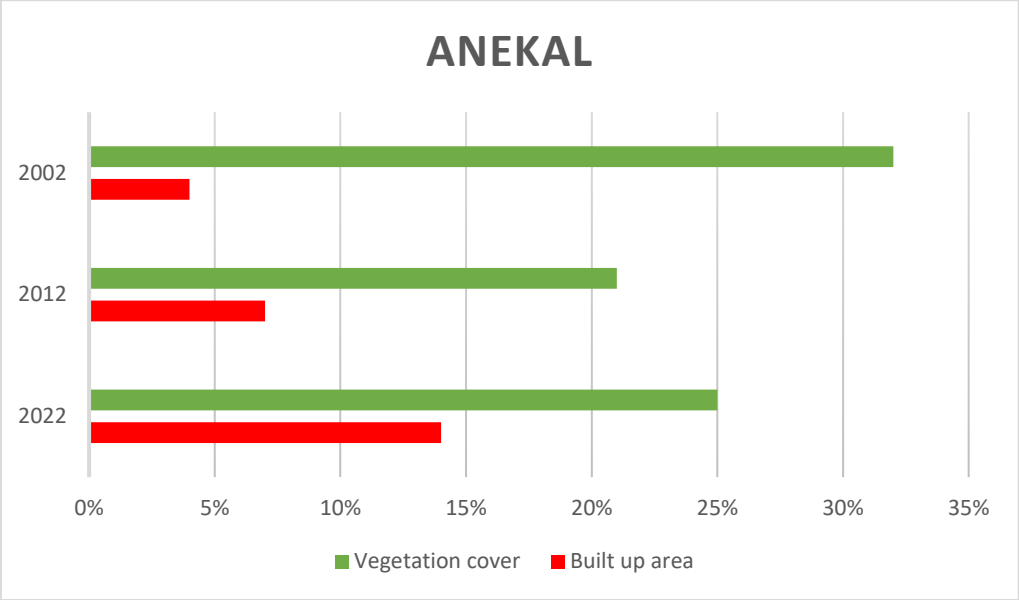


Figure 5.21. Graph showing tehsil level change for Anekal region

Hoskote Tehsil

Having an area of 546.08 sq. km. it is the only tehsil of the study region that is a part of Bangalore Rural district which itself comes with a unique research scope. The following graph shows the variation between urban spaces and green spaces over three time periods for the region. The 3% increase in green cover can be due to the increased efforts of Hoskote Lake protection or the rise of green themed properties in the region.

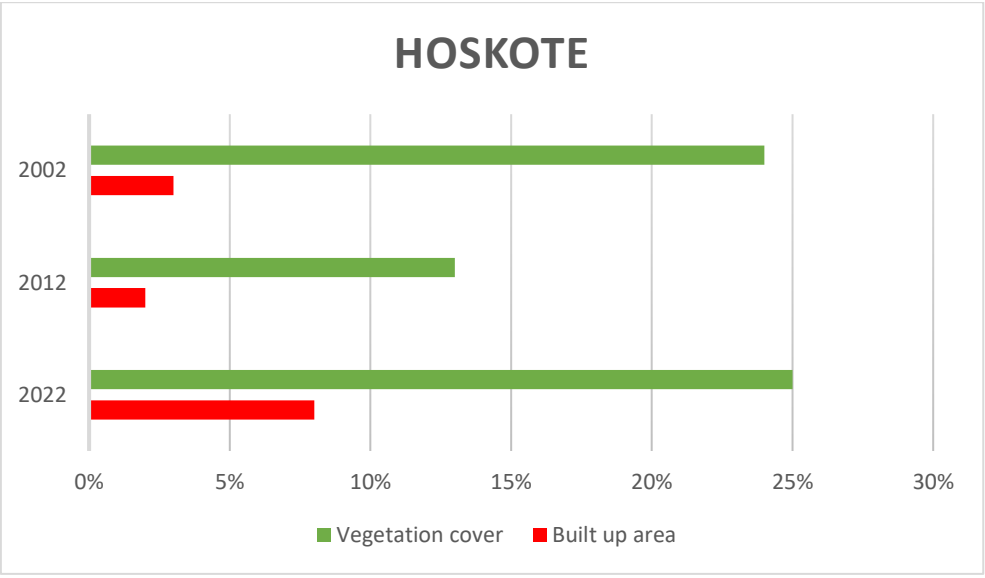


Figure 5.22. Graph showing tehsil level change for Hoskote region

5.4. 1955 classification for baseline map

One of the inputs for InVEST Habitat Quality Model is a baseline map. According to InVEST documentation, a baseline map is one from a time when extensive land use changes were not happening. As an example, they had suggested a map from 1800s. For the present study, the oldest map that was available was the 1:250,000 India and Pakistan Series U502, U.S. Army Map Service, 1955. Four different toposheets (ND 43-12, ND 43-16, ND 44-9A and ND 44-13A) had the following operations done on them to obtain the resulting classified map as shown below:

- Georeferencing
- Mosaicking
- Mask Extraction
- Classification

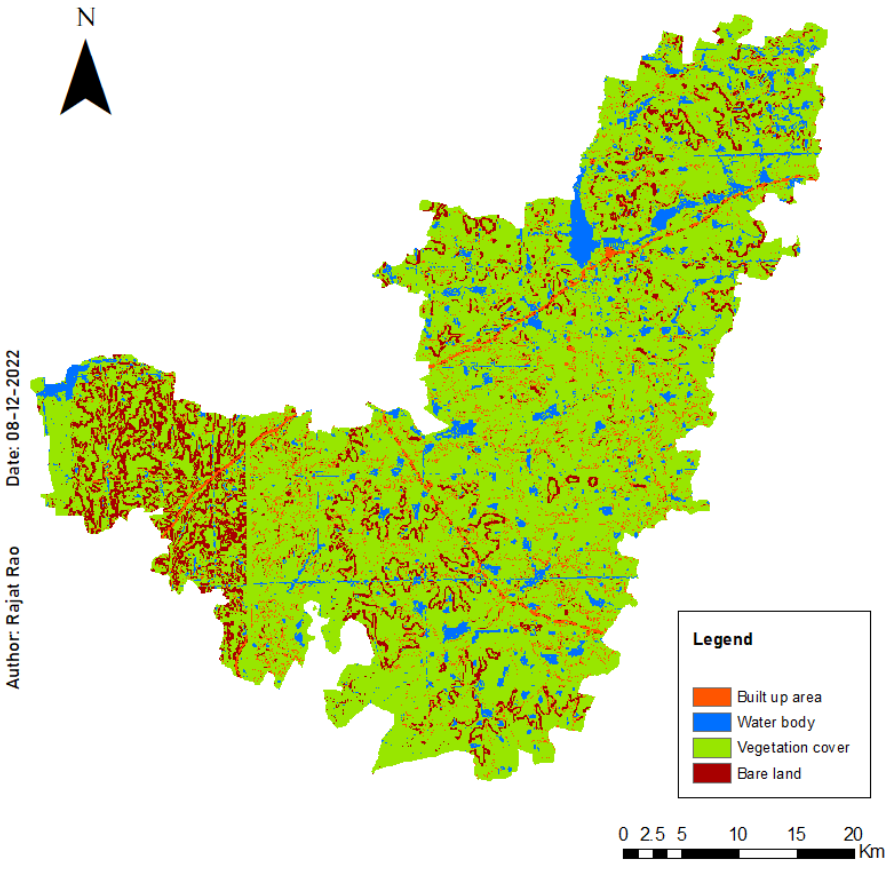


Figure 5.23. 1955 classified data. The above map shows the different classes of the basemap. The map need not be accurate because a baseline map does not adhere to land use plans.

5.5. 2032 future map

The following map is the output of MOLUSCE plugin for prediction of future map. The prediction is done with 55% accuracy.

% of Correctness	54.46593
Kappa (overal)	0.33814
Kappa (histo)	0.73998
Kappa (loc)	0.45696

Figure 5.24. Showing verification of accuracy for future map

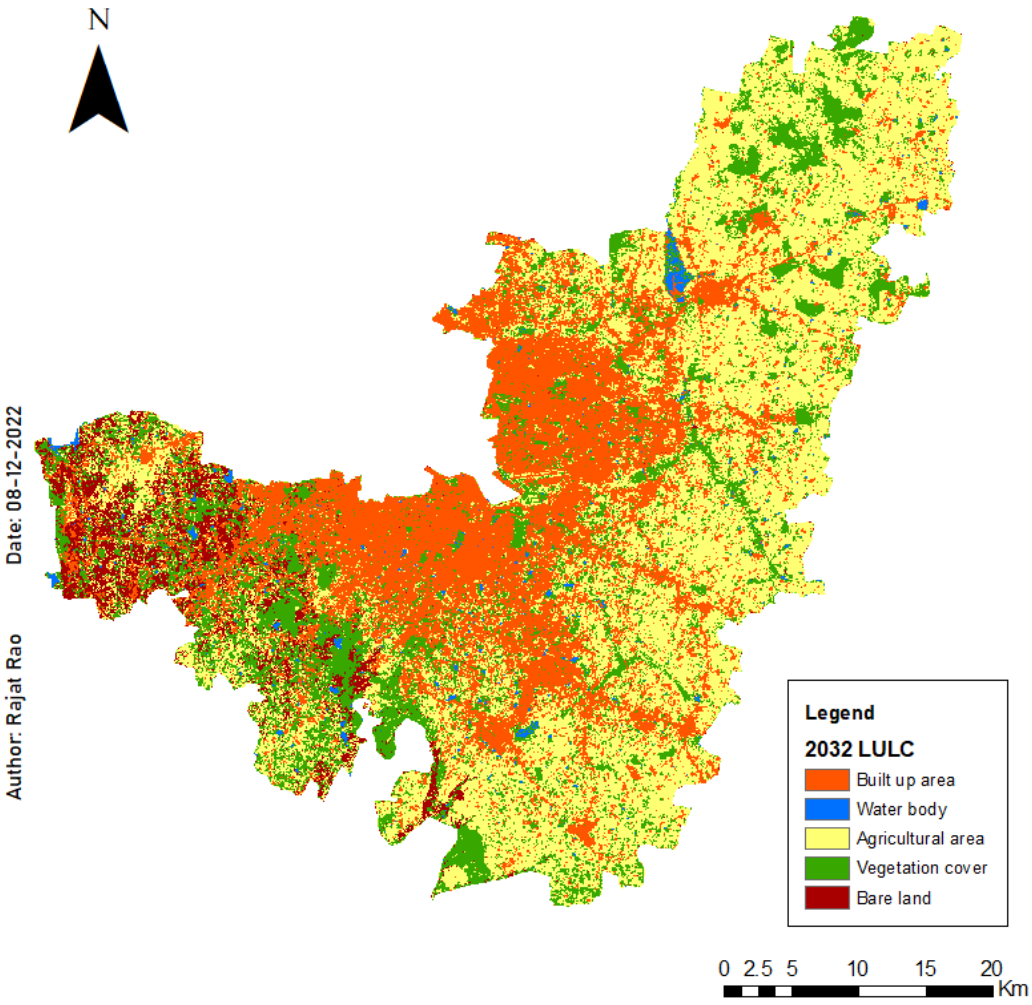


Figure 5.25. Showing 2032 future map. The future map is of 10 years in the future where the fragmentation seems to have only increased and the build up area also has increased.

5.6. Habitat quality and rarity

InVEST Habitat Quality model requires quite a bit of pre-preparation of data before the actual model is executed and the results obtained. Once all the data is prepared in the required format. It is added into the model and executed. It is observed that habitat quality is preserved in certain hotspots within the study region. Similarly, habitat rarity is restricted to small pockets and shows a pattern of reduction in the future maps. The following are the outputs obtained with respect to habitat quality and rarity:

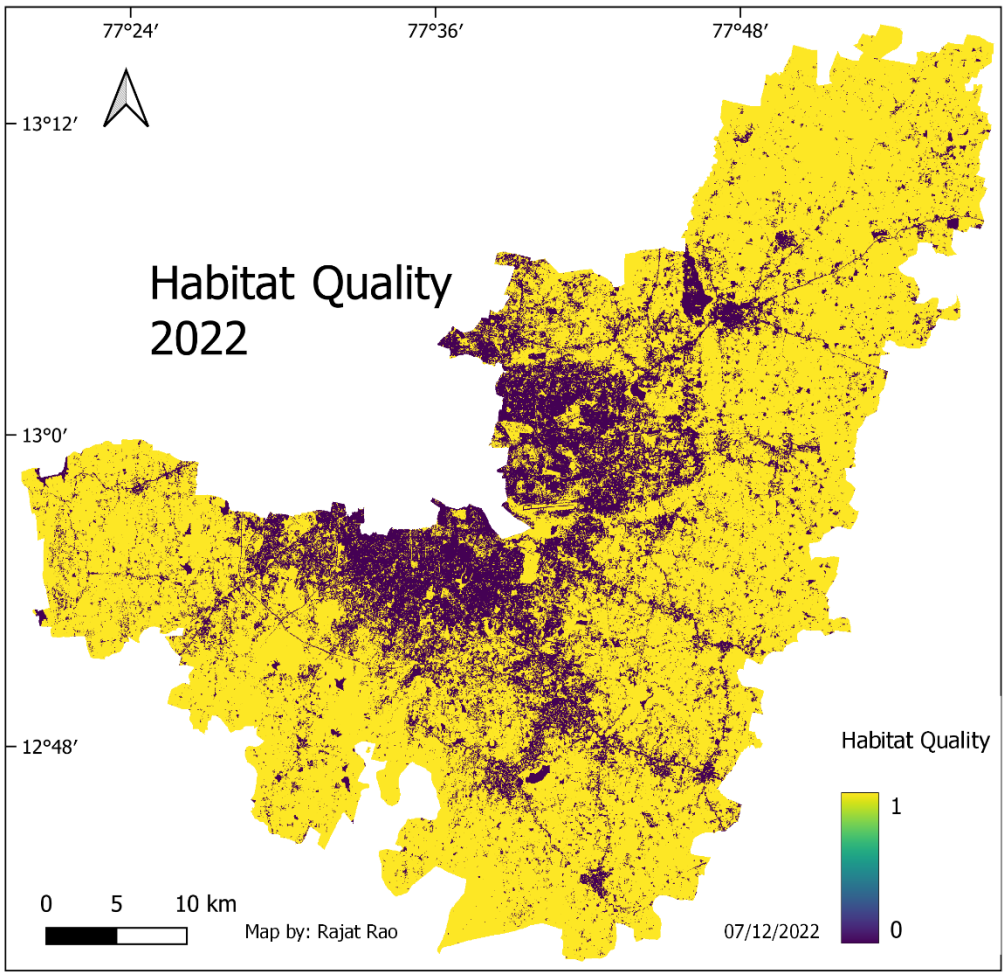


Figure 5.26. Habitat quality for the year 2022. The map shows habitat quality with protected areas having the highest habitat quality. In this map agricultural areas are also considered as habitat since they form a good abode for snakes and birds.

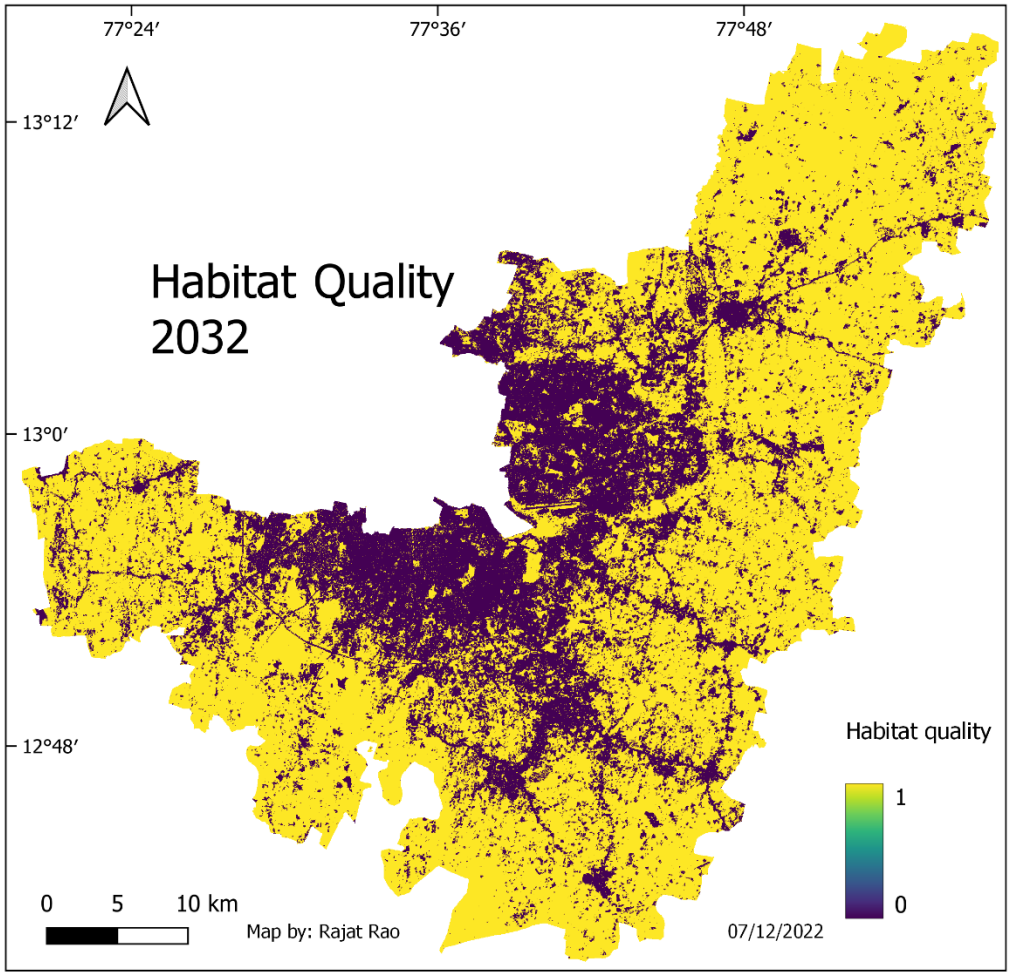


Figure 5.27. Habitat quality for the year 2032. The future habitat quality map shows a decrease in habitat quality with increase in urban regions. The main threat to habitats is considered to be purely built up areas since it does not allow any sustenance of species.

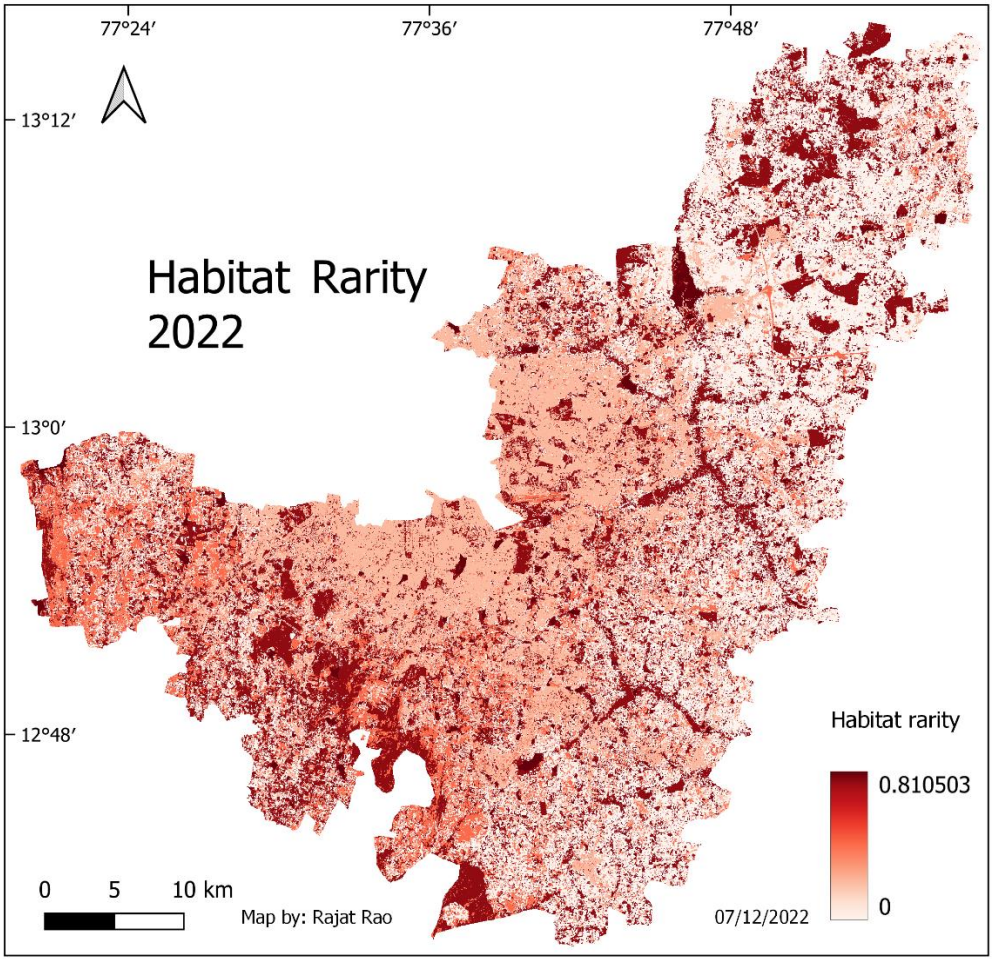


Figure 5.28. Habitat rarity for the year 2022. *The map shows habitat rarity where areas that are within built up areas but are green and protected have more rarity due to not being harmed despite being in urban spaces.*

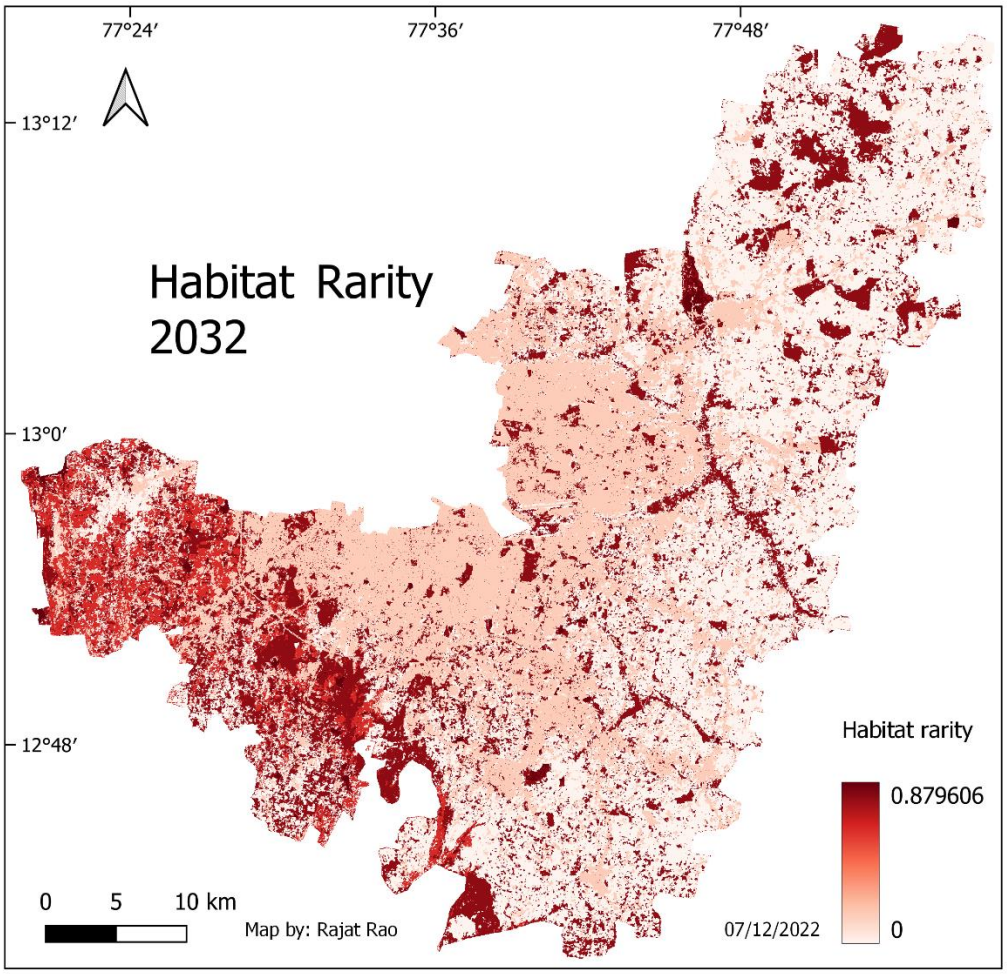


Figure 5.29. Habitat rarity for the year 2032. The rarity of future map shows that it has reduced by quite a bit but also the spaces which are rare the rarity values have increased because of the increase in uniqueness of the region.

5.7. Contextual maps for discussion

In this section we explore different maps as a part of the results, that provide contextual grounds for further discussion based on the area and habitat change data that was obtained in the previous sections. The following is the map of ecological assets in the study region.

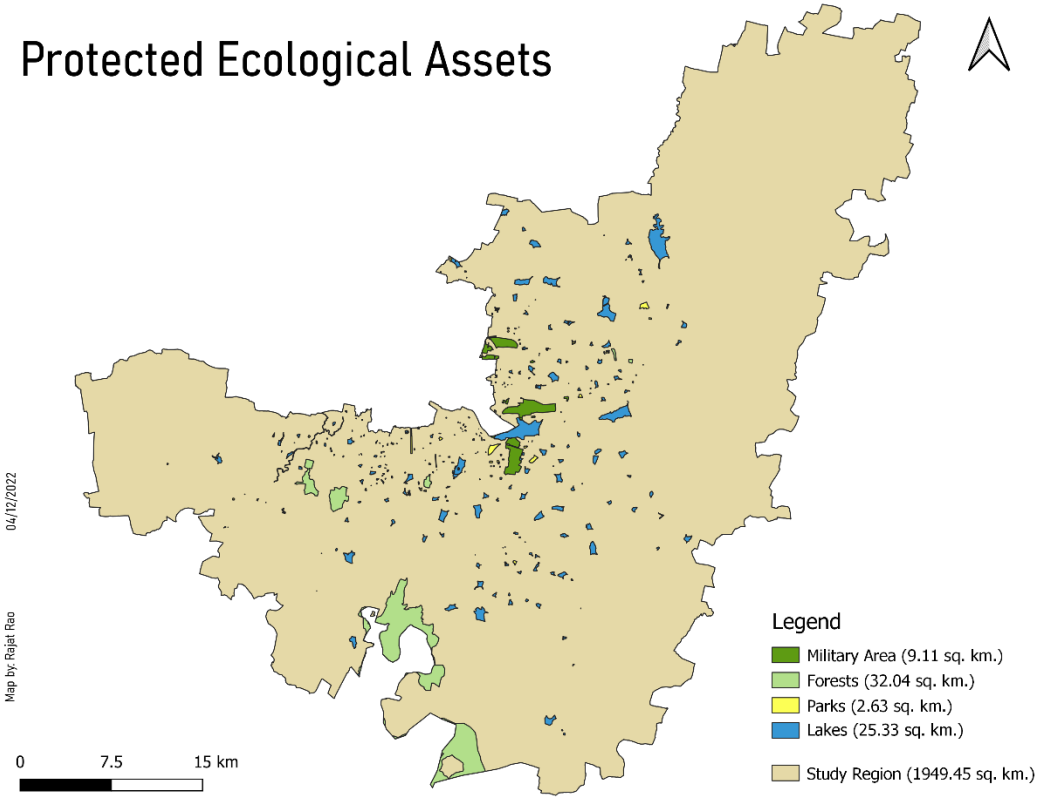


Figure 5.30. Map of protected ecological assets. The map shows regions that are protected under government laws. These are the regions which have a higher probability of not degrading over time.

The following map shows the 449 different Tech. Parks and IT companies that were documented as a part of the city-wide surveys held during the beginning of the study.

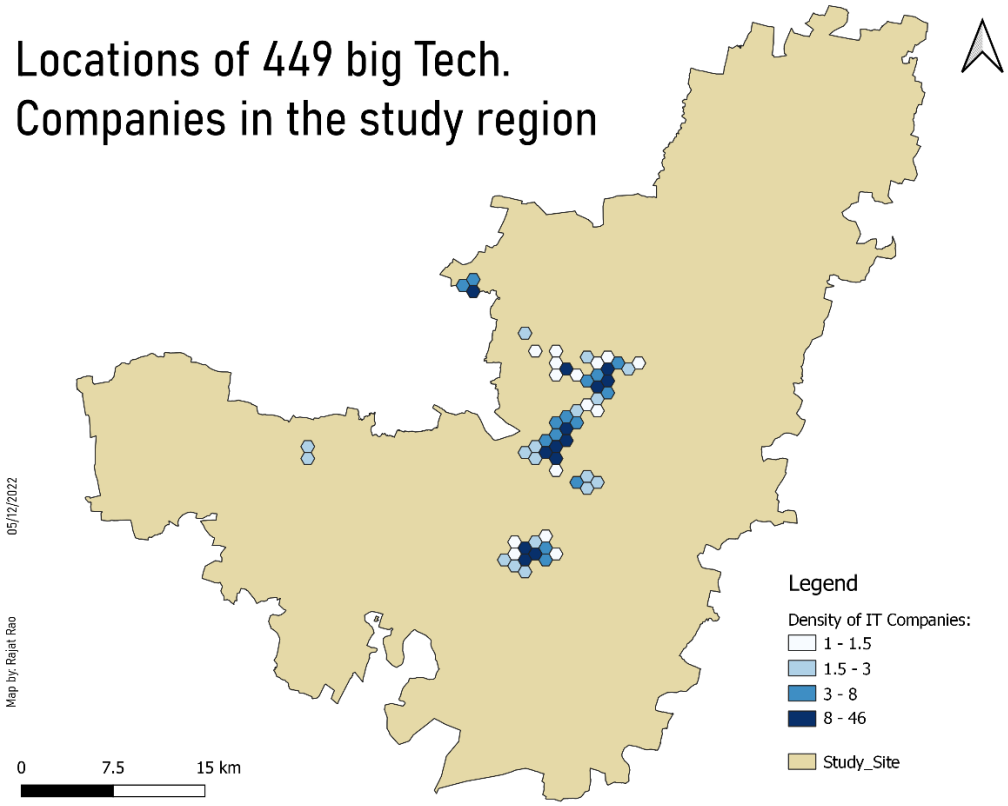


Figure 5.31. Map of big Technological companies in the study region. The density map shows the distribution of the number of offices and Tech. Parks in the hexagon.

The next map shows the distribution of high-rise condominium complexes distributed across the outskirts of the city. In the most recent survey done during the thesis, many more are under construction and there is expansion happening by occupying low lying areas (mostly flood plains of lakes).

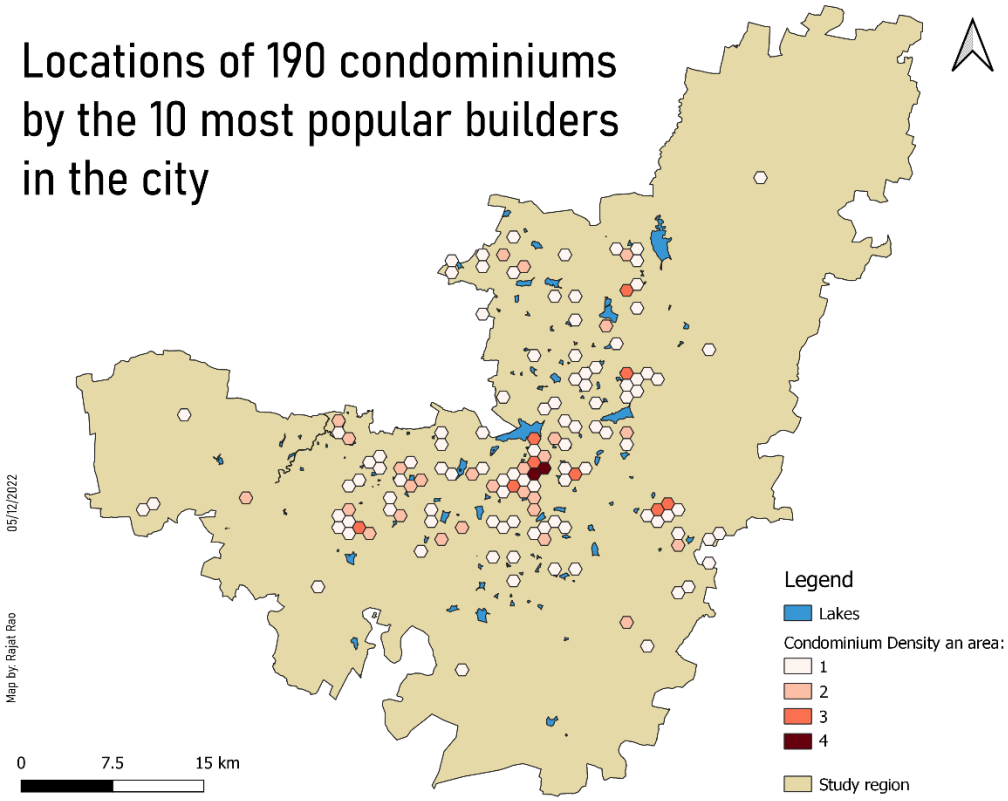


Figure 5.32. Map of condominium complexes in the study region. *This map shows the spread of condominium complexes within the study region with each hexagon representing 1 – 4 high rise apartment complexes or gated communities.*

These maps set context for the following chapter, which explores scope for further thesis discussion.

6. DISCUSSION

6.1. Inherent nature of the study: degradation caused by IT expansion

Bengaluru has seen two phases of IT industry in the past 25 years. The growth phase (1998 – 2010) and the expansion phase (2011 – now) (137). There is an effort made in this study to document this journey through an ecological purview. The documented evidence shows an increase in urban regions in the outskirts of the city, from 9% of the area in 2002 to 21% in 2022. The vegetation cover has diminished from 29% in 2002 to 25% in 2022. The habitat quality (Figure 5.26.)

and rarity (Figure 5.28.) maps also show that there is intense fragmentation of vegetation cover in the study region. With the vast number of technological parks shown in Figure 5.31., Bengaluru is still looking to set up more parks in the outskirts of the city (138).

Unlike cities like Mumbai or Delhi who have their growths nearing saturation (139,140), Bengaluru does not have a dearth of space and looks to go on expanding since it is surrounded by an agricultural region. The presence of the IT industry in this belt is exasperating the rate of expansion and putting the whole resource distribution system in disarray. Centralization of industry sets forth a chain reaction which will eventually lead to complete loss of habitat and concretizing a once green region.

6.2. A complex flux of biodiversity and development. What is the solution?

While most similar studies (139,141) have focused on forest areas as a separate classification, it is not considered that way over here. The effort is to classify all green spaces, including agricultural areas as a habitat, because, despite them being around urban spaces, they still house many birds, butterfly and reptile population in the city (142,143). A recent news article hints at the outskirts of Bengaluru including Bannerghatta National Park, being home for nearly 70 different leopard individuals (144). This is a comment on the type of prime habitat in the region The region in the South of the study region shows fragmented green spaces which has common leopard sightings.

Figure 5.30., is a map of the study region showing different areas that come under the government, municipality, military, or the forest department protection. The study has brought out findings that there are plenty more green spaces and avenues still present in the region that are not given an institutional protection as the ones shown in the map. Figures 5.31 and 5.32., show the distribution of IT parks and high-rise condominium complexes in the region. When this is looked at with respect to the land use land cover of the year 2022, tells a story of urban growth and habitat loss of the region.

The IT companies act as the catalysts for change, it all starts with construction of an SEZ (Special economic zone) in the outskirts of the city, this attracts plenty of companies because SEZs do not have to pay taxes for imports of technology while getting other benefits as well(145). Now the people working in these offices need a place to stay and opt for staying close to their workplace instead of staying far off and commuting in the city's infamous traffic. This leads to construction of many high rise buildings, most of which are built by destroying the flood plains of lakes or the forest cover of the region (146). It puts an immense stress on the underground water tables since each condominium has multiple borewells dug for the lack of supply of municipality waters to these regions due to them being out of the core of the city.

The high-rise apartments also lead to development of slums in the outer peripheries where people come from faraway places looking for job prospects with respect to housekeeping and maintenance of these condominium complexes. A chain reaction of development like this, takes place often by ignoring the biodiversity aspect of the region, it also explains the sudden increase in leopards trespassing into these high-rise apartment complexes more frequently in the past few years (147).

6.3. Habitat quality of the region

The maps obtained from InVEST Habitat Quality Model show the habitat quality and habitat rarity of the present scenario along with a future scenario (Figures 5.26., 5.27., 5.28. and 5.29.) The interpretative maps show that habitat quality is exponentially better as one moves away from urban spaces. There is also a good number of habitats (ecological assets) available in the main cluster of built-up area as shown in the image below.

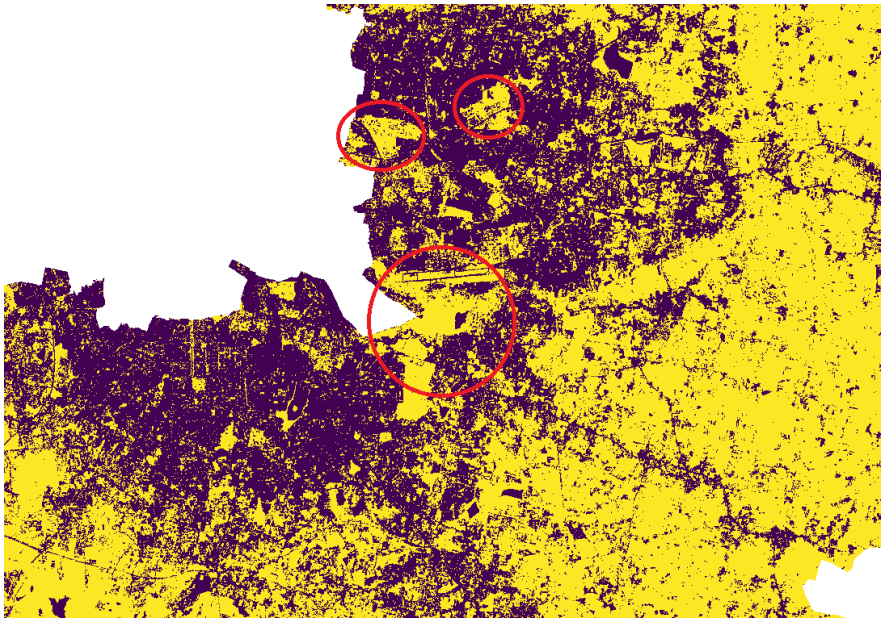


Figure 6.1. Image of ecological assets within the urban setting.

These are observed because of the presence of protected regions such as the Iblur firing range of the Indian Army along with many other spaces. Habitat quality is also observed to be lowest along the roads and better as the distance increases. These observations are in line with the evidence seen on field.

Habitat rarity shows the uniqueness of habitats as compared to its baseline data. Here, the clusters with more legal protection were considered the rarest and correspondingly it decreased. The rareness of the land use decreases in the spaces that have now become agricultural, showing that even though these regions house a good number of species, it does not serve as a prime habitat for endemic species. Parallels could be drawn to the fact that monoculture and fertilizer use limits the spread of species even though it houses a fair diversity compared to an urban setting. Rarity is more in urban regions since a green space within a quickly degrading landscape increases the rareness value of itself.

InVEST generically describes habitat as something that species can thrive in for a long period of time. To give a context of the species that are recorded and delve in the study region, there is a list of species in the Appendix section of this documentation, obtained from data provided by IUCN and Bird Life International. After the data extraction processes, 72 species of mammals, 18 species of frogs, 64 species of scaled reptiles and 303 species of birds were observed in the study region.

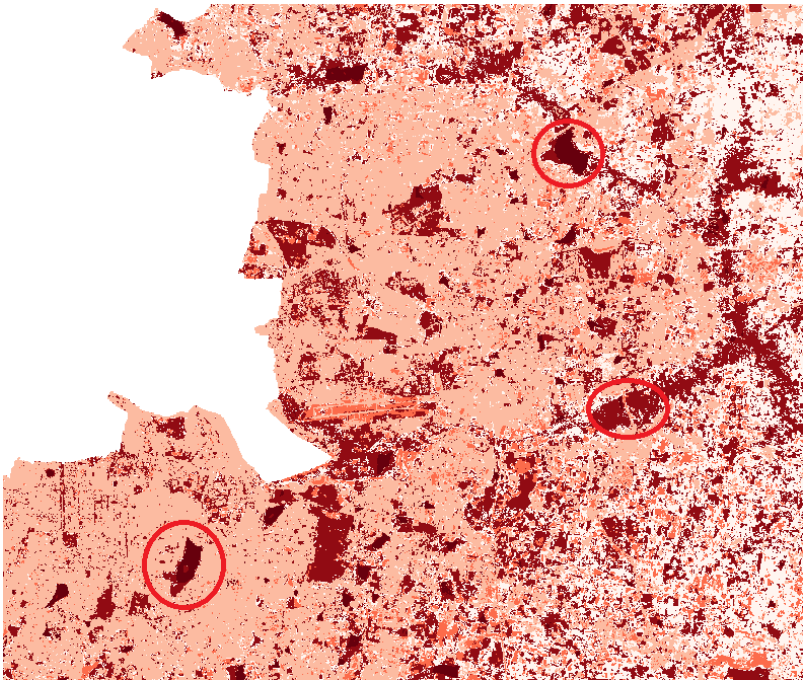


Figure 6.2. Habitat rarity within a built-up setting.

6.4. Limitations and scope for future implementation

The study is designed with a coarse filter conservation outlook, in which it looks at the conservation of the region that species occupy rather than the species themselves. This is different from the type of conservation practiced by many NGOs in India today, where they have one species as a flagship species, on the image of which conservation efforts are built. In urban – rural – wilderness interaction regions and studies based on fringe areas, a study like this holds more impact than going species specific. Moreover, due to immense fragmentation, even if diversity is documented in the region, it may not have a long-term viability if connectivity of habitats is not maintained through green spaces and avenues. Having said that, a few limitations are there that need to be discussed.

An aspect of InVEST model involves the use of prediction maps for understanding future habitat quality. This is obtained as an output of artificial neural network of the MOLUSCE plugin of QGIS. In the present study, the prediction of the future map is 55% accurate which according to Table 4.9. means moderate agreement with data. The model can be more accurate in creation future habitat quality if the Kappa co-efficient of the future map is made better.

Another limitation to the study is that it is a purely scientific study based on quantitative data. As a future implementation for this study, a stakeholder interview process can be established where people from all walks of life – IT employees, day wage workers, drivers, businessmen, municipality employees, farmers and forest department officials are interviewed and their vision for the future of the city's green spaces are recorded. It brings in a social touch and can be combined with the present study to holistically drive the conservation issue to a cognizable solution.

7. CONCLUSION

When it comes to protection of wilderness areas, forests are the first to be considered all the time. While the they face their own threats, there are a vast expansion of unprotected lands which are lost every day to farming, industrial growth and urbanization. This study looked to assess the impact of urban growth on the outskirts of Bengaluru city along its IT belt. While most regions in the outskirts sees upscaled development, with wealthy neighbourhoods and condominium complexes, these are putting a stress on the limited resources and the biodiversity of the region.

The citizens in the past few years have begun to appreciate the local biodiversity of birds, butterflies, and reptiles. There are multiple birding groups who have enthusiastic citizens taking park in nature walks and trails around the vegetation covers of the city. There is reason to believe that the citizens themselves are turning over a new leaf and understanding the importance of these green spaces, but they alone cannot instigate change without having relevant data. Hence, a study like this is needed to make the authorities understand about what is at stake and how development plans can be done in a decentralized approach by looking at alternate cities for IT expansion.

7.1. Conservation action plan

As part of the conservation action plan for this region, a further study needs to be carried out which looks to document the social aspect of the study like recording the vision for the city, from people of all walks of life. This would make the study a social and an ecological one. The combined study is to be then presented to the Forest department and the Department of Town Planning of BBMP (the municipality) and a complete multistakeholder body is to be formed (BBMP's monthly citizen's meet is a good platform for initiation) that looks at making it a win-win situation for all parties involved.

The conservation action plan can include three main stakeholders along with citizens forming the ancillary stakeholders –

The municipal council for whom short workshops are to be prepared, highlighting the outcomes of the study and suggestions to go forward with an ecosystem services approach for a green town planning having indigenous plantations and increasing tree avenues and forest parks.

The builder lobby, need to be educated about ways in which they can make money out of having green facades and biodiversity centric condominium complexes rather than making it entirely concrete. The IT companies, under the Corporate social responsibility (CSR) scheme, this data can be used for an awareness program about the houses that they choose to live in and the amount of stress they are causing on the ecology.

It is a known fact that conservation issues can never be solved when it is looked through a lens of only black and white. Addressing the greys of the problem is the only way forward and this can be done only through a multi-stakeholder approach having people from every walk of life participating in these interactions to arrive at greener, more amicable solution.

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APPENDIX

Anura:

Sl. No.	Common Name	Binomial Nomenclature	Legend	Category
1	Asian common toad	*Duttaphrynus melanostictus*	Extant (resident)	LC
2	Ferguson's toad	*Duttaphrynus scaber*	Extant (resident)	LC
3	Indian marbled toad	*Duttaphrynus stomaticus*	Extant (resident)	LC
4	Banded bullfrog	*Kaloula pulchra*	Extant (resident)	LC
5	Indian painted frog	*Uperodon taprobanicus*	Extant (resident)	LC
6	Ornate narrow-mouthed frog	*Microhyla ornata*	Extant (resident)	LC
7	Narrow-mouthed frog Sp.	*Microhyla rubra*	Extant (resident)	LC
8	Eluru dot frog	*Uperodon variegatus*	Extant (resident)	LC
9	Indian balloon frog	*Uperodon globulosus*	Extant (resident)	LC
10	Marbled balloon frog	*Uperodon systoma*	Extant (resident)	LC
11	Indian skipper frog	*Euphlyctis cyanophlyctis*	Extant (resident)	LC
12	Indian green frog	*Euphlyctis hexadactylus*	Extant (resident)	LC
13	Boie's wart frog	*Fejervarya limnocharis*	Extant (resident)	LC
14	Jerdon's bullfrog	*Hoplobatrachus crassus*	Extant (resident)	LC
15	Indian bullfrog	*Hoplobatrachus tigerinus*	Extant (resident)	LC
16	Fungoid frog	*Hydrophylax malabaricus*	Extant (resident)	LC
17	Indian burrowing frog	*Sphaerotheca breviceps*	Extant (resident)	LC
18	Indian tree frog	*Polypedates maculatus*	Extant (resident)	LC

Table A1. IUCN data of frogs documented in the study region.

Mammals:

Sl. No.	Common Name	Binomial Nomenclature	Legend	Category
1	Lesser short-nosed fruit bat	*Cynopterus brachyotis*	Extant (resident)	LC
2	Asian elephant	*Elephas maximus*	Extant (resident)	EN
3	Dhole	*Cuon alpinus*	Extant (resident)	EN
4	Four-horned antelope	*Tetracerus quadricornis*	Extant (resident)	VU
5	Stripe-necked mongoose	*Herpestes vitticollis*	Extant (resident)	LC
6	Madras treeshrew	*Anathana ellioti*	Extant (resident)	LC
7	Ruddy mongoose	*Herpestes smithii*	Extant (resident)	LC
8	Indian roundleaf bat	*Hipposideros lankadiva*	Extant (resident)	LC
9	Gaur	*Bos gaurus*	Extant (resident)	VU
10	Chital	*Axis axis*	Extant (resident)	LC
11	Leopard	*Panthera pardus*	Extant (resident)	VU
12	Rufous horseshoe bat	*Rhinolophus rouxii*	Extant (resident)	LC
13	Peyton's whiskered bat	*Myotis peytoni*	Extant (resident)	DD
14	Black-footed gray langur	*Semnopithecus hypoleucos*	Extant (resident)	LC
15	Gray slender loris	*Loris lydekkerianus*	Extant (resident)	NT
16	Tufted gray langur	*Semnopithecus priam*	Extant (resident)	NT
17	Indian flying fox	*Pteropus giganteus*	Extant (resident)	LC
18	Lesser bandicoot rat	*Bandicota bengalensis*	Extant (resident)	LC
19	Kelaart's pipistrelle bat	*Pipistrellus ceylonicus*	Extant (resident)	LC
20	Asiatic long-tailed climbing mouse	*Vandeleuria oleracea*	Extant (resident)	LC
21	Indian pipistrelle bat	*Pipistrellus coromandra*	Extant (resident)	LC
22	Dusky leaf-nosed bat	*Hipposideros ater*	Extant (resident)	LC
23	Lesser Asiatic yellow bat	*Scotophilus kuhlii*	Extant (resident)	LC
24	Leschenault's rousette	*Rousettus leschenaultii*	Extant (resident)	NT
25	Greater short-nosed fruit bat	*Cynopterus sphinx*	Extant (resident)	LC
26	Greater Asiatic yellow bat	*Scotophilus heathii*	Extant (resident)	LC
27	Indian muntjac	*Muntiacus vaginalis*	Extant (resident)	LC
28	Indian hare	*Lepus nigricollis*	Extant (resident)	LC
29	Greater bandicoot rat	*Bandicota indica*	Extant (resident)	LC
30	Small Indian civet	*Viverricula indica*	Extant (resident)	LC
31	Sambar deer	*Rusa unicolor*	Extant (resident)	VU
32	Indian bush rat	*Golunda ellioti*	Extant (resident)	LC
33	Rock-loving mouse	*Mus saxicola*	Extant (resident)	LC
34	Fulvus roundleaf bat	*Hipposideros fulvus*	Extant (resident)	LC

35	Nilgai	*Boselaphus tragocamelus*	Extant (resident)	LC
36	Schneider's leaf-nosed bat	*Hipposideros speoris*	Extant (resident)	LC
37	Bengal fox	*Vulpes bengalensis*	Extant (resident)	LC
38	Flat-haired mouse	*Mus platythrix*	Extant (resident)	LC
39	Bonnet macaque	*Macaca radiata*	Extant (resident)	VU
40	Phillips's mouse	*Mus phillipsi*	Extant (resident)	LC
41	Cutch rat	*Cremnomys cutchicus*	Extant (resident)	LC
42	Kolar leaf-nosed bat	*Hipposideros hypophyllus*	Extant (resident)	CR
43	Blanford's rat	*Madromys blanfordi*	Extant (resident)	LC
44	Earth-colored mouse	*Mus terricolor*	Extant (resident)	LC
45	Indian pangolin	*Manis crassicaudata*	Extant (resident)	EN
46	Little Indian field mouse	*Mus booduga*	Extant (resident)	LC
47	Rusty-spotted cat	*Prionailurus rubiginosus*	Extant (resident)	NT
48	Blackbuck	*Antilope cervicapra*	Extant (resident)	LC
49	Soft-furred rat	*Millardia meltada*	Extant (resident)	LC
50	Sloth bear	*Melursus ursinus*	Possibly Extant (resident)	VU
51	Indian palm squirrel	*Funambulus palmarum*	Extant (resident)	LC
52	Indian Grey Wolf	*Canis lupus*	Extant (resident)	LC
53	Golden jackal	*Canis aureus*	Extant (resident)	LC
54	Indian gerbil	*Tatera indica*	Extant (resident)	LC
55	Indian grey mongoose	*Herpestes edwardsii*	Extant (resident)	LC
56	Lesser mouse-tailed bat	*Rhinopoma hardwickii*	Extant (resident)	LC
57	Greater false vampire bat	*Lyroderma lyra*	Extant (resident)	LC
58	Blyth's horseshoe bat	*Rhinolophus lepidus*	Extant (resident)	LC
59	Wild boar	*Sus scrofa*	Extant (resident)	LC
60	Wrinkle-lipped free-tailed bat	*Chaerephon plicatus*	Extant (resident)	LC
61	Long-winged tomb bat	*Taphozous longimanus*	Extant (resident)	LC
62	Asian palm civet	*Paradoxurus hermaphroditus*	Extant (resident)	LC
63	Indian crested porcupine	*Hystrix indica*	Extant (resident)	LC
64	Jungle cat	*Felis chaus*	Extant (resident)	LC
65	Tiger	*Panthera tigris*	Extinct	EN
66	Asian house shrew	*Suncus murinus*	Extant (resident)	LC
67	Striped hyena	*Hyaena hyaena*	Extant (resident)	NT
68	Cheetah	*Acinonyx jubatus*	Extinct	VU
69	Honey badger	*Mellivora capensis*	Extant (resident)	LC
70	Egyptian free-tailed bat	*Tadarida aegyptiaca*	Extant (resident)	LC

71	Black rat	*Rattus rattus*	Extant & Introduced (resident)	LC
72	House mouse	*Mus musculus*	Extant & Origin Uncertain (resident)	LC

Table A2. IUCN data of mammals documented in the study region.

Birds:

Sl. No.	Common Name	Binomial Nomenclature	Presence	Origin	Seasonal
1	Streak-throated swallow	*Petrochelidon fluvicola*	1	1	3
2	Thick-billed warbler	*Arundinax aedon*	1	1	4
3	Indian blue robin	*Larvivora brunnea*	1	1	4
4	Rain quail	*Coturnix coromandelica*	1	1	1
5	Ashy woodswallow	*Artamus fuscus*	1	1	1
6	Tickell's leaf warbler	*Phylloscopus affinis*	1	1	4
7	Mongolian short-toed lark	*Calandrella dukhunensis*	1	1	3
8	Yellow-legged buttonquail	*Turnix tanki*	1	1	1
9	Crested treeswift	*Hemiprocne coronata*	1	1	1
10	Indian spotted eagle	*Clanga hastata*	1	1	1
11	White-browed wagtail	*Motacilla maderaspatensis*	1	1	1
12	Chestnut-tailed starling	*Sturnia malabarica*	1	1	3
13	Blue-capped rock thrush	*Monticola cinclorhyncha*	1	1	4
14	Indian paradise flycatcher	*Terpsiphone paradisi*	1	1	3
15	Black-rumped flameback	*Dinopium benghalense*	1	1	1
16	Indian peafowl	*Pavo cristatus*	1	1	1
17	Red-naped ibis	*Pseudibis papillosa*	1	1	1
18	Jungle babbler	*Argya striata*	1	1	1
19	Indian cormorant	*Phalacrocorax fuscicollis*	1	1	1
20	Small pratincole	*Glareola lactea*	1	1	1
21	Rufous treepie	*Dendrocitta vagabunda*	1	1	1
22	Grey-headed canary-flycatcher	*Culicicapa ceylonensis*	1	1	3
23	Painted stork	*Mycteria leucocephala*	1	1	1

24	Yellow-crowned woodpecker	*Leiopicus mahrattensis*	1	1	1
25	Black-bellied tern	*Sterna acuticauda*	1	1	1
26	Dusky crag martin	*Ptyonoprogne concolor*	1	1	1
27	Western spotted dove	*Spilopelia suratensis*	1	1	1
28	Richard's pipit	*Anthus richardi*	1	1	3
29	Red-necked falcon	*Falco chicquera*	1	1	1
30	Indian white-eye	*Zosterops palpebrosus*	1	1	1
31	Common woodshrike	*Tephrodornis pondicerianus*	1	1	1
32	Yellow-eyed babbler	*Chrysomma sinense*	1	1	1
33	Yellow-footed green pigeon	*Treron phoenicopterus*	1	1	1
34	Clamorous reed warbler	*Acrocephalus stentoreus*	1	1	3
35	Citrine wagtail	*Motacilla citreola*	1	1	3
36	Asian green bee-eater	*Merops orientalis*	1	1	1
37	Red-wattled lapwing	*Vanellus indicus*	1	1	1
38	Purple sunbird	*Cinnyris asiaticus*	1	1	1
39	Glossy ibis	*Plegadis falcinellus*	1	1	3
40	Blyth's pipit	*Anthus godlewskii*	1	1	3
41	Bluethroat	*Cyanecula svecica*	1	1	3
42	Ruddy shelduck	*Tadorna ferruginea*	1	1	3
43	Red-whiskered bulbul	*Pycnonotus jocosus*	1	1	1
44	Black redstart	*Phoenicurus ochruros*	1	1	3
45	Great grey shrike	*Lanius excubitor*	1	1	3
46	Rock dove	*Columba livia*	1	1	1
47	Lesser whitethroat	*Curruca curruca*	1	1	4
48	Black-tailed godwit	*Passer domesticus*	1	1	1
49	Eurasian collared dove	*Streptopelia decaocto*	1	1	1
50	Eurasian bittern	*Botaurus stellaris*	1	1	3
51	Baillon's crake	*Zapornia pusilla*	1	1	4
52	Short-eared owl	*Asio flammeus*	1	1	3
53	Eurasian woodcock	*Scolopax rusticola*	1	1	3
54	White-rumped vulture	*Gyps bengalensis*	4	1	1
55	Ashy drongo	*Dicrurus leucophaeus*	1	1	3
56	Red-headed vulture	*Sarcogyps calvus*	4	1	1
57	Jungle myna	*Acridotheres fuscus*	1	1	1
58	Steppe eagle	*Aquila nipalensis*	1	1	3
59	Eurasian hoopoe	*Upupa epops*	1	1	1
60	Shikra	*Accipiter badius*	1	1	1
61	Barred buttonquail	*Turnix suscitator*	1	1	1

62	Eurasian teal	*Anas crecca*	1	1	3
63	Northern pintail	*Anas acuta*	1	1	3
64	Eurasian wigeon	*Mareca penelope*	1	1	3
65	Great tit	*Parus major*	1	1	1
66	Tufted duck	*Aythya fuligula*	1	1	3
67	Eurasian sparrowhawk	*Accipiter nisus*	1	1	3
68	White wagtail	*Motacilla alba*	1	1	3
69	Northern shoveler	*Spatula clypeata*	1	1	3
70	Spotted redshank	*Tringa erythropus*	1	1	3
71	White-throated kingfisher	*Halcyon smyrnensis*	1	1	1
72	Eurasian wryneck	*Jynx torquilla*	1	1	3
73	Temminck's stint	*Calidris temminckii*	1	1	3
74	Pheasant-tailed jacana	*Hydrophasianus chirurgus*	1	1	3
75	Common kestrel	*Falco tinnunculus*	1	1	3
76	Little ringed plover	*Charadrius dubius*	1	1	1
77	Indian roller	*Coracias benghalensis*	1	1	1
78	Oriental skylark	*Alauda gulgula*	1	1	1
79	Rose-ringed parakeet	*Alexandrinus krameri*	1	1	1
80	Jacobin cuckoo	*Clamator jacobinus*	1	1	1
81	Montagu's harrier	*Circus pygargus*	1	1	3
82	Intermediate egret	*Ardea intermedia*	1	1	1
83	Paddyfield pipit	*Anthus rufulus*	1	1	1
84	Cotton pygmy goose	*Nettapus coromandelianus*	1	1	1
85	Pin-tailed snipe	*Gallinago stenura*	1	1	3
86	Asian brown flycatcher	*Muscicapa dauurica*	1	1	3
87	King quail	*Synoicus chinensis*	1	1	1
88	Bonelli's eagle	*Aquila fasciata*	1	1	1
89	Black kite	*Milvus migrans*	1	1	1
90	Pacific swift	*Apus pacificus*	1	1	3
91	Common moorhen	*Gallinula chloropus*	1	1	1
92	Little grebe	*Tachybaptus ruficollis*	1	1	1
93	Great egret	*Ardea alba*	1	1	1
94	Black-crowned night heron	*Nycticorax nycticorax*	1	1	1
95	Laughing dove	*Spilopelia senegalensis*	1	1	1
96	Peregrine falcon	*Falco peregrinus*	1	1	3
97	Barn owl	*Tyto alba*	1	1	1
98	Eurasian coot	*Fulica atra*	1	1	1
99	Grey heron	*Ardea cinerea*	1	1	1
100	Common snipe	*Gallinago gallinago*	1	1	3

101	Grey wagtail	*Motacilla cinerea*	1	1	3
102	Garganey	*Spatula querquedula*	1	1	3
103	Blue rock thrush	*Monticola solitarius*	1	1	3
104	Amur falcon	*Falco amurensis*	1	1	4
105	Long-tailed shrike	*Lanius schach*	1	1	1
106	Brahminy kite	*Haliastur indus*	1	1	1
107	Marsh sandpiper	*Tringa stagnatilis*	1	1	3
108	Barn swallow	*Hirundo rustica*	1	1	3
109	Wire-tailed swallow	*Hirundo smithii*	1	1	1
110	Wood sandpiper	*Tringa glareola*	1	1	3
111	Whiskered tern	*Chlidonias hybrida*	1	1	3
112	Osprey	*Pandion haliaetus*	1	1	3
113	Greater painted-snipe	*Rostratula benghalensis*	1	1	1
114	Common greenshank	*Tringa nebularia*	1	1	3
115	Common sandpiper	*Actitis hypoleucos*	1	1	3
116	Purple heron	*Ardea purpurea*	1	1	1
117	Western swamphen	*Porphyrio porphyrio*	1	1	1
118	Cattle egret	*Bubulcus ibis*	1	1	1
119	Western Yellow Wagtail	*Motacilla flava*	1	1	3
120	Western marsh harrier	*Circus aeruginosus*	1	1	3
121	Pallid harrier	*Circus macrourus*	1	1	3
122	Common redshank	*Tringa totanus*	1	1	3
123	Long-billed pipit	*Anthus similis*	1	1	1
124	Green sandpiper	*Tringa ochropus*	1	1	3
125	Common kingfisher	*Alcedo atthis*	1	1	1
126	Tawny eagle	*Aquila rapax*	1	1	1
127	Pied kingfisher	*Ceryle rudis*	1	1	1
128	Striated heron	*Butorides striata*	1	1	1
129	Fulvous whistling duck	*Dendrocygna bicolor*	1	1	1
130	Little egret	*Egretta garzetta*	1	1	1
131	Black-winged stilt	*Himantopus himantopus*	1	1	1
132	Great cormorant	*Phalacrocorax carbo*	1	1	1
133	Zitting cisticola	*Cisticola juncidis*	1	1	1
134	Asian palm swift	*Cypsiurus balasiensis*	1	1	1
135	Thick-billed flowerpecker	*Dicaeum agile*	1	1	1
136	Asian koel	*Eudynamys scolopaceus*	1	1	1
137	Crested honey buzzard	*Pernis ptilorhynchus*	1	1	1
138	Woolly-necked stork	*Ciconia episcopus*	1	1	1
139	Short-toed snake eagle	*Circaetus gallicus*	1	1	1

140	White-rumped Shama	*Kittacincla malabarica*	1	1	1
141	Lesser yellownape	*Picus chlorolophus*	1	1	1
142	Crested serpent eagle	*Spilornis cheela*	1	1	1
143	Rufous woodpecker	*Micropternus brachyurus*	1	1	1
144	Slaty-breasted rail	*Lewinia striata*	1	1	1
145	Ruddy-breasted crake	*Zapornia fusca*	1	1	1
146	Red-rumped swallow	*Cecropis daurica*	1	1	3
147	Common quail	*Coturnix coturnix*	1	1	3
148	Chestnut-bellied sandgrouse	*Pterocles exustus*	1	1	1
149	Laggar falcon	*Falco jugger*	1	1	1
150	River tern	*Sterna aurantia*	1	1	1
151	White-breasted waterhen	*Amaurornis phoenicurus*	1	1	1
152	Swinhoe's snipe	*Gallinago megala*	1	1	3
153	Large-billed crow	*Corvus macrorhynchos*	1	1	1
154	Oriental magpie-robin	*Copsychus saularis*	1	1	1
155	Pied harrier	*Circus melanoleucos*	1	1	3
156	Red collared dove	*Streptopelia tranquebarica*	1	1	1
157	Common myna	*Acridotheres tristis*	1	1	1
158	White-rumped munia	*Lonchura striata*	1	1	1
159	Brown shrike	*Lanius cristatus*	1	1	3
160	Orange minivet	*Pericrocotus flammeus*	1	1	1
161	Golden-headed cisticola	*Cisticola exilis*	1	1	1
162	Orange-headed thrush	*Geokichla citrina*	1	1	1
163	Olive-backed pipit	*Anthus hodgsoni*	1	1	3
164	Hair-crested drongo	*Dicrurus hottentottus*	1	1	1
165	Common tailorbird	*Orthotomus sutorius*	1	1	1
166	Greater coucal	*Centropus sinensis*	1	1	1
167	Scaly-breasted munia	*Lonchura punctulata*	1	1	1
168	Plain prinia	*Prinia inornata*	1	1	1
169	Eurasian spoonbill	*Platalea leucorodia*	1	1	1
170	Green warbler	*Phylloscopus nitidus*	1	1	4
171	Painted sandgrouse	*Pterocles indicus*	1	1	1
172	Lesser florican	*Sypheotides indicus*	4	1	3
173	Nilgiri wood pigeon	*Columba elphinstonii*	1	1	1
174	White-cheeked barbet	*Psilopogon viridis*	1	1	1
175	Red spurfowl	*Galloperdix spadicea*	1	1	1
176	Painted spurfowl	*Galloperdix lunulata*	1	1	1

177	Loten's sunbird	*Cinnyris lotenius*	1	1	1
178	Large cuckooshrike	*Coracina macei*	1	1	1
179	Jungle owlet	*Glaucidium radiatum*	1	1	1
180	Brown-capped pygmy woodpecker	*Picoides nanus*	1	1	1
181	Indian golden oriole	*Oriolus kundoo*	1	1	3
182	Rosy starling	*Pastor roseus*	1	1	3
183	Indian grey hornbill	*Ocyceros birostris*	1	1	1
184	Brown-headed barbet	*Psilopogon zeylanicus*	1	1	1
185	Mottled wood owl	*Strix ocellata*	1	1	1
186	Lesser cuckoo	*Cuculus poliocephalus*	1	1	4
187	Grey junglefowl	*Gallus sonneratii*	1	1	1
188	Painted francolin	*Francolinus pictus*	1	1	1
189	Indian blackbird	*Turdus simillimus*	1	1	3
190	White-bellied minivet	*Pericrocotus erythropygius*	1	1	1
191	Rock bush quail	*Perdicula argoondah*	1	1	1
192	Sykes's warbler	*Iduna rama*	1	1	3
193	Sirkeer malkoha	*Taccocua leschenaultii*	1	1	1
194	Plum-headed Parakeet	*Himalayapsitta cyanocephala*	1	1	1
195	Yellow-wattled lapwing	*Vanellus malabaricus*	1	1	1
196	Indian courser	*Cursorius coromandelicus*	1	1	1
197	Yellow-throated sparrow	*Gymnoris xanthocollis*	1	1	1
198	Indian pitta	*Pitta brachyura*	1	1	4
199	Large grey babbler	*Argya malcolmi*	1	1	1
200	White-naped tit	*Machlolophus nuchalis*	1	1	1
201	Eastern Orphean warbler	*Curruca crassirostris*	1	1	3
202	Egyptian vulture	*Neophron percnopterus*	1	1	1
203	Lesser whitethroat	*Curruca curruca*	1	1	3
204	Indian scops owl	*Otus bakkamoena*	1	1	1
205	Crested hawk-eagle	*Nisaetus cirrhatus*	1	1	1
206	Brown boobook	*Ninox scutulata*	1	1	1
207	Velvet-fronted nuthatch	*Sitta frontalis*	1	1	1
208	Indian pond heron	*Ardeola grayii*	1	1	1
209	Common buttonquail	*Turnix sylvaticus*	1	1	1
210	Horsfield's bush lark	*Mirafra javanica*	1	1	1
211	Tree pipit	*Anthus trivialis*	1	1	3
212	Booted eagle	*Hieraaetus pennatus*	1	1	3
213	Knob-billed duck	*Sarkidiornis melanotos*	1	1	1

214	Chestnut-winged cuckoo	*Clamator coromandus*	1	1	3
215	Spot-billed pelican	*Pelecanus philippensis*	1	1	1
216	Black-headed ibis	*Threskiornis melanocephalus*	1	1	1
217	Bar-winged flycatcher-shrike	*Hemipus picatus*	1	1	1
218	Small minivet	*Pericrocotus cinnamomeus*	1	1	1
219	Lesser whistling duck	*Dendrocygna javanica*	1	1	1
220	Coppersmith barbet	*Psilopogon haemacephalus*	1	1	1
221	Common iora	*Aegithina tiphia*	1	1	1
222	Watercock	*Gallicrex cinerea*	1	1	1
223	Banded bay cuckoo	*Cacomantis sonneratii*	1	1	1
224	Black-naped monarch	*Hypothymis azurea*	1	1	1
225	Black drongo	*Dicrurus macrocercus*	1	1	1
226	Baya weaver	*Ploceus philippinus*	1	1	1
227	Little Cormorant	*Microcarbo niger*	1	1	1
228	Streaked weaver	*Ploceus manyar*	1	1	1
229	Black-hooded oriole	*Oriolus xanthornus*	1	1	1
230	Greater racket-tailed drongo	*Dicrurus paradiseus*	1	1	1
231	Oriental darter	*Anhinga melanogaster*	1	1	1
232	Bronzed drongo	*Dicrurus aeneus*	1	1	1
233	Long-toed stint	*Calidris subminuta*	1	1	3
234	Pied bush chat	*Saxicola caprata*	1	1	1
235	Red avadavat	*Amandava amandava*	1	1	1
236	Blue-tailed bee-eater	*Merops philippinus*	1	1	3
237	Yellow-throated bulbul	*Pycnonotus xantholaemus*	1	1	1
238	Pied thrush	*Geokichla wardii*	1	1	4
239	Black-winged kite	*Elanus caeruleus*	1	1	3
240	Blue-faced malkoha	*Phaenicophaeus viridirostris*	1	1	1
241	Asian emerald cuckoo	*Chrysococcyx maculatus*	1	1	3
242	Ultramarine Flycatcher	*Ficedula superciliaris*	1	1	3
243	Painted bush quail	*Perdicula erythrorhyncha*	1	1	1
244	Malabar parakeet	*Nicopsitta columboides*	1	1	1
245	Streak-throated woodpecker	*Picus xanthopygaeus*	1	1	1
246	Ashy minivet	*Pericrocotus divaricatus*	1	1	3
247	Indian cuckoo	*Cuculus micropterus*	1	1	1

248	Lesser fish eagle	*Icthyophaga humilis*	1	1	1
249	Orange-breasted green pigeon	*Treron bicinctus*	1	1	1
250	Indian eagle-owl	*Bubo bengalensis*	1	1	1
251	Indian bush lark	*Mirafra erythroptera*	1	1	1
252	Red-vented bulbul	*Pycnonotus cafer*	1	1	1
253	Indian spot-billed duck	*Anas poecilorhyncha*	1	1	1
254	Indian nightjar	*Caprimulgus asiaticus*	1	1	1
255	White-browed fantail	*Rhipidura aureola*	1	1	1
256	Grey-breasted prinia	*Prinia hodgsonii*	1	1	1
257	Asian openbill	*Anastomus oscitans*	1	1	1
258	Golden-fronted leafbird	*Chloropsis aurifrons*	1	1	1
259	Indian thick-knee	*Burhinus indicus*	1	1	1
260	Blyth's reed warbler	*Acrocephalus dumetorum*	1	1	3
261	Bar-headed goose	*Anser indicus*	1	1	3
262	Wood snipe	*Gallinago nemoricola*	1	1	3
263	Blue-throated blue flycatcher	*Cyornis rubeculoides*	1	1	4
264	Ashy-crowned sparrow-lark	*Eremopterix griseus*	1	1	1
265	Brahminy starling	*Sturnia pagodarum*	1	1	1
266	Common hawk-cuckoo	*Hierococcyx varius*	1	1	1
267	Ashy prinia	*Prinia socialis*	1	1	1
268	Jungle bush quail	*Perdicula asiatica*	1	1	1
269	White-spotted fantail	*Rhipidura albogularis*	1	1	1
270	Indian vulture	*Gyps indicus*	4	1	1
271	Indian scimitar babbler	*Pomatorhinus horsfieldii*	1	1	1
272	Tickell's blue flycatcher	*Cyornis tickelliae*	1	1	1
273	White-naped woodpecker	*Chrysocolaptes festivus*	1	1	1
274	Jungle nightjar	*Caprimulgus indicus*	1	1	1
275	Tawny-bellied babbler	*Dumetia hyperythra*	1	1	1
276	Rufous-tailed lark	*Ammomanes phoenicura*	1	1	1
277	Jungle prinia	*Prinia sylvatica*	1	1	1
278	Oriental turtle dove	*Streptopelia orientalis*	1	1	3
279	Pale-billed flowerpecker	*Dicaeum erythrorhynchos*	1	1	1
280	Purple-rumped sunbird	*Leptocoma zeylonica*	1	1	1
281	Oriental hobby	*Falco severus*	1	1	3

282	White-browed bulbul	*Pycnonotus luteolus*	1	1	1
283	Booted warbler	*Iduna caligata*	1	1	3
284	Jerdon's leafbird	*Chloropsis jerdoni*	1	1	1
285	White-bellied drongo	*Dicrurus caerulescens*	1	1	1
286	Jerdon's bush lark	*Mirafra affinis*	1	1	1
287	Grey-bellied cuckoo	*Cacomantis passerinus*	1	1	1
288	Brown fish owl	*Ketupa zeylonensis*	1	1	1
289	White-eyed buzzard	*Butastur teesa*	1	1	1
290	Indian robin	*Saxicoloides fulicatus*	1	1	1
291	Bay-backed shrike	*Lanius vittatus*	1	1	1
292	House crow	*Corvus splendens*	1	1	1
293	Common babbler	*Argya caudata*	1	1	1
294	Spotted owlet	*Athene brama*	1	1	1
295	Grey francolin	*Francolinus pondicerianus*	1	1	1
296	Indian silverbill	*Euodice malabarica*	1	1	1
297	Yellow-billed babbler	*Argya affinis*	1	1	1
298	Tricoloured munia	*Lonchura malacca*	1	1	1
299	Savanna nightjar	*Caprimulgus affinis*	1	1	3
300	Great Indian bustard	*Ardeotis nigriceps*	4	1	1
301	Sykes's lark	*Galerida deva*	1	1	1
302	Large-billed leaf warbler	*Phylloscopus magnirostris*	1	1	4
303	Black-headed cuckooshrike	*Lalage melanoptera*	1	1	1

Table A3. Bird Life International data of birds documented in the study region.

Scaled Reptiles:

Sl. No.	Common Name	Binomial Nomenclature	Legend	Category
1	Saw-scaled Viper	*Echis carinatus*	Extant (resident)	LC
2	Streaked Kukri Snake	*Oligodon taeniolatus*	Extant (resident)	LC
3	Indian rat snake	*Ptyas mucosa*	Extant (resident)	LC
4	Common house gecko	*Hemidactylus frenatus*	Extant (resident)	LC
5	Bangalore slender gecko	*Hemiphyllodactylus jnana*	Extant (resident)	LC
6	Whitaker's termite hill gecko	*Hemidactylus whitakeri*	Extant (resident)	LC

7	Hemidactylus graniticolus	*Hemidactylus graniticolus*	Extant (resident)	LC
8	Reticulate Leaf-toed Gecko	*Hemidactylus reticulatus*	Extant (resident)	LC
9	South Indian rock agama	*Psammophilus dorsalis*	Extant (resident)	LC
10	Yellow-collared wolf snake	*Lycodon flavicollis*	Extant (resident)	LC
11	Giant Leaf-toed Gecko	*Hemidactylus giganteus*	Extant (resident)	LC
12	Common Bridal Snake	*Lycodon nympha*	Extant (resident)	LC
13	Yellow-green cat snake	*Boiga flaviviridis*	Extant (resident)	LC
14	Beaked worm snake	*Grypotyphlops acutus*	Extant (resident)	LC
15	Treutler's gecko	*Hemidactylus treutleri*	Extant (resident)	LC
16	Allapalli grass skink	*Eutropis allapallensis*	Extant (resident)	LC
17	Scarce bridle snake	*Lycodon gracilis*	Extant (resident)	DD
18	Southern banded kukri	*Oligodon arnensis*	Extant (resident)	LC
19	Travancore wolf snake	*Lycodon travancoricus*	Extant (resident)	LC
20	Leschenault's snake-eye	*Ophisops leschenaultii*	Extant (resident)	LC
21	Leschenault's leaf-toed gecko	*Hemidactylus leschenaultii*	Extant (resident)	LC
22	Beddome's skink	*Eutropis beddomei*	Extant (resident)	LC
23	Bamboo pit viper	*Trimeresurus gramineus*	Extant (resident)	LC
24	Slender coral snake	*Calliophis melanurus*	Extant (resident)	LC
25	Blanford's rock agama	*Psammophilus blanfordanus*	Extant (resident)	LC
26	Indian three-banded skink	*Eutropis trivittata*	Extant (resident)	LC
27	Common bronzeback	*Dendrelaphis tristis*	Extant (resident)	LC
28	Split keelback	*Atretium schistosum*	Extant (resident)	LC
29	Keeled Indian mabuya	*Eutropis carinata*	Extant (resident)	LC
30	Gunther's supple skink	*Lygosoma guentheri*	Extant (resident)	LC
31	Roux's forest lizard	*Monilesaurus rouxii*	Extant (resident)	LC

32	Jerdon's many-toothed snake	*Sibynophis subpunctatus*	Extant (resident)	LC
33	Green vine snake	*Ahaetulla nasuta*	Extant (resident)	LC
34	Brahminy blind snake	*Indotyphlops braminus*	Extant (resident)	LC
35	Common dotted garden skink	*Lygosoma punctata*	Extant (resident)	LC
36	Common trinket snake	*Coelognathus helenae*	Extant (resident)	LC
37	Spectacled cobra	*Naja naja*	Extant (resident)	LC
38	Common krait	*Bungarus caeruleus*	Extant (resident)	LC
39	Calamaria reed snake	*Liopeltis calamaria*	Extant (resident)	LC
40	Forsten's cat snake	*Boiga forsteni*	Extant (resident)	LC
41	Indian chameleon	*Chamaeleo zeylanicus*	Extant (resident)	LC
42	Indian wolf snake	*Lycodon aulicus*	Extant (resident)	LC
43	Green keelback snake	*Rhabdophis plumbicolor*	Extant (resident)	LC
44	Common Sand Boa	*Eryx conicus*	Extant (resident)	NT
45	Banded Racer	*Argyrogena fasciolata*	Extant (resident)	LC
46	Indian gamma snake	*Boiga trigonata*	Extant (resident)	LC
47	Jerdon's cabrita lizard	*Ophisops jerdonii*	Extant (resident)	LC
48	Brown Sand Boa	*Eryx johnii*	Extant (resident)	NT
49	Indian rock python	*Python molurus*	Extant (resident)	NT
50	Yellow-belly gecko	*Hemidactylus flaviviridis*	Extant (resident)	LC
51	Russell's Viper	*Daboia russelii*	Extant (resident)	LC
52	Barred wolf snake	*Lycodon striatus*	Extant (resident)	LC
53	Indian Flying Snake	*Chrysopelea ornata*	Extant (resident)	LC
54	Bronze grass skink	*Eutropis macularia*	Extant (resident)	LC
55	Slender Worm Snake	*Indotyphlops porrectus*	Extant (resident)	LC
56	Buff striped keelback	*Amphiesma stolatum*	Extant (resident)	LC

57	Mourning gecko	*Lepidodactylus lugubris*	Extant (resident)	LC
58	Checkered keelback	*Fowlea piscator*	Extant (resident)	LC
59	Common Indian monitor lizard	*Varanus bengalensis*	Extant (resident)	NT
60	Indian garden lizard	*Calotes versicolor*	Extant (resident)	LC
61	Brooke's house gecko	*Hemidactylus brookii*	Extant (resident)	LC
62	Travancore Kukri Snake	*Oligodon travancoricus*	Extant (resident)	DD
63	Mysore Day Gecko	*Cnemaspis mysoriensis*	Extant (resident)	LC
64	Bangalore Geckoella	*Cyrtodactylus srilekhae*	Extant (resident)	NT

Table A4. IUCN data of scaled reptiles documented in the study region.